

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



February 21, 2007

MATERIALS LABORATORY FACTUAL REPORT

Report No. 06-091

A. ACCIDENT

Place : Boston, Massachusetts
Date : July 10, 2006
Vehicle : Connector tunnel to Ted Williams Tunnel
NTSB No. : HWY06MH024
Investigator : Mark Bagnard

B. COMPONENTS EXAMINED

Twenty adhesive anchors from the failure area (ceiling module EB081, middle1 row),
Laser scanned images of the twenty failure anchor holes and selected others,
Additional adhesive anchors (57) and core samples, not from the failure area.

C. BACKGROUND

The adhesive anchoring system used to secure the ventilation ducting to the roof of the tunnel included 8 inch long by 5/8-inch diameter, 11 threads per inch, stainless steel threaded rods (anchors) inserted into $\frac{3}{4}$ inch diameter holes in the concrete roof. The anchors were secured in the holes using epoxy injected into the hole before insertion of the anchor. Also, a red polyethylene cap¹ was inserted in the open end of the hole before insertion of the anchor. The anchor was to be inserted through the cap, which would remain in position around the anchor at the lower end of the hole.² Typically, 3 inches of the threaded anchor protruded from below the tunnel roof, and the ventilation duct hanger system attached to this portion of the rod. Figure 1 shows a drawing of a typical anchor installation on a ceiling fitting.

The failure anchor holes were from the "M1" row of module 081 of the eastbound (EB) tunnel. The label applied to an individual anchor hole reflects the tunnel (EB), the module (such as module 081), the row (such as the "M1" row), the hanger or hangers associated with the ceiling fitting at the anchor (numbered from east to west), and the position of the anchor in the ceiling fitting. Thus, anchor hole "EB081-M1-1-N" is the hole for the north anchor associated with hanger 1 (the most eastern hanger) in the M1 row of

¹ The caps contained markings indicating that they were W-10 caps manufactured by Caplugs.

² For a more detailed description of the entire anchor installation procedure see the Structures Group Chairman's Factual Report.

module 081 from the eastbound tunnel, and anchor hole “EB081-M1-9/10/11-SE” is the hole for the southeast anchor associated with the ceiling fitting with the ninth, tenth, and eleventh hangers in the M1 row of module 081 of the east-bound tunnel.

All 20 of the adhesive anchors from the failure area reportedly remained within ceiling fittings immediately following the accident, trapped between the flange on the cap above the fittings and the nut and washers on the anchor below the fittings (see figure 1). During the recovery process, 16 of the 20 anchors were removed from ceiling fittings. There was no record made of the fittings from which these 16 anchors were removed. The Massachusetts State Police labeled these 16 anchors 1-1, 1-2, 1-3, 1-4, 2-1, 2-2, 2-3, 2-4, 3-1, 3-2, 3-3, 3-4, 4-2³, 5-1, 5-2, 5-3 as they were recovered from the accident scene. Of the other four adhesive anchors, two remained in a two-anchor ceiling fitting, one remained in another two-anchor fitting, and one remained in a four-anchor fitting. All three of these fittings had been separated from other structure during the recovery process. The two-anchor fittings were not labeled, and it is not known which of the six two-anchor fitting they were. The four-anchor fitting was labeled as being the EB081-M1-9/10/11 fitting.

At some point between the initial on-scene examination and August 11, 2006, the anchor retained in the four-anchor fitting was removed from its ceiling fitting and labeled as anchor E1-1. The three anchors in the two-anchor fittings were not removed from ceiling fittings during this time period.

The 17 anchors from the failure area that had been removed from their fittings were transported by the undersigned (James Wildey) from the Boston area to the NTSB Materials Laboratory on August 11, 2006. Representatives of the Massachusetts State Police hand delivered the three remaining anchors from the failure area, still contained within their ceiling fittings, to the Materials Laboratory at a later date. The labels 6-1, 6-2, and 6-3 were applied to these final three anchors once they were received in the laboratory. An additional 57 anchors not from the failure area⁴ were also examined. Of these additional anchors, 25 were transported to the Materials Laboratory with the bulk of the anchors from the failure area, 31 were delivered by a Massachusetts State Police Trooper, and 1 was sent using a delivery service.

Polymer characterization testing was performed on epoxy samples from most of the anchors from the failure area and from selected anchors outside of the failure area. The results of this testing are presented in a separate report.

³ An adhesive anchor labeled 4-1 was also recovered from the accident scene and received in the Materials Laboratory. However, this anchor was of a smaller diameter than the 5/8 inch diameter anchors from the ceiling hanger system, and it was not further examined.

⁴ Federal Highway Administration personnel removed 49 of these 57 additional anchors and many others from the EB and west-bound (WB) tunnel roofs as part of the on-scene portion of the accident investigation. An additional 4 anchors were associated with core samples removed from sustained load test areas, and a private contractor, Advance Testing, removed the final 4 anchors after the on-scene portion of the accident investigation. FHWA found the methods and procedures used by Advance Testing were in general conformance with the methods and procedures used by FHWA.

D. EXAMINATION OF ANCHORS FROM THE FAILURE AREA

The majority of the anchors from the failure area (the 13 anchors labeled 1-1, 1-2, 1-3, 1-4, 2-1, 2-2, 2-3, 2-4, 3-1, 3-2, 3-3, 3-4, and 4-2) did not have red polyethylene caps when received in the Materials Laboratory. Examination of other anchors removed from the tunnel roof as part of the accident investigation showed that the caps were sufficiently tight around the anchor that it was difficult to move the caps relative to the anchors. Also, photographs taken before significant recovery efforts showed that the flanges on the red polyethylene caps were generally sufficient to retain the anchors on the ceiling fitting after separation from the ceiling. It was reported that some of the anchors were removed from ceiling fittings by application of force to the upper end of the anchor, pushing the embedment length of the anchor through the cap, and allowing the anchor to be removed from the ceiling fitting.

The remaining anchors from the failure area (the 7 anchors labeled E1-1, 5-1, 5-2, 5-3, 6-1, 6-2, and 6-3) were received with their red polyethylene caps in place around the anchor. On anchor E1-1, the cap was displaced across most of the embedment length and was located near the upper end of the anchor, as shown in figure 2. On the other six anchors, the cap was approximately in its installed position on the anchor. Based on the presence of an intact flange on the cap on anchors E1-1, 5-1, 5-2, and 5-3, it was apparent that the anchor nut had been removed from the bottom of the anchor, allowing the anchor to be removed from the ceiling fitting by being pulled upward. This method was also used in the Materials Laboratory to remove anchors 6-1, 6-2, and 6-3 from their fittings. Photographs of the 20 anchors from the failure location are shown in Appendix 1, pages 1 through 20 after removal of the caps. Each anchor is shown from four views equally spaced around the anchor circumference.

All of the anchors were stamped with "316" on each end, consistent with a marking for 316 stainless steel. X-ray energy dispersive spectroscopic analysis of anchor 6-2 generated a spectrum consistent with a 316 stainless steel alloy, with major peaks for iron, chromium and nickel, and smaller peaks for manganese, carbon, and molybdenum. The undamaged surfaces of the anchors generally were smooth and reflective, and no evidence of corrosion was noted.

The anchors from the failure area were examined for (1) the presence of damage on the anchors, including the location and amount of any bending deformation, (2) the extent of epoxy coverage on the anchor and (3) the epoxy's condition. Features associated with the epoxy included the following conditions.

Void areas are unbonded epoxy surfaces along the embedment length of the anchor, created where epoxy did not completely fill the gap between the anchor and hole wall. The void areas were very smooth with a greenish tint. In locations where void areas were deformed because they were adjacent to fractured or displaced epoxy, the color of the void surface changed to white, approximately the color of the fractured regions of epoxy. Figure 3 shows an example of a large void area on anchor 5-3.

Fractured epoxy is epoxy along the embedment length of the anchor that has been sheared or otherwise separated, typically resulting in exposure of the crowns of the anchor threads or creation of a pattern corresponding to the thread spacing. An example of fractured epoxy on anchor 2-1 is shown in figure 4.

Adhesive failure describes areas along the embedment length of the anchor where the epoxy appears to have separated from the concrete surface of the anchor hole and then pulled out of the hole, remaining attached to the anchor. The adhesive failures were thicker areas of epoxy usually with a smooth white surface. Figure 4 also shows an example of adhesive failure.

Yellowed epoxy describes fractured (or, in some cases, adhesively failed) epoxy that had a distinct yellow tint. Figure 5 shows an example of yellowed epoxy on anchor 1-2. As can be seen in figure 5, the epoxy over most of the length of the anchor was fractured, and the fractured epoxy closer to the installed position of the red polyethylene cap had a yellow appearance. A region of yellowed epoxy, when it was present on an anchor, was always located on the lower portion of the embedment length, and an area of fractured epoxy without the yellow discoloration was found above the discolored area. The boundaries between the yellowed epoxy and regions without the yellow discoloration were usually not distinct. The yellow appearance typically faded over a distance of $\frac{1}{4}$ to $\frac{1}{2}$ inch. In some cases, the boundary of the yellowed epoxy region was not circumferential and was slightly above the edge of a void area, as shown in figure 6 for anchor 5-3.

Overflow epoxy is epoxy that extended down below the position of the cap and onto the threads of the anchor. It was also documented if the nut threads engaged this overflow epoxy (based on crushing and damage to the overflow epoxy) and if the anchor had a black mark on the anchor threads just below the cap. Figure 7 shows an example of overflow epoxy from anchor 3-3 after removal of the nut and lock washer. The area of overflow epoxy shown in figure 7 had been contacted by the anchor nut, and the overflow epoxy contained a black ink mark just below the cap position.

In order to estimate the fraction of the cylindrical surface of potential bond area between the anchor and hole wall attributed to void area, adhesive failure, and missing epoxy, these features were mapped out on graph paper. The area of each individual feature was measured using a graphics and analysis software program. The void area fraction was in relation to the embedment length of the anchor and typically did not include the smooth surface of the epoxy created where the epoxy was in contact with the red polyethylene cap. In some cases, void areas did extend down into the cap area, and these void areas are included in the void area total. No attempt was made to measure the void volume fraction. The area fraction value listed for the cap includes both the cylindrical portion of the cap and the area of the four flaps that extend upward against the anchor as the anchor is inserted through the cap. The fraction of the embedment length listed for the

cap is reduced by the amount of any void areas found in the cap area. See figure 7 for an example of a void area extending down into the cap position for anchor 3-3.

For each anchor examined, Table 1 lists the calculated area fraction of embedment covered by the cap, occupied by void areas, with adhesive failure, and from which the epoxy was missing. The remaining percent not listed represents the area fraction with fractured epoxy. It should be noted that much of the epoxy originally retaining the anchor within the tunnel roof remained within the anchor holes or could have been lost from the anchors during the accident or during recovery and removal of the anchors from the structure. Because evidence of voids might have been located in this epoxy that remained within the hole or was otherwise missing from the anchor, the area fraction of void areas listed in table 1 should be considered a minimum. Similarly, areas of adhesive failure or fractured epoxy could have been present in areas where the epoxy was missing from the anchor or damaged. Table 1 also lists the maximum upward extent of the yellowed epoxy, as measured from the bottom of the cap, and the distance above which only non-yellowed epoxy was found⁵, also measured from the bottom of the cap.

The details of the examination of each of the anchors from the failure area are presented in the section below. Unless otherwise stated, anchor nuts that could not be moved by hand forces were unthreaded by placing two similar nuts on the lower end of the anchor, locking these nuts together, and then using a wrench to unthread the anchor nut relative to the added nut pair until the anchor nut was loose. As previously stated, photographs of each anchor are contained in appendix 1.

Anchor E1-1. The red polyethylene cap for this anchor was retained around the anchor's upper end, as shown in figure 2, indicating that the anchor had been removed from its fitting plate by removal of the anchor nut. The anchor was not visibly bent. Small areas of impact damage were noted on the anchor's lower end. Epoxy was missing from major areas over the lower two thirds of the embedment length. A void area, largely surrounded by missing epoxy, was noted above the position of the cap. The void area had a length of about 1.5 inches and extended about half way around the anchor's circumference, as indicated in figure 2. Epoxy at the top of the anchor had been fractured, and yellowed epoxy was noted below about 3 ½ inches above the position corresponding to the bottom of the red polyethylene cap. This anchor was received with (starting from the lowest item) a nut, a lock washer, a sealing (rubber/metal) washer, and a second lock washer assembled on the lower end of the anchor. A part of the rubber portion of the sealing washer had separated from the remainder of this washer, and this separated portion was compressed into and remained trapped on the anchor threads above the as-received location of the sealing washer. This was the only anchor received with two lock washers, and it was noted that the orientation of the sealing washer was reversed from the typical

⁵ These numbers are the same for anchors that had epoxy present at the transition between the yellowed and non-yellowed epoxy, and are different if epoxy was missing from the transition area. An "EM" in table 1 indicates that epoxy was missing from the lower portion of the embedment length; therefore, the upper extent of yellowed epoxy, if present, could not be determined.

orientation⁶, indicating that it had been removed then replaced incorrectly. Overflow epoxy extended 0.5 inch below the cap, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was noted on the overflow epoxy.

Anchor 1-1. The red polyethylene cap for this anchor was not received with the anchor. The anchor was bent 23 degrees with the center of the bend at about 3.3 inches above the anchor's lower end. Severe mechanical damage was noted on the embedded portion of the anchor threads on the outside of the bend above the center of the bend, with heavier damage in the lower portion of the damaged length. Epoxy was largely missing from the lower two thirds of the embedment length as well as from some areas in the top one third. No void areas were noted in the epoxy remaining on the anchor, and the epoxy at the top of the anchor was fractured. No yellowed epoxy was noted. The anchor was received with (starting from the lowest item) a nut, a lock washer, and a sealing washer with only small pieces of the rubber portion. Overflow epoxy extended approximately 1 inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was noted on the overflow epoxy.

Anchor 1-2. The red polyethylene cap for this anchor was not received with the anchor. The anchor was not visibly bent, and it did not exhibit any significant mechanical damage. Epoxy was present over most of the embedment length. Void areas were noted above the cap area and along one side of the anchor along the top half of the embedment length. In other areas the epoxy was fractured. Yellowed epoxy extended up to about 3.8 inches above the position corresponding to the bottom of the red polyethylene cap, with white fractured epoxy above this position. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The lock washer was partially compressed between the sealing washer and nut, and the nut could not be turned on the anchor using hand forces. Overflow epoxy extended approximately 3/4 inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was noted on the overflow epoxy.

Anchor 1-3. The red polyethylene cap for this anchor was not received with the anchor. The anchor was bent 3 to 4 degrees with the center of the bend at about 3.5 inches above the anchor's lower end. Some mechanical damage was noted on the threads at the top of the anchor on the outside of the bend. Epoxy was almost entirely missing over the lower one half of the embedment length and intermittently for another inch above the mid point of the embedment length. In one small area that did contain epoxy in the lower half of the embedment length, a void area was found. Void areas were found along one side of the anchor along the upper portion of the embedment length. In other areas the epoxy was fractured. No yellowed epoxy was noted. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. Overflow epoxy extended approximately 3/4 inch below the cap position, and there was no evidence

⁶ The typical arrangement was a nut, a lock washer, and a sealing washer, with the rubber portion of the sealing washer facing up.

that the nut had been threaded onto the overflow epoxy. A black ink mark was not found on the overflow epoxy, but the overflow epoxy directly adjacent to the cap position was missing.

Anchor 1-4. The red polyethylene cap for this anchor was not received with the anchor. The anchor was bent 16 degrees with the center of the bend at about 3.4 inches above the anchor's lower end. Minor mechanical damage was noted on the upper end of the anchor, and this damage was not centered on the outside of the bend. Epoxy was missing from the cap area and within several inches above the cap. Small linear void areas were noted on one side of the upper portion of the embedment length. In other areas the epoxy was fractured. Yellowed epoxy extended up to about 2 inches above the position corresponding to the bottom of the red polyethylene cap, with white fractured epoxy above this position. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The nut could be moved by hand but did not turn smoothly on the anchor threads. Overflow epoxy extended down to the nut area, and the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 2-1. The red polyethylene cap for this anchor was not received with the anchor. The anchor was bent 2 degrees, with the center of the bend approximately 4 inches above the anchor's lower end. The anchor contained a heavy impact mark on the corner of its upper end on the outside of the bend. Epoxy was almost entirely missing over the lower 2 to 2.5 inches of the embedment length, including the cap area. A few linear void areas were noted on one side of the upper portion of the embedment length. In most other areas, the epoxy was fractured, but one region of what appeared to be adhesive failure was also noted, as shown in figure 4. No evidence of yellowed epoxy was noted on the epoxy remaining on the anchor. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The nut could be moved by hand, and it turned smoothly on the anchor threads. Overflow epoxy extended about $\frac{3}{4}$ inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 2-2. The red polyethylene cap for this anchor was not received with the anchor. The anchor was bent 42 degrees, with the center of the bend approximately 3.3 inches above the anchor's lower end. The anchor contained mechanical damage and what appeared to be remnants of concrete on the threads adjacent to the upper end of the anchor on the outside of the bend. Threads were also damaged on the outside of the bend in the epoxy overflow region and on the inside of the bend at the center of the bend. Epoxy was almost entirely missing over the entire embedment length of the anchor, including the cap area. A few threads were filled with fractured epoxy near the upper end of the anchor and one small patch with a void surface was found about 2 inches from the upper end. No evidence of yellowed epoxy was noted on the epoxy remaining on the anchor. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped

on the anchor threads. The lock washer was partially compressed between the sealing washer and nut, and the nut could not be turned on the anchor using hand forces. Overflow epoxy extended about $\frac{3}{4}$ inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A very small black ink mark was found on the overflow epoxy.

Anchor 2-3. The red polyethylene cap for this anchor was not received with the anchor. The anchor was bent 8 degrees, with the center of the bend approximately 5 inches above the anchor's lower end. The anchor contained mechanical damage and what appeared to be remnants of concrete on the threads adjacent to the upper end of the anchor, although this damage was not directly on the outside of the bend. Epoxy was missing over the lower 2 inches of the embedment depth, including the cap area, and over about half of the anchor circumference for several inches above this area. Voids in the epoxy were noted on one side of the anchor in a slightly spiral pattern, and in another small area near the top of the anchor. The remainder of the epoxy was fractured. No evidence of yellowed epoxy was noted on the epoxy remaining on the anchor. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The lock washer was partially compressed between the sealing washer and nut, and the nut could not be turned on the anchor using hand forces. Overflow epoxy extended about $\frac{3}{4}$ inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A small black ink mark was found on the overflow epoxy.

Anchor 2-4. The red polyethylene cap for this anchor was not received with the anchor. The anchor was bent 36 degrees, with the center of the bend approximately 3.5 inches above the anchor's lower end. Severe mechanical damage was noted on the embedded portion of the anchor threads on the outside of the bend above the center of the bend, with heavier damage in the lower portion of the damaged length. Epoxy was missing over the lower 2.5 inches of the embedment depth, including the cap area, mostly missing for another 1.5 inches above this area, and mostly present on the top 1 inch. Some void areas were noted in the epoxy near the top of the anchor, and the remainder of the epoxy was fractured. Some of the epoxy at about 2.5 inches above the position corresponding to the bottom of the red polyethylene cap was yellowed. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The lock washer was partially compressed between the sealing washer and nut, and the nut could not be turned on the anchor using hand forces. Overflow epoxy extended about 1 inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 3-1. The red polyethylene cap for this anchor was not received with the anchor. The anchor was not visibly bent, and it did not exhibit any significant mechanical damage. The epoxy was missing from about half the thread area for about 2 inches above the cap position, and from several individual threads above this area. Void areas, predominantly on one side of the anchor, were noted through most of the embedment length, and the remainder of the epoxy was fractured. An area of yellowed epoxy was noted up to about

1.2 inches above the position corresponding to the bottom of the red polyethylene cap. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The lock washer was partially compressed between the sealing washer and nut, and the nut could not be turned on the anchor using hand forces. Overflow epoxy extended about 1 inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 3-2. The red polyethylene cap for this anchor was not received with the anchor. The anchor was bent 7 to 8 degrees, with the center of the bend approximately 3.4 inches above the anchor's lower end. The anchor contained mechanical damage and what appeared to be remnants of concrete on the threads adjacent to the upper end of the anchor on the outside of the bend. Mechanical damage was also found on the threads adjacent to the lower end of the anchor, and a portion of the threads adjacent to the lower end of the anchor contained a rubber-like material. The epoxy was mostly missing from about 2.7 inches of the lower portion of its embedment length, including the cap area, and some epoxy was missing in the upper portion of the embedment length. Void areas covered most of the epoxy remaining in the lower portion of the embedment length, and an axially oriented void area, centered about 2 inches from the anchor's upper end was also noted. The remainder of the epoxy was fractured. No evidence of yellowed epoxy was noted on the anchor. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The lock washer was partially compressed between the sealing washer and nut, and the nut could not be turned on the anchor using hand forces. The lower end of the anchor was clamped in a vise, and the nut unthreaded toward the lower end of the anchor. Overflow epoxy extended about 1 inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 3-3. The red polyethylene cap for this anchor was not received with the anchor. The anchor was not visibly bent, and it did not exhibit any significant mechanical damage. The epoxy was missing from a portion of the cap area and from several threads on one side of the anchor approximately midway along the embedment length. A void area was found near the top of the anchor, for several inches above the cap, and in a portion of the cap area. The remaining epoxy was fractured, except for a small area at the top of the anchor, opposite from the void, that contained an adhesive failure area. The fractured epoxy was yellowed up to about 3 inches above the position corresponding to the bottom of the red polyethylene cap. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The lock washer was partially compressed between the sealing washer and nut, and the nut could not be turned on the anchor using hand forces. Overflow epoxy extended almost 2 inches below the cap position, and the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 3-4. The red polyethylene cap for this anchor was not received with the anchor. The anchor was not visibly bent, and it did not exhibit any significant mechanical damage. The epoxy was missing from various areas along the length of the anchor, including a portion of the cap area. A large void area, which extended around two thirds of the anchor circumference at its widest, was found above the cap position up to 2.3 inches above the bottom of the cap. The void area intersected an area of missing epoxy its upper end, and another smaller void area roughly in line with this larger void area was observed intersecting the upper end of the area of missing epoxy. This smaller void extended up to 3.1 inches above the bottom of the cap. Another area with multiple small voids was found on one side near the top of the anchor. The remaining epoxy was fractured. The fractured epoxy was yellowed up to 1.3 inches above the position corresponding to the bottom of the red polyethylene cap. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The lock washer was partially compressed between the sealing washer and nut, and the nut could not be turned on the anchor using hand forces. Overflow epoxy extended $\frac{3}{4}$ inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 4-2. The red polyethylene cap for this anchor was not received with the anchor. The anchor was bent 6 degrees, with the center of the bend approximately 3.4 inches above the anchor's lower end. The anchor contained mechanical damage and what appeared to be remnants of concrete on the threads adjacent to the upper end of the anchor on the outside of the bend. The epoxy was missing from some areas in the lower 2 inches (including a portion of the cap area) and upper 1 inch of the embedment length. Void areas were noted in the cap area (this void extended around $\frac{3}{4}$ of the anchor's circumference), middle portion and upper end. Most of the remainder of the epoxy areas were fractured, but one area of adhesive failure was noted near the top of the anchor. The fractured epoxy was yellowed up to about 3 inches above the position corresponding to the bottom of the red polyethylene cap. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The lock washer was partially compressed between the sealing washer and nut, and the nut could not be turned on the anchor using hand forces. Overflow epoxy extended $\frac{3}{4}$ inch below the cap position, and what appeared to be a small amount of damaged epoxy was noted in the anchor threads below the nut. A black ink mark was found on the overflow epoxy.

Anchor 5-1. The red polyethylene cap for this anchor was received with the anchor, and the cap was located on the anchor approximately in its installed position. The anchor was not visibly bent, and it did not exhibit any significant mechanical damage. An area of missing epoxy was noted on one side of the anchor from about 1.4 to 2.7 inches from the anchor's upper end. Void areas were found over about three fourths of the anchor within 2 inches above the cap, and a void area extended down through the cap position. The remainder of the anchor, including the upper 2.5 inches of the embedment length, contained fractured epoxy, except for a void area within about 1 inch of the top. The fractured epoxy was yellowed up to about 4 inches above the position corresponding to the

bottom of the red polyethylene cap. The anchor was received with (starting from the lower item) a nut and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The nut was loose on the threads. Overflow epoxy extended $\frac{3}{4}$ inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 5-2. The red polyethylene cap for this anchor was received with the anchor, and the cap was located on the anchor approximately in its installed position. The anchor was bent 5 to 6 degrees, with the center of the bend approximately 3.5 inches above the anchor's lower end. The anchor did not contain significant mechanical damage. Most of the epoxy was missing from the anchor, with a few circumferential areas of epoxy remaining. Void areas were noted in most of the epoxy areas that were present, and the remaining epoxy was fractured. Yellowed epoxy was noted in the epoxy just above the cap position, extending up to and intersecting an area of missing epoxy 0.8 inch above the position corresponding to the bottom of the cap. No yellowed epoxy was noted in the next lowest area of epoxy located 2.4 inches above the position corresponding to the bottom of the cap. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. The nut could be turned on the anchor using hand forces. Overflow epoxy extended $\frac{3}{4}$ inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 5-3. The red polyethylene cap for this anchor was received with the anchor, and the cap was located on the anchor approximately in its installed position. The anchor was not visibly bent, and it did not exhibit any significant mechanical damage. There were no areas where the epoxy was completely missing from this anchor. Large void areas were noted on this anchor, including a void area on one side that extended from above the cap almost to the top of the anchor. In most of the remaining areas, the epoxy was fractured, but there were also several areas where adhesive fracture was present. The fractured epoxy was yellowed in fractured areas around the circumference up to about 2.4 inches above the position corresponding to the bottom of the red polyethylene cap, but in areas adjacent to voids, the yellow discoloration extended further upward, see figure 6. The anchor was received with a sealing washer, which was located on the anchor threads well below its normal position. Overflow epoxy extended about 1 inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 6-1. This anchor was received installed in a ceiling fitting with anchor 6-2. The red polyethylene cap for this anchor was received with the anchor, and the cap was located on the anchor approximately in its installed position. After removal of the nut and washers and disassembly of the anchor from the fitting, it was noted that the anchor was not visibly bent, and it did not exhibit any significant mechanical damage. An area of missing epoxy was found above the cap position. One larger void was found on one side of the anchor. This void extended 3.3 inches up from the bottom of the cap. A large portion of the remainder of

the epoxy had adhesive failure, and the remainder of the epoxy was fractured. The epoxy in the fractured and adhesive failure areas was yellowed up to within about 3.2 inches above the position corresponding to the bottom of the red polyethylene cap. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. Overflow epoxy extended about 1 inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 6-2. This anchor was received installed in a ceiling fitting with anchor 6-1. The red polyethylene cap for this anchor was received with the anchor, and the cap was located on the anchor approximately in its installed position. After removal of the nut and washers and disassembly of the anchor from the fitting, it was noted that the anchor was bent 11 to 12 degrees, with the center of the bend approximately 3 inches from the anchor's lower end. The anchor did not contain any obvious mechanical damage. No missing epoxy was found, and there were only a few small voids adjacent to the cap position. The epoxy present was fractured, with the exception of one area of adhesive separation about midway along the embedment length. A piece of epoxy that had separated from the anchor also showed adhesive failure on its outer surface and showed fractured epoxy on its inner surface. No yellowed epoxy was noted. The anchor was received with (starting from the lowest item) a nut, lock washer, and sealing washer. The rubber portion of the sealing washer had been compressed into and remained trapped on the anchor threads. Overflow epoxy extended about $\frac{3}{4}$ inch below the cap position, and there was no evidence that the nut had been threaded onto the overflow epoxy. A black ink mark was found on the overflow epoxy.

Anchor 6-3. This anchor was received installed in a ceiling fitting. The red polyethylene cap for this anchor was received with the anchor, and the cap was located on the anchor approximately in its installed position. After removal of the nut and washers and disassembly of the anchor from the fitting, it was noted that the anchor was bent 15 degrees, with the center of the bend approximately 2.2 inches from the anchor's lower end. The anchor contained mechanical impact damage at three positions, on the threads at the anchor's lower end on the outside of the bend, on the threads between 3 and 3.5 inches above the lower end on the outside of the bend, and at the center of the bend on the inside of the bend. The locations of the damage areas are consistent with an impact to the lower end of the anchor after the anchor had pulled out of the ceiling, causing the anchor to rotate within the ceiling fitting and to contact opposite sides of the hole in the fitting. Missing epoxy was found in areas associated with the impact damage and at the top of the anchor. The anchor contained two large voids that were connected to each other. The larger of the two voids extended from the bottom of the cap to a position 3.5 inches above the bottom of the cap, and at its widest, was approximately half the anchor circumference. The smaller void was about 1.1 inches long along the length of the anchor, and was adjacent to the widest portion of the larger void. The remaining epoxy was fractured. Slightly yellowed epoxy was noted up to about 1.8 inches above the position corresponding to the bottom of the red polyethylene cap. An area of fractured epoxy near the top of anchor 6-3 had a light

brown tint to the surface, as shown in figure 8. This was the only anchor from the failure area with this type of feature.

E. EXAMINATION OF ANCHOR HOLES FROM THE FAILURE AREA

Representatives of the NTSB Materials Laboratory, including Derek Nash, Carl Schultheisz, and the undersigned, performed hand-held borescope inspections of the failure anchor holes on August 9 and 10, 2006. Initial visual and borescope examinations of the anchor holes showed the presence of significant loose epoxy that, in some cases, nearly completely blocked the hole. After removal of the loose epoxy with a slender wooden rod, it was noted that all of the holes had some amount of epoxy remaining in the hole above the top of the anchor. The epoxy above the end of the anchors had separated cleanly from the anchors leaving the positive mold of the characters "316" that had been stamped on the ends of the anchors. The epoxy removed from each hole was collected in small glass jars and provided to the Massachusetts State Police.

Although the borescope inspections generated video images that showed the features within the anchor holes, because the inspections were conducted by hand, it was difficult to precisely measure the location or area fraction of features or to generate an overall image of the entire hole surface. Therefore, a contract was developed with Laser Techniques Company to generate laser-scanned images of the holes. These images were acquired on September 28 and 29, 2006.

The laser scanned images were generated by projecting a laser beam up through a head, and reflecting the beam in a 45-degree mirror so that the beam, as it exited the head, was perpendicular to the hole surface. The diameter of the projected laser was approximately 0.002 inch.

The head was inserted into the hole, and then the head was rotated through 360 degrees, incremented one step (0.125 millimeter, or 0.005 inch) upward, and rotated back again through 360 degrees. The rotary sampling resolution was set to 471 samples per revolution, which resulted in a circumferential sampling resolution of approximately 0.125 mm or 0.005 inch as well. This process was continued through 4.5 inches of travel. Because the scans were initiated slightly into the hole, the very bottom of the hole, at the corner between the hole and the tunnel roof, was not captured. Also, because of the physical limitations of the head, the very top of the hole also was not captured. The image of the interior of the hole was generated by recording the brightness of the dispersed beam at each point on the hole surface. The images were then recorded digitally. The images were processed⁷ to best show the various features within the holes. The processed images are shown in appendix 2, pages 1 through 20. The vertical scale in all of the laser images represents distance upward from the starting point of the scan. The horizontal scale represents a circumferential distance for a $\frac{3}{4}$ inch diameter hole. In order to more easily compare the generated images with the anchors from the failure area, the images

⁷ Processing involved adjusting the voltage range over which the data was collected, adjusting the gray scale contrast over this range, and adjusting the horizontal scale (originally in rotational degrees) to correspond to a 0.75 inch diameter hole.

are displayed as if viewed looking radially inward. That is, the clockwise direction looking up at a hole is to the right in the image of the hole.

In general, the laser images showed the features within the holes with adequate clarity, with the exception of the anchor hole at the EB081-M1-6-S anchor position. The image from this hole was blurred in the mid length of the hole, possibly suggesting that epoxy fragments interfered with the laser beam, and the laser equipment apparently shifted at about the 0.8-inch position into the hole, resulting in a discontinuity in the image at this position. Some of the images contain dark, circumferentially streaked areas (for example, see the dark area between 1.5 inches and 3.7 inches in depth for the hole at anchor position “EB081-M1-2/3/4-SE”; appendix 2, page 5). These are apparently areas where the laser head was so close to the epoxy surface that very little reflected light reached the recording sensor.

The laser images of the anchor holes showed the following characteristic areas.

Void areas are unbonded surfaces in the epoxy that appear relatively dark and very smooth.

Cap areas are smooth epoxy surfaces that had been in contact with portions of the red polyethylene cap, including both the cylindrical portion of the cap and the area of the four flaps that extend upward against the anchor as the anchor is inserted through the cap.

Fractured epoxy areas are areas where a thickness of epoxy extended from the surface of the hole with a rough appearance.

Adhesive failure surfaces are areas where the epoxy has separated from the concrete or rebar cut surface, leaving behind minimal or no epoxy. The areas of adhesive failure at the concrete surfaces of the holes are areas in which the darker appearing aggregate is readily identifiable from the lighter appearing cement in the concrete. In areas of adhesive failure where the rebar intersected the hole, the hole surface appeared uniform and reflective with circumferential drilling marks.

Representative examples of the different features defined in the laser images are shown for the holes for anchor “EB081-M1-2/3/4-NE” and “EB081-M1-6-N” in figure 9. Drilled rebar surface and concrete with adhesive failure, void areas, and fractured epoxy areas are labeled in this figure. A graphing and analysis software program was used to calculate the area fraction of void areas, cap areas, fractured epoxy, and adhesive failure surfaces in areas of concrete or rebar in the laser scanned images from the 20 anchor holes from the failure area. Table 2 contains a summary of this information. The following considerations apply to the information in table 2.

- The area fractions of a specific feature are in relation to the area of the scanned surface, which covered 4.5 inches of the typical 5-inch embedment length.

- The circumferential size of an area of interest on a laser image depended on how close the laser head was to that particular area when the image was generated, so there is an amount of uncertainty in the percentages listed in table 2. This uncertainty would potentially be higher for thin, longitudinally oriented areas, and minimal for areas that extended completely around the hole circumference. For example, calculations show that if the rotational centerline of the laser head was offset from the centerline of the hole by $\frac{1}{4}$ the radius, the apparent width of an object on the concrete wall of the hole could be increased by as much as 33 percent or decreased by as much as 20 percent.
- The fraction of void area present in the anchor when it was installed could be higher than the fractions listed in the table, since it is possible that evidence of voids could have been lost or obscured as the anchor pulled out of the hole.
- An obvious imprint of the red polyethylene cap was noted on only three of the laser-scanned images. This was partly due to the fact that the laser scanning started slightly up into the anchor hole and partly due to the damage created to the lower portion of the anchor hole as the entire embedment length of the anchor moved across this area.
- No attempt was made to determine a void volume fraction.

All but one of the anchor holes from the failure area contained some amount of void area. The void areas were generally oriented longitudinally along the length of the hole, as shown in appendix 2.

F. MATCHING FAILURE ANCHOR HOLES TO FAILURE ANCHORS

Based on the characteristics of the anchor holes from the failure area, as shown in the laser scanned images, and the direct visual examination of the anchors from the failure area, an attempt was made to match anchors to anchor holes. The matching was primarily done using the sizes, shapes, and locations of void areas and, when possible, the presence of adhesive failure areas on the anchor and anchor holes.

Several possible matches were made, including

anchor hole "EB081-M1-5-N" with	anchor 6-3,
anchor hole "EB081-M1-8-S" with	anchor 3-3,
anchor hole "EB081-M1-1-N" with	anchor 3-4,
anchor hole "EB081-M1-7-N" with	anchor 1-2, and
anchor hole "EB081-M1-2/3/4-NW" with	anchor 1-3.

Figure 10 shows a sketch of anchor 6-3 and the laser scanned image of anchor hole "EB081-M1-5-N", along with possible matching features. In figure 10, the sketch and laser image are aligned so that the 0.25 inch location on the sketch corresponds to the 0 inch location on the laser image, and that the 4.75 inch location on the sketch corresponds to the 4.5 inch location on the laser image. The features for this anchor and anchor hole provided the best correlation of any of the pairs listed above. However, none of the void areas, for example, matched exactly from an anchor to an anchor hole, and the typical matching feature was the general pattern of voids or other features.

As previously mentioned, anchors 6-1 and 6-2 were received in a two-anchor fitting. Therefore, if anchor 6-1 had been installed in one anchor hole of any particular fitting, anchor 6-2 would have been installed in the other anchor hole of the fitting. The features on anchors 6-1 and 6-2 could not be positively matched with any pair of holes for any particular fitting.

As previously discussed, for a period of time after the accident, anchor E1-1 was retained in a four-anchor fitting labeled as the EB081-M1-9/10/11 position. Nevertheless, the features on anchor E1-1 could not be positively matched with the features in any of the four anchor holes at the EB081-M1-9/10/11 position. Therefore, based on the uncertainty in the possible matches, the problems with matching anchors 6-1 and 6-2 with a pair of anchor holes from one fitting, and the relatively few number of possible matches, no additional effort was expended to further define relationships between the anchor holes and anchors.

G. COMPARISON OF ANCHOR HOLES OUTSIDE OF THE FAILURE AREA WITH KNOWN ANCHORS

Laser scan images were generated from four anchor holes outside of the failure area (positions EB082-M1-12-N and -S, and EB081-N-10-N and -S). Processed images from these holes are shown in appendix 3, pages 1, 3, 5, and 7. The same data collected from the failure area anchor holes was collected from these additional holes, using the same techniques, and this data is also presented in table 2.

The four anchors from positions EB082-M1-12-N and -S, and EB081-N-10-N and -S were among the 57 anchors outside of the failure area that were also examined in the Safety Board's Materials Laboratory. Images of these four anchors are shown in appendix 3, pages 2, 4, 6, and 8 for comparison with the laser scan images also shown in this appendix. As can be seen in appendix 3, there was a very high degree of correlation between the laser scan image of the hole and the appearance of the anchor for the "EB082-M1-12-N" position (pages 1 and 2 in appendix 3) and the "EB081-N-10-S" position (pages 7 and 8 in appendix 3), much better than the correlation between the laser scan image of any of the anchor holes from the failure area and the failure area anchors. There was poor correlation between the laser scan image of the hole and the appearance of the anchor for the "EB082-M1-12-S" position (pages 3 and 4 in appendix 3) and the "EB081-N-10-N" position (pages 5 and 6 in appendix 3). Additional information regarding the features on these four anchors is discussed in the following section along with the remaining anchors examined.

H. EXAMINATION OF ADDITIONAL ANCHORS

In addition to the 20 anchors directly from the failure area (the EB-081-M1 row of anchors), the Materials Laboratory received 57 anchors that had been removed from the roof of the east-bound and west-bound (WB) tunnels. Some of these additional anchors were chosen because they had a specific condition of interest, and some were chosen at random from those that had been removed. Table 3 contains a summary of the features

found during the laboratory examination of these anchors. The data on the area fraction of embedment length associated with specific features was gathered in the same manner used for the anchors from the failure area. The final column in table 3 contains displacement data (the distance that the anchor had moved downward, or displaced, from its originally installed position, as determined by visual indicators on the anchor and tunnel roof before removal of the anchor⁸) for the specific anchors for reference purposes. Displacement data for all of the anchors in the three tunnels is documented in Materials Laboratory Report No. 06-079. The following portion of this section contains a description of specific features noted on the anchors examined.

“EB080-N-4/5/6-NE” Cap in place. Most of the surface of the epoxy on this anchor has a unique smooth appearance with a diameter less than the hole and therefore was classified as void. Surface had a glassy or “wet” appearance that was very soft when removed from the ceiling and that remained somewhat soft as of the date of this report. Darker brown epoxy material found at the top of the anchor. Anchor originally placed in a plastic bag because of the soft material. No yellowed epoxy. See photographs in figure 11.

“EB080-N-4/5/6-NW” Cap in place. Void area on bottom portion of embedment length. No yellowed epoxy.

“EB080-N-4/5/6-SE” Cap in place. Some void area, primarily along one side of the anchor. Some adhesive failure. No yellowed epoxy.

“EB080-N-4/5/6-SW” Cap in place. No voids. Some adhesive failure in the bottom of the embedment length. No yellowed epoxy.

“EB080-M1-1-N” Cap in place. Anchor slightly bent. Some voids and adhesive failure, with light brown discoloration in adhesive failure area near the top of the anchor. No yellowed epoxy.

“EB080-M1-1-S” Cap missing. Significant missing epoxy in lower portion of embedment length. Long void area on one side of anchor. Yellowed epoxy up to 2.6 inches.

“EB080-M1-2-N” Cap displaced upward 0.6 inch. Significant voids. Yellowed epoxy up to 2.2 inches.

“EB080-M1-3/4/5-NE” Cap in place. Long void on one side of anchor. Yellowed epoxy up to 1.2 to 1.4 inches.

“EB080-M1-3/4/5-NW” Cap in place. Large void primarily on one side of the anchor. Yellowed epoxy up to 1.2 inches.

“EB080-M1-3/4/5-SE” Cap in place. Significant voids primarily on one side of the anchor. No yellowed epoxy.

“EB080-M1-3/4/5-SW” Cap in place. Scattered voids. No yellowed epoxy.

“EB081-N-10-N” Cap in place, but cut off in laboratory. Some voids, especially near bottom of embedment length. No Yellowed epoxy. See photographs, appendix 3 page 6.

“EB081-N-10-S” Cap in place. Long void on one side of anchor. Anchor slightly bent. No yellowed epoxy. See photographs, appendix 3, page 8.

“EB081-M2-2/3/4-SE” Cap in place. Few voids, some missing epoxy. No yellowed epoxy.

⁸ There are differences between the FHWA measurements of displacement and the NTSB measurements. These differences are primarily the result of interpretation of the position of the polyethylene cap, and both the instrument and method used to quantify displacement.

- “EB081-M2-2/3/4-SW” Cap in place. Few voids, some missing epoxy. No yellowed epoxy.
- “EB081-S-8-S” Cap in place. Larger areas of adhesive failure, especially the lower portion of the embedment depth. Dark brown to light brown discoloration of the epoxy noted in various areas, see figure 12. Yellowed epoxy noted up to 0.7 inch, slightly above cap area.
- “EB081-S-9-S” Cap in place. Large void, predominantly on one side of the anchor. Yellowed epoxy up to 0.8 inch.
- “EB081-S-10-S” Cap missing. Large amount of adhesive failure throughout embedment length, with dark brown, almost black discolored epoxy, especially near the top of the embedment length, see figure 13. Anchor was originally placed in a plastic bag because discolored epoxy was soft and fluid. Discolored epoxy became much harder over time. Yellowed epoxy up to 1.6 inches.
- “EB082-M1-8/9/10-NE” Cap in place. Some voids, adhesive failure, and areas of missing epoxy. No yellowed epoxy.
- “EB082-M1-8/9/10-NW” Cap in place. Some voids. Yellowed epoxy up to 1.2 inches.
- “EB082-M1-8/9/10-SW” Cap in place. Some voids. Yellowed epoxy up to 0.9 inch.
- “EB082-M1-11-S”. Cap in place but cut off. Some voids, primarily on one side of the anchor. Yellowed epoxy up to 1.6 inches.
- “EB082-M1-12-N” Cap in place. Anchor is slightly bent. Significant voids, primarily on one side of the anchor. See photographs, appendix 3, page 2. No yellowed epoxy.
- “EB082-M1-12-S” Cap displaced upward 2.0 inches, cut off in lab. Voids only below received position of the cap. Yellowed epoxy up to 1.5 inches.
- “EB082-M2-12-N” Cap in place. Large voids, especially near bottom of embedment length. No yellowed epoxy.
- “EB082-S-7.5-S” Cap in place. Significant adhesive failure, with dark brown discoloration. Void area near lower portion of embedment length. Yellow discoloration of the adhesive failure area up to 0.9 inch. See photos figure 14.
- “EB083-N-3-N” Cap in place. Some voids around cap, some missing epoxy. No yellowed epoxy.
- “EB083-N-3-S” Cap in place. Fractured epoxy throughout except for small void adjacent to the cap. No yellowed epoxy.
- “EB083-M1-1-N” Cap in place. A few voids, some missing epoxy. No yellowed epoxy.
- “EB083-M1-1-S” Cap in place. A few voids. No yellowed epoxy.
- “EB083-M1-2-N” Cap in place. Some voids, primarily on one side of the anchor. No yellowed epoxy.
- “EB083-M1-2-S” Cap in place, but cut off in laboratory. Significant voids, primarily on the bottom portion of the embedment depth. No yellowed epoxy.
- “EB083-M2-1-N” Cap in place. Large void spiraling from bottom to top. Adhesive failure. No yellowed epoxy.
- “EB083-M2-7/8/9-SE” Cap in place. Void, primarily on one side of the anchor. Significant adhesive failure. No yellowed epoxy.
- “EB083-M2-7/8/9-SW” Cap in place. A few voids. Significant adhesive failure on the lower portion of the embedment depth. No yellowed epoxy.
- “EB083-M2-11-N” Cap in place. Significant adhesive failure with brown discoloration. Some voids predominantly on one side of the anchor. Yellowed epoxy up to 0.8 inch.

- "EB083-S-1-S" Cap in place. Significant adhesive failure, especially over the bottom portion of the embedment length. Dark brown deposits noted at and above the top of the anchor. Anchor was originally placed in a plastic bag because discolored epoxy was soft and fluid. Discolored epoxy became much harder over time. See figure 15.
- "WB082-M-2/3/4-NW" Cap in place. A few voids. Yellowed epoxy up to 1 inch.
- "WB082-M-2/3/4-SW" Cap in place. A few voids. Yellowed epoxy up to 0.9 inch.
- "WB082-M-5-S" Cap displaced upward about 0.3 inch. Cap cut off in laboratory. Some voids and areas of missing epoxy. Yellowed epoxy up to 1.5 inches.
- "WB082-M-6-S" Cap displaced upwards about 0.5 inch. Cap cut off in laboratory. Fractured epoxy in the upper portion of the embedment length. Some voids, primarily in the bottom of the embedment length and cap area. Very little adhesive failure. Epoxy in cap area is intact, but large void chipped out in order to measure void area. Yellowed epoxy up to 1.5 inches.
- "WB082-M-6A-N" Cap in place. Some voids. No yellowed epoxy.
- "WB082-M-6A-S" Cap in place. Voids at mid length and around cap. Yellowed epoxy up to 1.2 inches.
- "WB082-M-7-N" Cap displaced slightly upward. Epoxy missing over large areas. Some voids at top and bottom of embedment length. Yellowed epoxy up to about 1 inch.
- "WB082-S-2-N" Cap in place. Few voids, some missing epoxy. No yellowed epoxy.
- "WB082-S-2-S" Cap in place. Few voids, some missing epoxy. No yellowed epoxy.
- "WB083-M-9/10/11-SE" Cap missing. Significant voids, primarily on one side of the anchor. Some missing epoxy. Yellowed epoxy up to 1.9 inches.
- "WB083-M-9/10/11-NW" Cap in place. Some voids adjacent to cap. No yellowed epoxy.
- "WB083-M-9/10/11-SW" Cap in place, but cut off in laboratory. A few voids, but there are additional flat regions not clearly fracture or voids. Yellowed epoxy extends higher on one side of the anchor, up to 1.8 to 2.5 inches.
- "WB083-S-1-N" Anchor and core sample from sustained load test location. Cap in place on the anchor but slightly displaced downward from the tunnel roof surface. Core was cut longitudinally, and a gap was noted between the top of the anchor and the epoxy above the anchor, but the sample crumbled before the gap was measured. There is adhesive failure, possibly created during cutting or removal of the core, near the top of the anchor, but no apparent fracture in the epoxy between the anchor threads and hole wall over most of the embedment length. Significant voids noted. No yellowing. Crack or void noted in the concrete above #11 rebar adjacent to the anchor, and epoxy extended about 2 inches from the anchor hole in this void.
- "WB083-S-3-N" Anchor and core sample from sustained load test location. Cap in place on anchor but displaced downward about 0.08 inch from the tunnel roof surface. Void surface noted above epoxy above the top of the anchor. Core was cut longitudinally, and a gap of about 0.05 inch was noted between the top of the anchor and the epoxy above the anchor. There is no apparent fracture in the epoxy between the anchor threads and hole wall over most of the embedment length. (Figure 16 shows the void surface above the epoxy above the top of the anchor, the gap above the top of the anchor, and the intact epoxy between the anchor threads and hole wall.) Significant voids noted. Epoxy near the top of the anchor is softer in

some locations; a darker streak was noted in the epoxy in the softer area. No yellowing.

“WB083-S-3-S” Anchor pulled from tunnel roof during sustained load testing. Core sample also examined. Cap in place on the anchor. Some voids. No yellowed epoxy.

“WB084-N-4/5/6-NE” Cap in place. Some voids. No yellowed epoxy.

“WB084-N-4/5/6-NW” Cap in place. Some voids. No yellowed epoxy.

“WB084-N-4/5/6-SE” Cap in place. Some voids and missing epoxy. No yellowed epoxy.

“WB084-N-4/5/6-SW” Cap in place. Very small void. Small adhesive failure. No yellowed epoxy.

“WB084-S-9-S” Anchor and core sample from sustained load test location. The epoxy above the top of anchor was not contained in the core sample. Cap in place on anchor but displaced downward about 0.35 inch from the tunnel roof surface. Core was cut longitudinally. The epoxy between the anchor threads and hole wall is fractured over the embedment length. Significant voids noted. No yellowing.

To demonstrate how the removal of an anchor from an anchor hole can affect the measured area fraction of voids or other features, the void area fraction measured on an anchor was compared to the void area fraction measured on the wall of the corresponding anchor hole at three positions: WB083-S-3-S, EB081-N-10-S, and EB082-M1-12-N. The anchors at these positions had been pulled from the tunnel roof either during sustained load testing (WB083-S-3-S), in which case a concrete core had also been removed and submitted for examination, or during pull testing (remaining anchors), after which a laser scan image of the hole wall was generated. The quantified void area fraction on anchor WB083-S-3-S was 8 percent, which contrasts with the 22 percent void area fraction quantified on the wall of the corresponding anchor hole. An overlay of the feature maps showed that most of the void area documented for the anchor was also present on the map of the anchor hole. The void area fraction for this position, based on the overlay of the two maps, rises to 24 percent. Similar comparisons were conducted at positions EB081-N-10-S and EB082-M1-12-N, and the results are shown in the following table.

Position	Void Area Fraction (Percent) From		
	Anchor	Hole	Overlay
WB083-S-3-S	8	22	24
EB081-N-10-S	21	31	31
EB082-M1-12-N	23	33	36

I. EVALUATION OF YELLOWED EPOXY

As previously discussed in this report, it was noted during the examination of the anchors from the failure area that yellow discoloration of the fractured epoxy occurred on at least 13 of these 20 anchors. The discoloration on the failure area anchors was located in the lower portion of the embedment depth, for a distance above the cap position, extending up to a maximum documented distance of 4 inches above the bottom of the embedment depth. Evaluation of other anchors removed from the tunnel roofs showed that similar discoloration was not found on any anchor that had not been displaced a significant amount, consistent with the yellow discoloration being related to some type of

environmental effect on the exposed portion of a displaced anchor while the remainder of the anchor was still in the anchor hole.

In order to document a possible relationship between the extent of yellowed epoxy and the amount of displacement for anchors in the remainder of the tunnel, the removed anchors in the east-bound and west-bound tunnels with greater amounts of displacement were among those chosen for examination in the laboratory. Table 4 contains a list of the 54 anchors from the three tunnels with displacement greater than 0.3 inch, starting with the anchor with the greatest displacement. This list was compiled from the displacement data in Materials Laboratory Report No. 06-079. Of these 54 anchors, 30 had been pulled from the tunnel roofs after the accident and were among the additional anchors examined in the Materials Laboratory (those with data under the "Upper Extent of Yellowed Epoxy" column). The other 24 anchors listed in table 4 (54 minus 30) remained in the tunnel roofs and were not available for examination. Table 4 also has a column noting how close a given anchor was to the failure area. Anchors noted as being adjacent to the failure area were the anchors in the N and M2 rows of section EB081, and the adjacent anchors in the M1 row of sections EB080 and EB082. Those anchors noted as being near the failure area were in the M1 rows of sections EB080 and EB082 and were one position (or in some cases two positions) removed from being directly adjacent.

As can be seen in table 4, yellowed epoxy was not found on any of the examined anchors with a displacement less than 0.375 inch and was found on all but one of the examined anchors with displacement of 0.5 inch and greater. The one anchor that was the exception to this pattern, anchor number 3 in table 4, EB081-N-10-N, had one of the greatest amounts of displacement and was directly adjacent to the failure area, as indicated in table 4. It is therefore possible that most or all of the displacement of this anchor occurred at the time of the accident and not over time before the accident.

The embedment length containing yellowed epoxy corresponded very well with the measured displacement length, with the yellowed appearance typically about 0.3 inch greater than the measured displacement length, see table 4. On some anchors, for example, anchor 5-3 from the failure area (see figure 6), the yellowed epoxy extended around the perimeter of a void, consistent with environmental exposure extending up into the hole through a void.

J. ANCHORS WITH DARK BROWN EPOXY

A significant amount of dark brown epoxy was found on six of the additional anchors received for examination, and a smaller amount of brown epoxy was noted on one of the anchors from the failure area, as shown on the table on the following page.

This dark brown epoxy was, for the most part, found to be soft and pliable immediately after each anchor was pulled from the tunnel roof, to the extent that it was felt necessary to place several of the anchors in plastic bags before being inserted into paper evidence bags. It was noted that in the months after the accident, the dark brown material became harder, although some soft areas remained. For the anchors not from the failure

area, there was a clear relationship between the presence of the dark brown epoxy and the amount of adhesive failure noted after removal of the anchors, as shown in the table below. In the failure area, anchor 6-3 contained a much smaller area of brown epoxy near the top of the anchor, as shown in figure 8.

Anchor Position	Area Fraction (Percent)		See Figure
	Void	Adhesive Failure	
EB080-N-4/5/6-NE	70*	0	Figure 11
EB081-S-8-S	0	52	Figure 12
EB081-S-10-S	5	75	Figure 13
EB082-S-7.5-S	16	61	Figure 14
EB083-M2-11-N	13	57	Not shown
EB083-S-1-S	1 ⁹	59	Figure 15
6-3	6	0	Figure 8

*The void surface on this anchor appeared different than others, having a glassy reflective appearance, a brown color, and a very soft surface.

A review of the data shown in table 4 shows that the four anchors with greatest displacement were either directly adjacent to or near the failure area, indicating the likelihood that their displacement was influenced by the failure area. Table 5 was therefore generated in order to evaluate the characteristics of the anchors with greatest displacement that were clearly not affected by the failure area. Table 5 shows information for anchors with the following characteristics:

- displacement of 0.3 inch or more,
- pulled from the tunnel roofs and examined in the laboratory, and
- not adjacent to or near the failure area.

Table 5, which contains a subset of table 4, lists information on the void fraction, the adhesive failure fraction, the extent of yellowed epoxy, and the displacement length for the anchors meeting the above criteria. As shown in table 5, five of the six anchors outside of the failure area with dark brown epoxy were in the group of anchors with greatest displacement (those in bold in table 5). Also, of the nine anchors with greatest displacement, four had high area fractions of adhesive failure and were anchors with dark brown epoxy. Many of the anchors with dark brown epoxy were among those chosen for polymer characterization testing to be documented in a separate report.

James F. Wildey II
Supervisory Metallurgist

⁹ The void was also associated with an additional 4% of soft, dark material at the top of the anchor.

Table 1
Summary of Anchor Conditions
Anchors From Failure Area

Anchor	Area Fraction (Percent)				Upper Extent of Yellowed Epoxy (Inch)	Beginning of Non-Yellowed Epoxy (Inch)
	Void	Cap	Adhesive Failure	Missing Epoxy		
E1-1	14	3	0	49	3.5	
1-1	0	0	2	82	EM	1.7
1-2	12	13	0	5	3.8	
1-3	4	0	0	43	EM	2.5
1-4	1	0	0	21	2	
2-1	1	0	5	47	EM	2
2-2	1 - 2	0	0	95+	EM	1
2-3	12	0	0	67	EM	2
2-4	2	0	0	75	2.5	
3-1	11	0	0	26	1.2	
3-2	6	0	0	56	EM	2.7
3-3	10	13	0	11	3	
3-4	12	13	0	23	1.3	
4-2	8	13	0	19	3	
5-1	27	14	0	2	4	
5-2	3	15	0	80	0.8	2.4
5-3	40	14	4	2	2.4	
6-1	15	12	29	10	3.2	
6-2	2	13	1	2	0	
6-3	26	6	0	16	1.8	

Note: "EM" in this table indicates that epoxy was missing from the lower portion of the embedment length, and the upper extent of yellowed epoxy could not be determined.

Table 2
Summary of Anchor Hole Conditions Based on Laser Scan Images

Failure Area Anchor Hole Positions	Area Fraction (Percent)				
	Void	Cap	Adhesive Failure	Drilled Surface	Fractured Epoxy
EB081-M1-1-N	38	6	0	6	50
EB081-M1-1-S	27	0	0	4	69
EB081-M1-2/3/4-NE	28	8	5	5	54
EB081-M1-2/3/4-NW	12	4	0	0	84
EB081-M1-2/3/4-SE	14	0	0	3	83
EB081-M1-2/3/4-SW	18	0	18	0	74
EB081-M1-5-N	16	0	6	6	72
EB081-M1-5-S	36	0	6	5	53
EB081-M1-6-N	9	0	49	0	42
EB081-M1-6-S	0	0	7	0	93
EB081-M1-7-N	20	0	0	0	80
EB081-M1-7-S	22	0	0	0	78
EB081-M1-8-N	23	0	2	0	75
EB081-M1-8-S	24	0	0	0	76
EB081-M1-9/10/11-NE	3	0	31	0	66
EB081-M1-9/10/11-NW	23	0	0	0	77
EB081-M1-9/10/11-SE	21	0	22	0	57
EB081-M1-9/10/11-SW	16	0	24	0	60
EB081-M1-12-N	37	0	0	0	63
EB081-M1-12-S	11	0	43	0	46
Other Anchor Hole Positions					
EB081-N-10-N	27	1	0	0	72
EB081-N-10-S	31	2	0	0	67
EB082-M1-12-N	33	2	0	0	65
EB082-M1-12-S	24	4	0	0	72

Table 3
Summary of Anchor Conditions
Anchors Not From Failure Area

Anchor Position	Area Fraction (Percent)				Upper Extent of Yellowed Epoxy (Inch)	Displaced Length (Inch)
	Void	Cap	Adhesive Failure	Missing Epoxy		
EB080-N-4/5/6-NE	70 ¹⁰	13	0	0	0	0.3
EB080-N-4/5/6-NW	16	13	0	4	0	0
EB080-N-4/5/6-SE	30	13	9	2	0	0.31
EB080-N-4/5/6-SW	0	11	13	2	0	0
EB080-M1-1-N	11	13	19	3	0	0.47
EB080-M1-1-S	15	0	0	32	2.6	2.65
EB080-M1-2-N	26	13	0	6	2.2	1.9
EB080-M1-3/4/5-NE	16	13	0	4	1.2 – 1.4	0.875
EB080-M1-3/4/5-NW	27	13	0	2	1.2	0.75
EB080-M1-3/4/5-SE	17	13	0	2	0	0.44
EB080-M1-3/4/5-SW	10	13	0	2	0	0.33
EB081-N-10-N	17	7	0	2	0	2.23
EB081-N-10-S	21	13	0	2	0	0
EB081-M2-2/3/4-SE	8	13	0	13	0	0
EB081-M2-2/3/4-SW	9	13	0	5	0	0
EB081-S-8-S	0	10	52	0	0.7	0.55
EB081-S-9-S	23	13	0	24	0.8	0.42
EB081-S-10-S	5	9	75	11	1.6	1.375
EB082-M1-8/9/10-NE	15	13	12	18	0	0.34
EB082-M1-8/9/10-NW	25	13	24	0	1.2	0.61
EB082-M1-8/9/10-SW	11	13	0	2	0.9	0.6
EB082-M1-11-S	18	13	0	2	1.6	1.25
EB082-M1-12-N	23	13	0	2	0	0.34
EB082-M1-12-S	18	8	0	9	1.5	2.5
EB082-M2-12-N	25	13	0	2	0	0.375
EB082-S-7.5-S	16	10	61	0	0.9	0.625
EB083-N-3-N	6	13	0	18	0	0
EB083-N-3-S	1	13	0	2	0	0
EB083-M1-1-N	6	13	0	7	0	0
EB083-M1-1-S	15	13	0	2	0	0
EB083-M1-2-N	26	13	0	2	0	0
EB083-M1-2-S	32	13	0	2	0	0
EB083-M2-1-N	28	13	33	2	0	0.125

¹⁰ The void surface on this anchor appeared different than others, having a glassy reflective appearance, a brown color, and a very soft surface.

EB083-M2-7/8/9-SE	14	13	31	2	0	0
EB083-M2-7/8/9-SW	5	13	32	6	0	0.125
EB083-M2-11-N	13	11	57	2	0.8	0.5
EB083-S-1-S	1 ¹¹	13	59	0	0	0
WB082-M-2/3/4-NW	12	13	0	0	1.0	0.375
WB082-M-2/3/4-SW	19	13	0	0	0.9	0.375
WB082-M-5-S	11	6	0	11	1.5	0.625
WB082-M-6-S	15	8	0	4	1.5	1.0
WB082-M-6A-N	23	13	0	0	0	0
WB082-M-6A-S	17	12	0	0	1.2	1.0
WB082-M-7-N	7	13	0	43	1.0	0.5
WB082-S-2-N	7	13	0	11	0	0
WB082-S-2-S	2	13	0	13	0	0
WB083-M-9/10/11-SE	16	13	3	7	1.9	1.625
WB083-M-9/10/11-NW	11	13	0	2	0	0.0625
WB083-M-9/10/11-SW	8	11	1	5	1.8 – 2.5	1.75
WB083-S-1-N (creep)	30	13	5	0	0	0
WB083-S-3-N (creep)	28	13	0	0	0	0
WB083-S-3-S (creep)	8	13	22	0	0	0
WB084-N-4/5/6-NE	11	13	0	2	0	0
WB084-N-4/5/6-NW	11	13	0	2	0	0
WB084-N-4/5/6-SE	8	13	0	6	0	0
WB084-N-4/5/6-SW	1	13	7	2	0	0
WB084-S-9-S (creep)	12	13	0	0	0	0

¹¹ The void was also associated with an additional 4% of soft, dark material at the top of the anchor.

Table 4
Summary of Anchor Conditions
Anchors With Greatest Displacement

Anchor Position	Notes	Upper Extent of Yellowed Epoxy (Inch)	Displaced Length (Inch)
1. EB080-M1-1-S	Adj. to failure area	2.6	2.65
2. EB082-M1-12-S	Adj. to failure area	1.5	2.5
3. EB081-N-10-N	Adj. to failure area	0	2.23
4. EB080-M1-2-N	Near failure area	2.2	1.9
5. WB083-M-9/10/11-SW		1.8 – 2.5	1.75
6. WB083-M-9/10/11-SE		1.9	1.625
7. WB083-M-8-N		N/A	1.5
8. EB081-S-10-S		1.6	1.375
9. EB082-M1-11-S	Near failure area	1.6	1.25
10. WB084-M-2/3/4-NW		N/A	1.25
11. WB084-M-2/3/4-NE		N/A	1.125
12. EB080-N-1-N		N/A	1.0
13. WB082-M-6-S		1.5	1.0
14. WB082-M-6A-S		1.2	1.0
15. WB084-S-10-S		N/A	0.875
16. EB080-M1-3/4/5-NE	Near failure area	1.2 – 1.4	0.875
17. WB084-M-12-S		N/A	0.75
18. WB084-M-1-S		N/A	0.75
19. EB080-M1-3/4/5-NW	Near failure area	1.2	0.75
20. WB084-M-1-N		N/A	0.625
21. WB084-M-6-S		N/A	0.625
22. WB084-M-8-N		N/A	0.625
23. WB084-M-9/10/11-NE		N/A	0.625
24. WB084-M-9/10/11-NW		N/A	0.625
25. WB082-M-5-S		1.5	0.625
26. EB082-S-7.5-S		0.9	0.625
27. EB082-M1-8/9/10-NW	Near failure area	1.2	0.61
28. EB082-M1-8/9/10-SW	Near failure area	0.9	0.6
29. EB081-S-8-S		0.7	0.55
30. WB083-M-6-S		N/A	0.5
31. WB083-M-7-S		N/A	0.5
32. WB083-M-12-N		N/A	0.5
33. WB084-S-9-N		N/A	0.5
34. WB082-M-7-N		1.0	0.5
35. EB083-M2-11-N		0.8	0.5
36. EB080-N-2-N		N/A	0.5
37. HOV080-N-10-S		N/A	0.5

38. HOV080-M-9/10/11-SE		N/A	0.5
39. HOV080-M-9/10/11-SW		N/A	0.5
40. HOV079-S-2-S		N/A	0.5
41. EB080-M1-1-N	Adj. to failure area	0	0.47
42. EB080-M1-3/4/5-SE	Near failure area	0	0.44
43. EB081-S-9-S		0.8	0.42
44. HOV080-M-8-N		N/A	0.375
45. EB080-M1-6-S		N/A	0.375
46. HOV079-S-10-S		N/A	0.375
47. EB082-M2-12-N		0	0.375
48. WB082-M-2/3/4-NW		1.0	0.375
49. WB082-M-2/3/4-SW		0.9	0.375
50. EB082-M1-8/9/10-NE	Near failure area	0	0.34
51. EB082-M1-12-N	Adj. to failure area	0	0.34
52. EB080-M1-3/4/5-SW	Near failure area	0	0.33
53. EB080-N-4/5/6-SE		0	0.31
54. EB080-N-4/5/6-NE		0	0.3

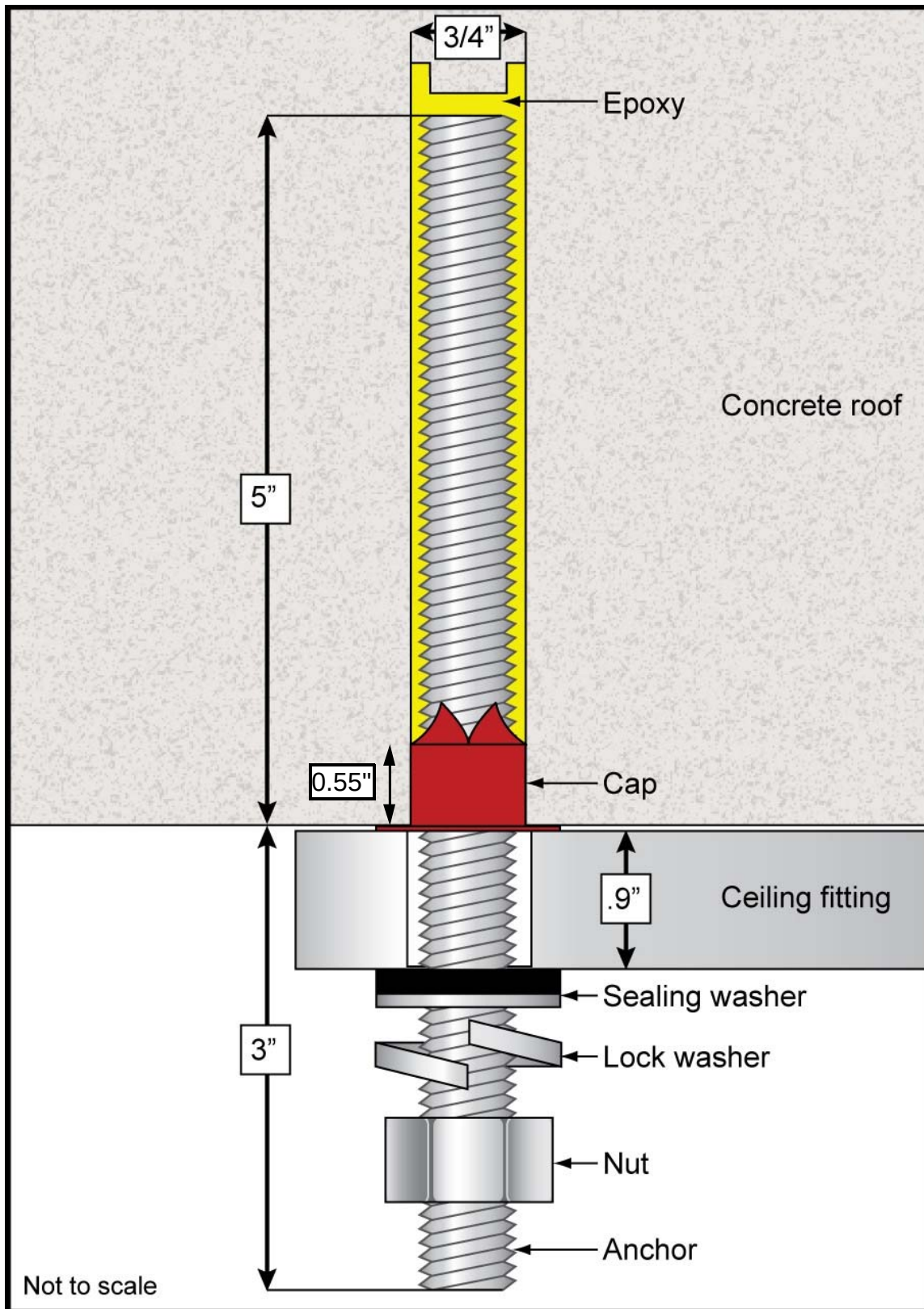
N/A – Anchor not available, remaining in tunnel roof

Table 5
Summary of Anchor Conditions
Examined Anchors With Greatest Displacement from Locations
Remote from the Failure Area

Anchor Position	Area Fraction (Percent)		Upper Extent of Yellowed Epoxy (Inch)	Displaced Length (Inch)
	Void	Adhesive Failure		
5. WB083-M-9/10/11-SW	8	1	1.8 – 2.5	1.75
6. WB083-M-9/10/11-SE	16	3	1.9	1.625
8. EB081-S-10-S	5	75	1.6	1.375
13. WB082-M-6-S	15	0	1.5	1.0
14. WB082-M-6A-S			1.2	1.0
25. WB082-M-5-S	11	0	1.5	0.625
26. EB082-S-7.5-S	16	61	0.9	0.625
29. EB081-S-8-S	0	52	0.7	0.55
35. EB083-M2-11-N	13	57	0.8	0.5
43. EB081-S-9-S	23	0	0.8	0.42
47. EB082-M2-12-N	25	0	0	0.375
48. WB082-M-2/3/4-NW			1.0	0.375
49. WB082-M-2/3/4-SW			0.9	0.375
53. EB080-N-4/5/6-SE	30	9	0	0.31
54. EB080-N-4/5/6-NE	70¹²	0	0	0.3

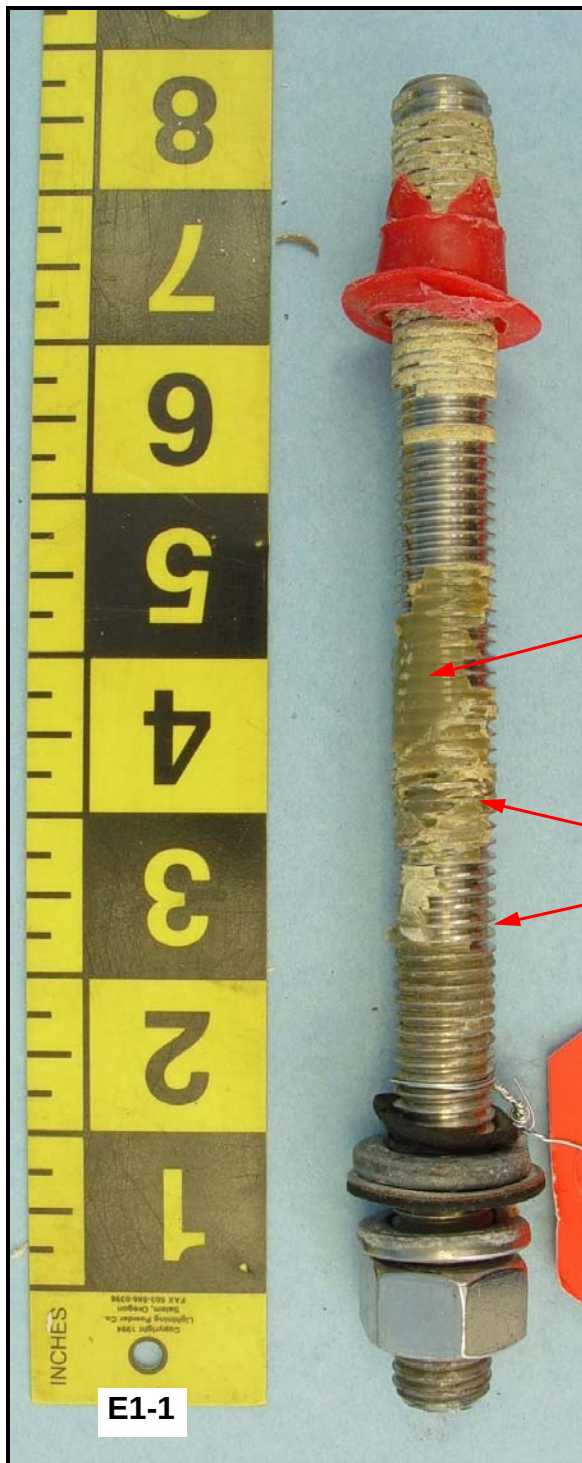
Anchors shown in bold indicate anchors with dark brown epoxy.

¹² The void surface on this anchor appeared different than others, having a glassy reflective appearance, a brown color, and a very soft surface.



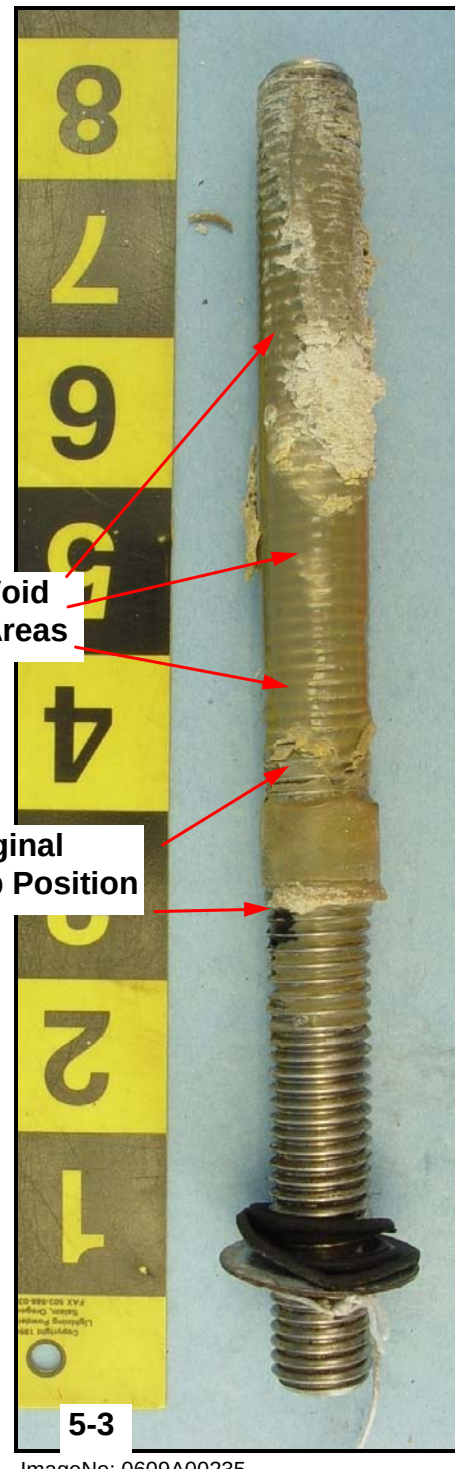
ImageNo:0701A00636, Project No:2006080009

Figure 1. Drawing showing the relationship between an installed anchor and the polyethylene cap, epoxy, and attaching hardware.



ImageNo:0609A00194, Project No:2006080009

Figure 2. Overall view of anchor E1-1 showing the cap on the upper portion of the embedment length, a void area, and the anchor nut, lock washers, and sealing washer on the anchor's lower end.



ImageNo: 0609A00235

Figure 3. An example of a larger epoxy void area, on anchor 5-3.

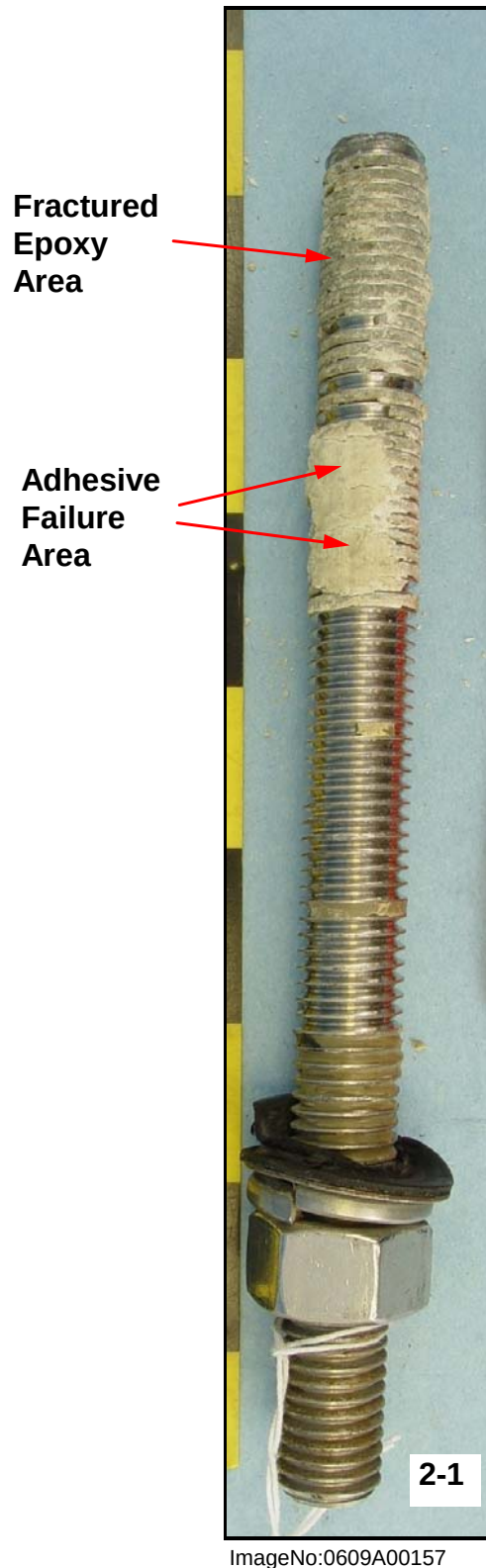


Figure 4. An example of fractured epoxy and adhesively failed epoxy on anchor 2-1.

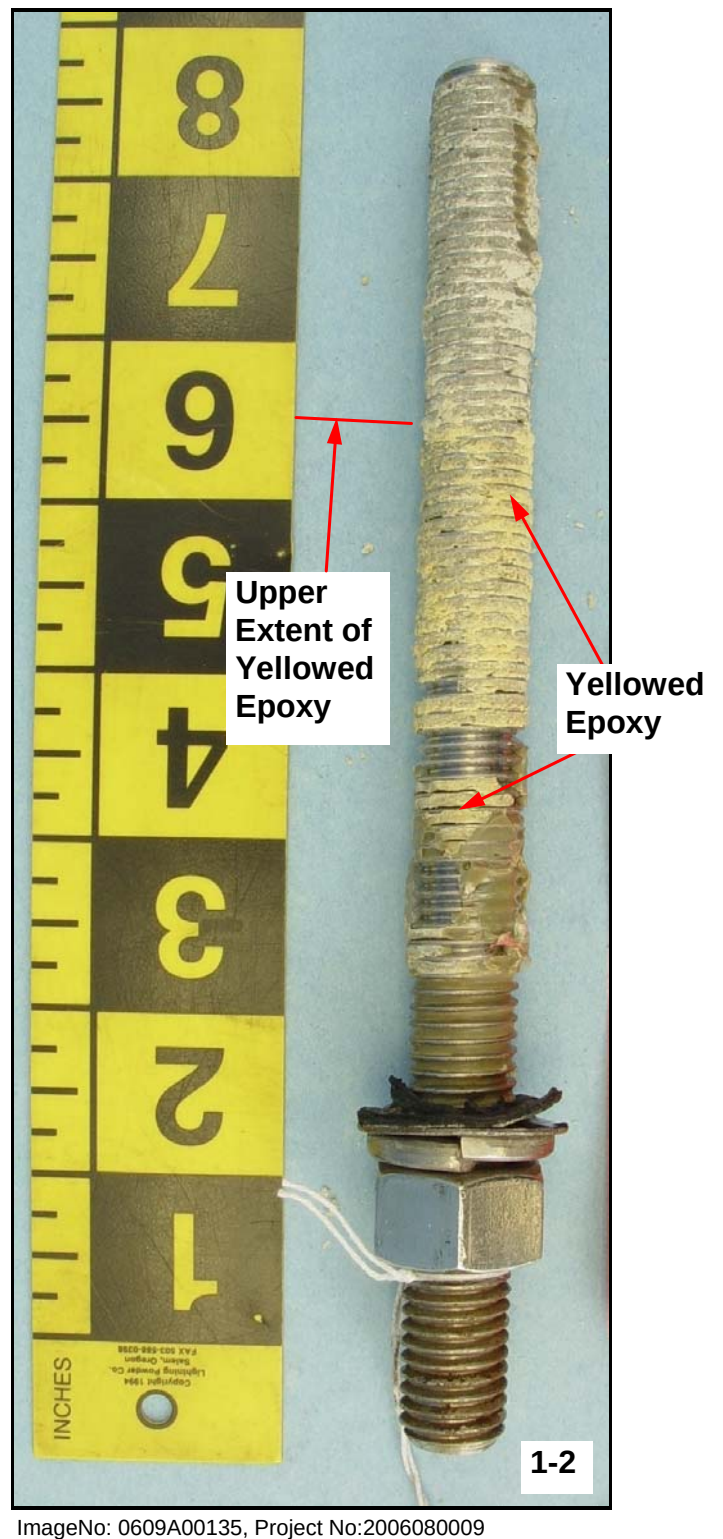


Figure 5. An example showing how a portion of the fractured epoxy on anchor 1-2 has a yellowed appearance.

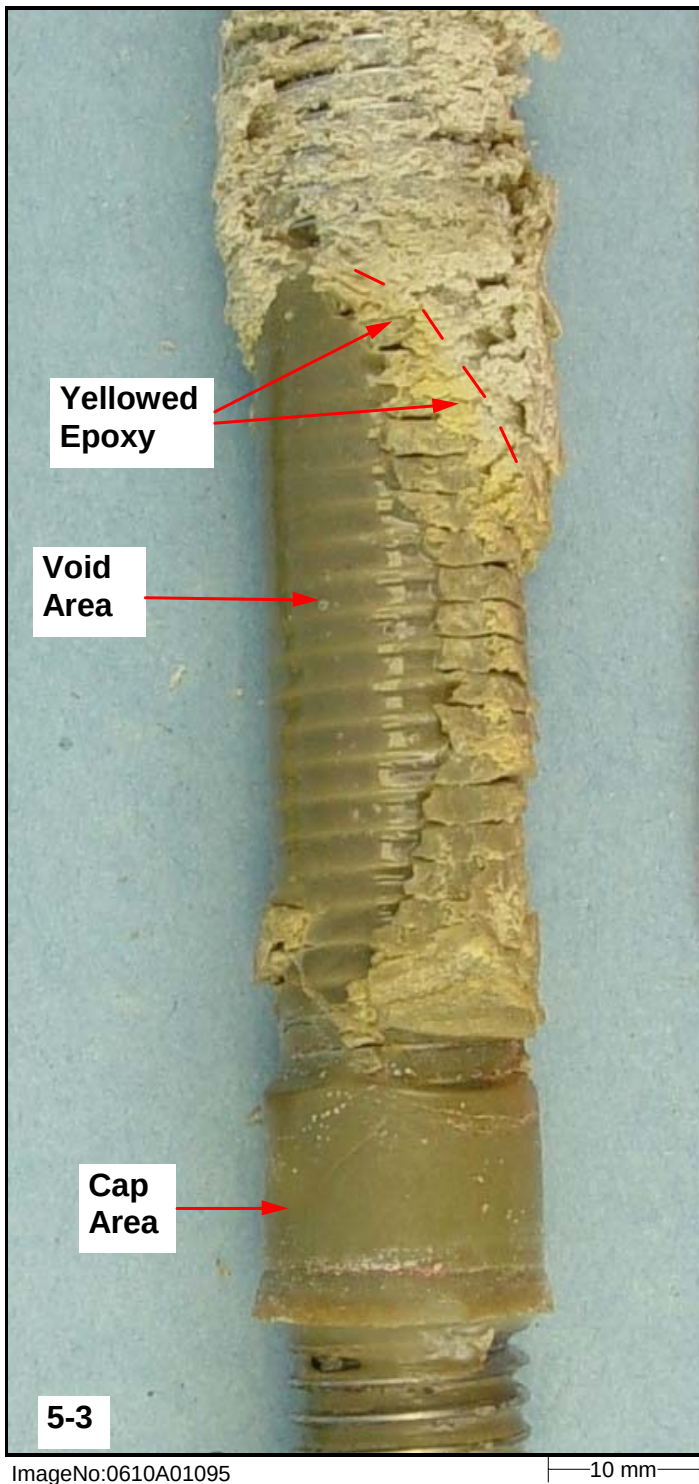


Figure 6. An example of a yellowed epoxy region slightly above the edge of a void area, on anchor 5-3. The dashed line indicates the boundary between the yellowed and non-yellowed epoxy.

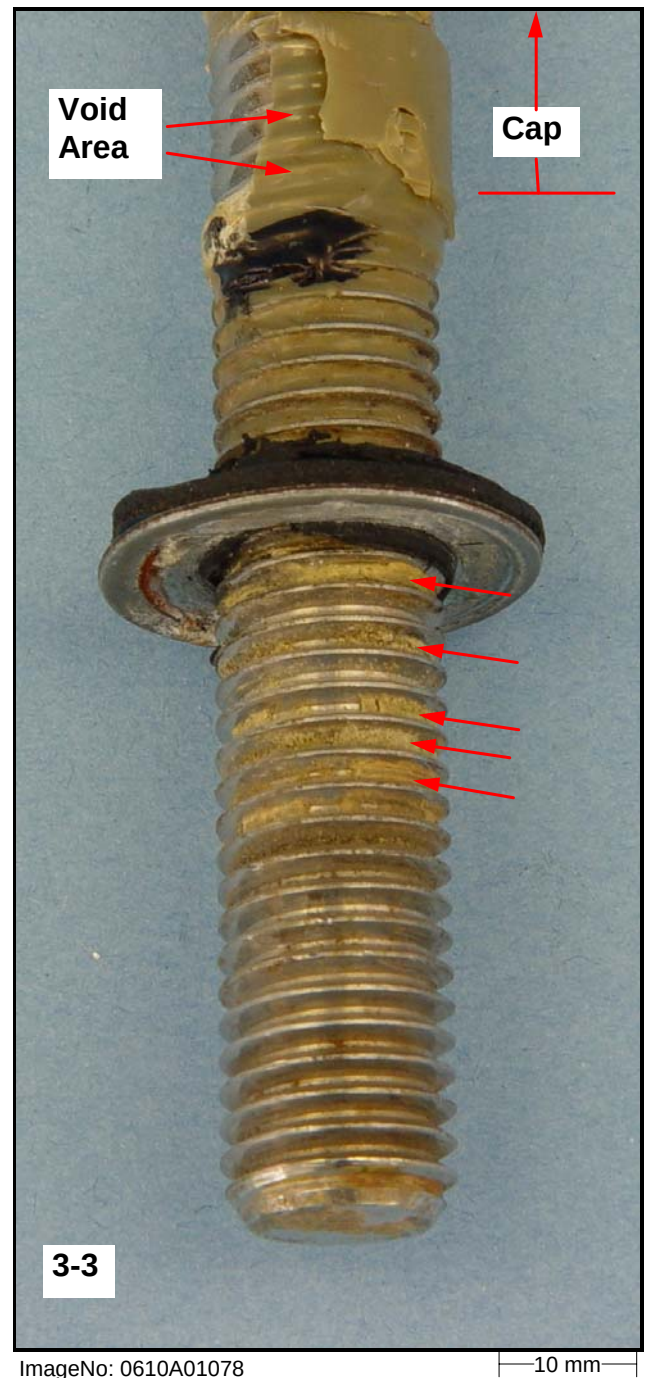


Figure 7. An example of overflow epoxy from anchor 3-3. The position corresponding to the cap is indicated, and arrows indicate where the overflow epoxy below the sealing washer had been contacted by the nut threads. An ink mark is also visible just below the position of an installed cap.

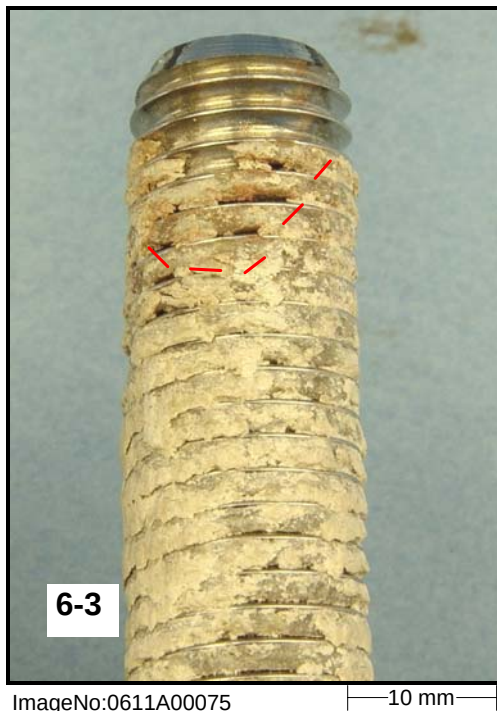


Figure 8. Closer view of an area of fractured epoxy near the top of anchor 6-3. The dashed line outlines an area with a light brown tint to the surface.

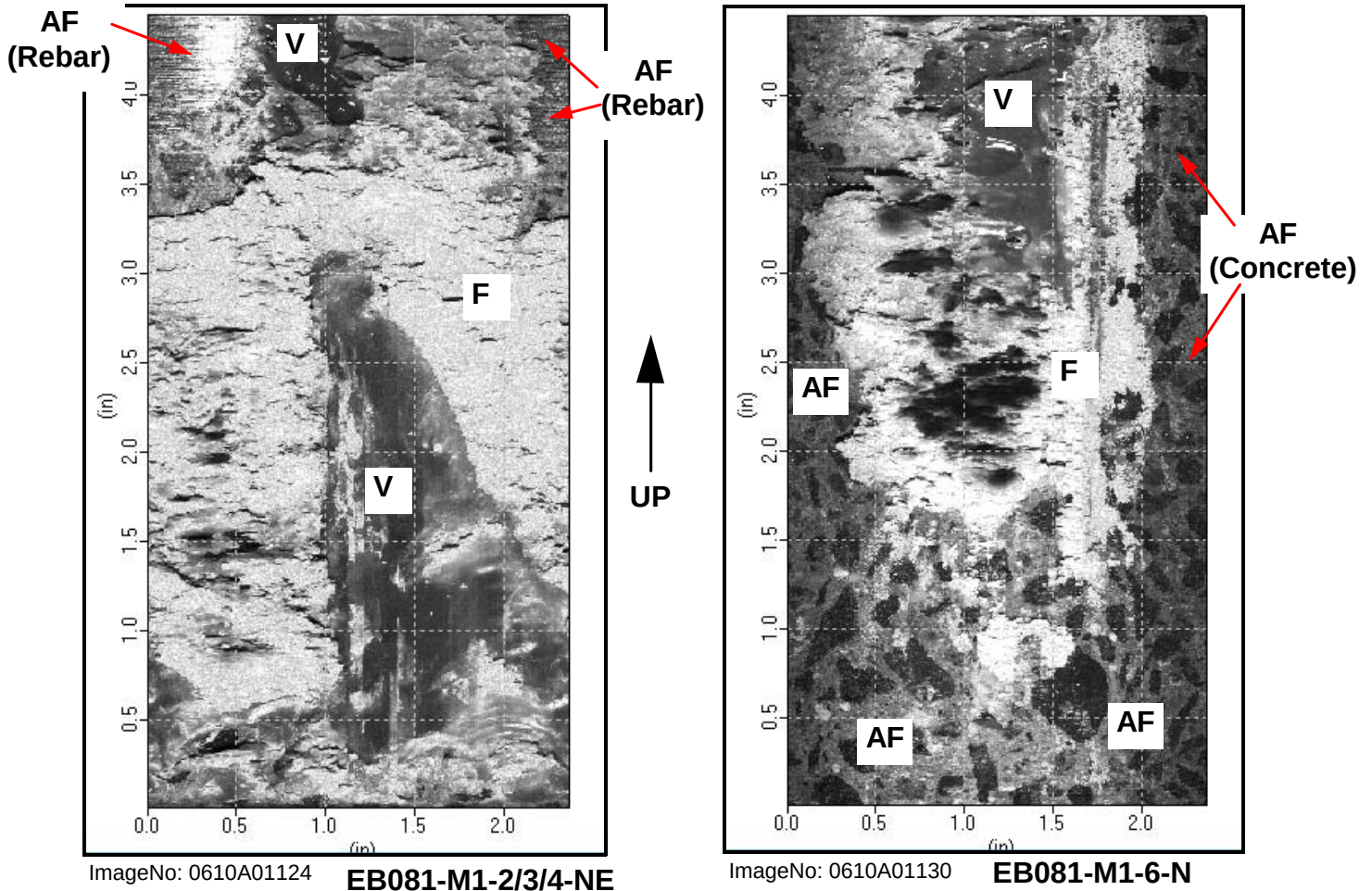
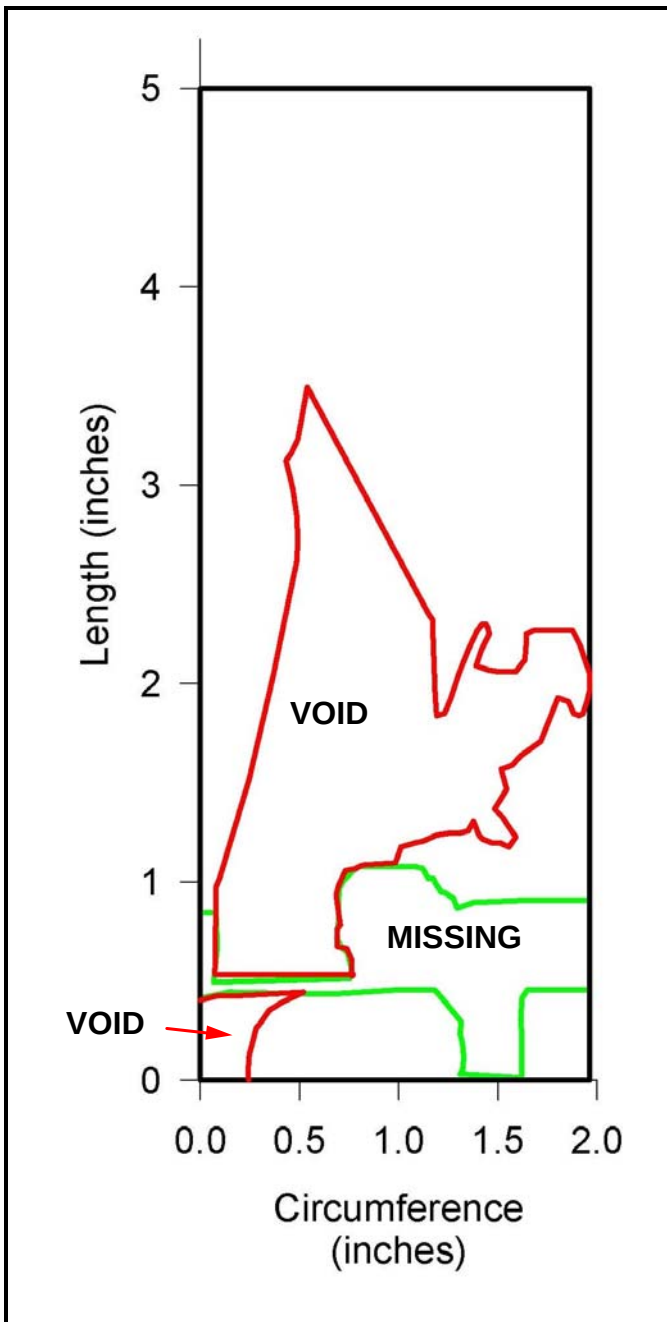
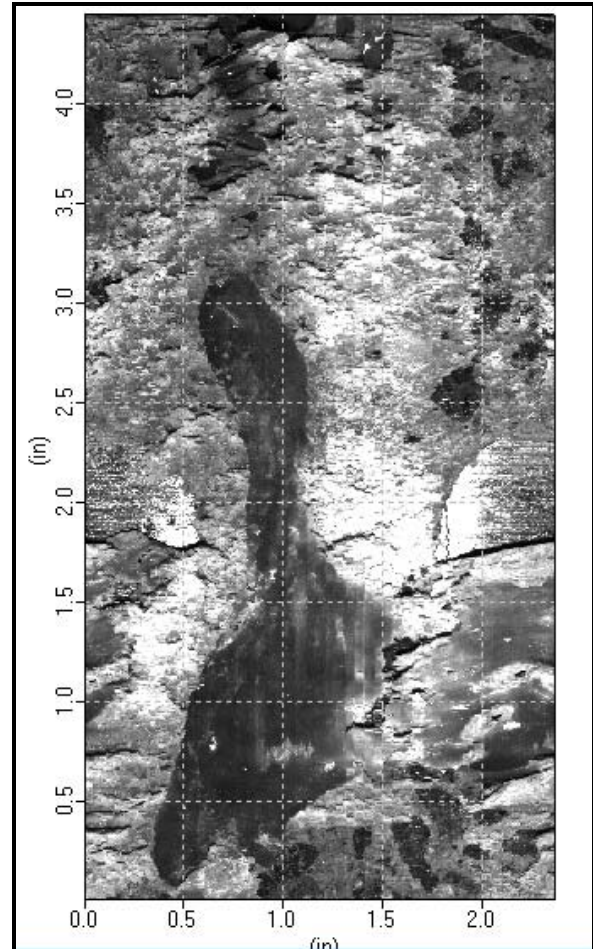


Figure 9. The laser-scanned images from the hole for anchor "EB081-M1-2/3/4-NE" (on the left), which has an area of adhesive failure ("AF") at rebar near the top of the image, and from the hole for anchor "EB081-M1-6-N" (on the right), which has adhesive failure surfaces in concrete areas over much of the anchor hole. Also labeled are areas with voids ("V") and fractured epoxy ("F").



ImageNo:0701A00368, Project No:2006080009



ImageNo: 0610A01128, Project No:2006080009

EB081-M1-5-N

Figure 10. Sketch of anchor 6-3 (on the left) and laser image of anchor hole EB081-M1-5-N (on the right).



ImageNo:0701A00429, Project No:2006080009

Figure 11. The anchor from position
EB080-N-4/5/6-NE.



ImageNo: 0612A00427, Project No:2006080009

Figure 12. The anchor from positon
EB081-S-8-S.



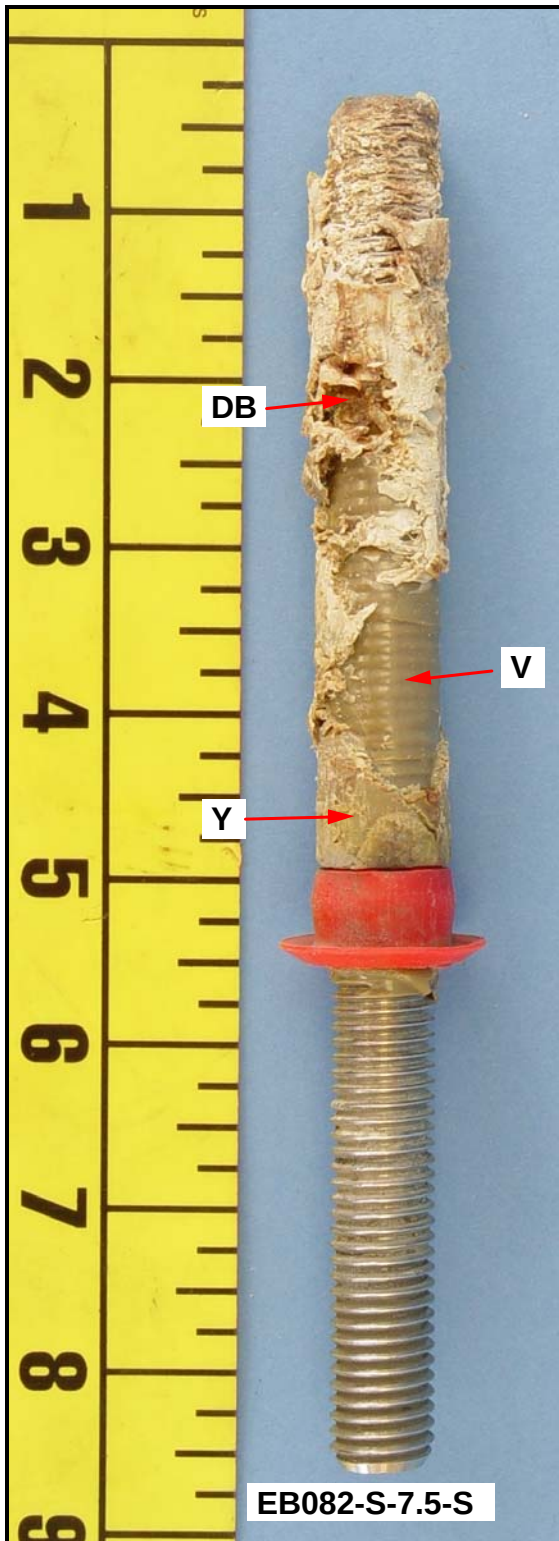
ImageNo:0612A00260, Project No:2006080009



ImageNo: 0612A00275, Project No:2006080009

10 mm

Figure 13. Overall view of the anchor from position EB081-S-10-S (left photograph) and a closer view of the soft epoxy at the top of this anchor (right photograph). The dark streaks along the shank of the anchor visible in the right photograph (indicated by the unlabeled arrows) are areas where the epoxy was apparently originally against the concrete surface of the hole, then moved radially inward at a void region.



ImageNo:0612A00399, Project No:2006080009

Figure 14. The anchor from position EB082-S-7.5-S, with void areas ("V"), dark brown epoxy ("DB"), and yellowed epoxy ("Y") indicated.



ImageNo: 0701A00433, Project No:2006080009

Figure 15. The anchor from position EB083-S-1-S. The dark brown epoxy extends above the top of the anchor and is also present in streaks along the anchor's length.

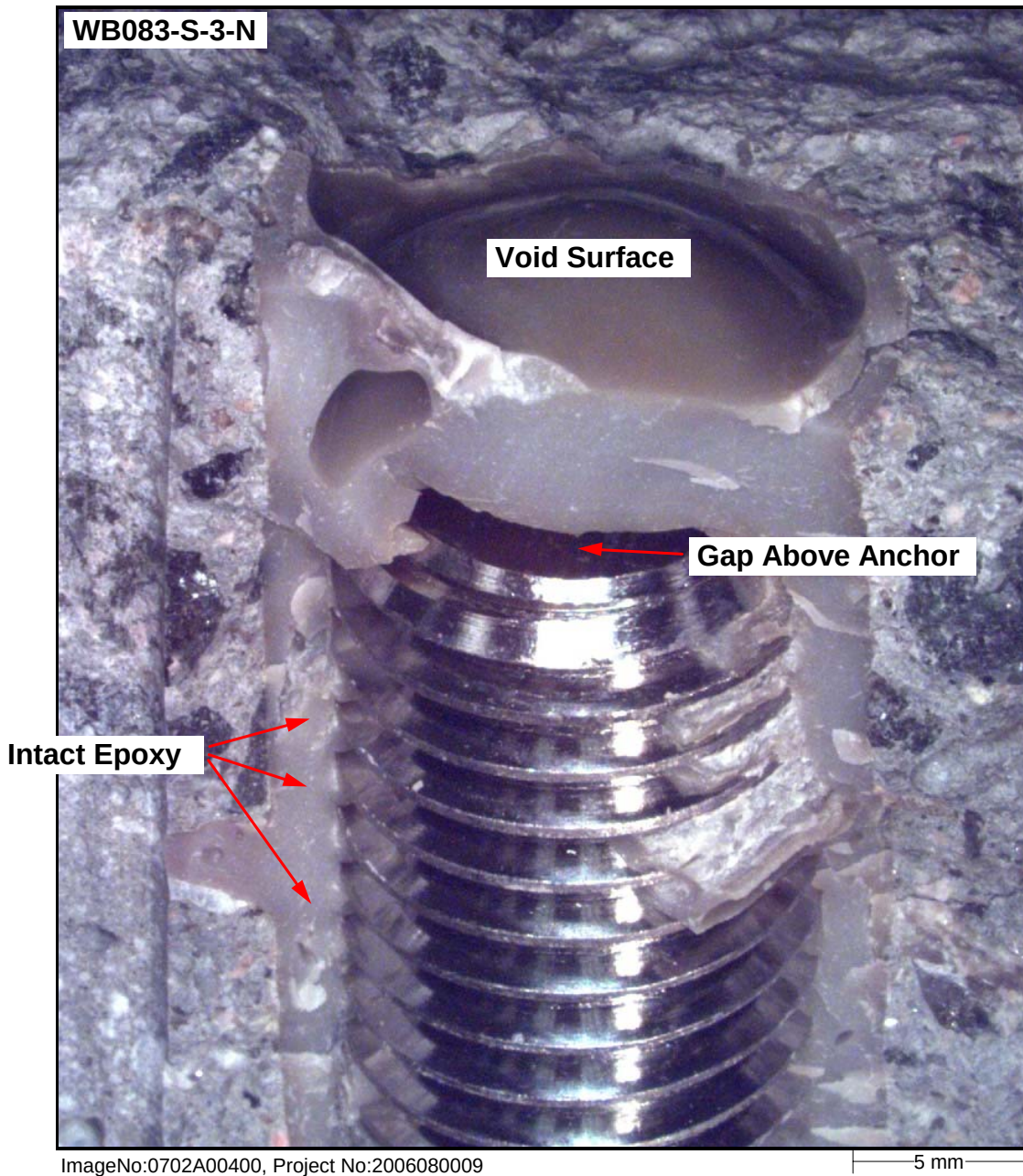


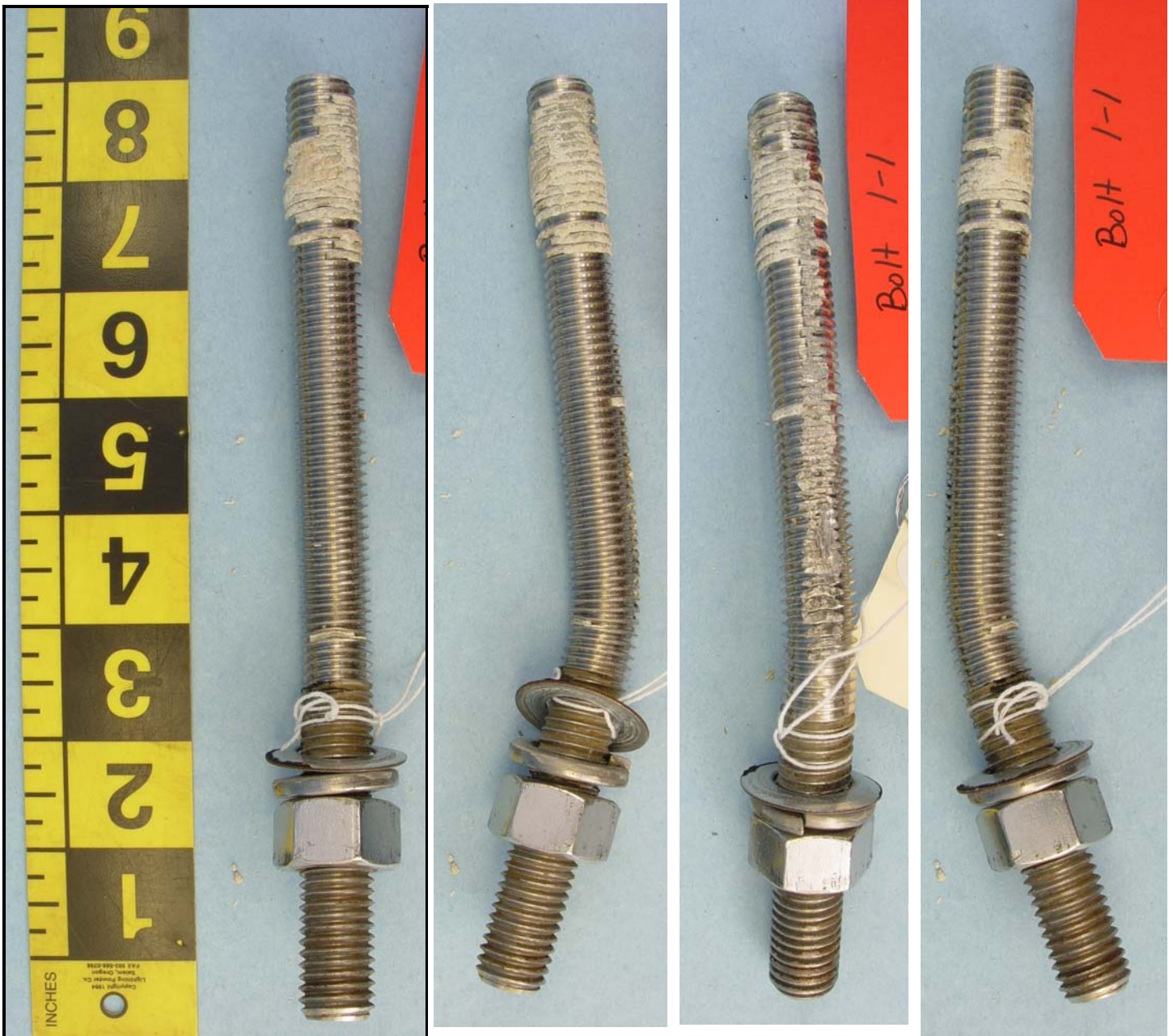
Figure 16. View of the upper end of anchor WB083-S-3-N, still contained within half of its core sample. The void surface above the epoxy above the top of the anchor, the gap above the top of the anchor, and the intact epoxy between the anchor threads and hole wall are indicated.

Appendix 1
Overall views of Adhesive Anchors from
the Failure Area



ImageNo: 0609A00197, Project No:2006080009

Four views of anchor E1-1 after removal of the red polyethylene cap.



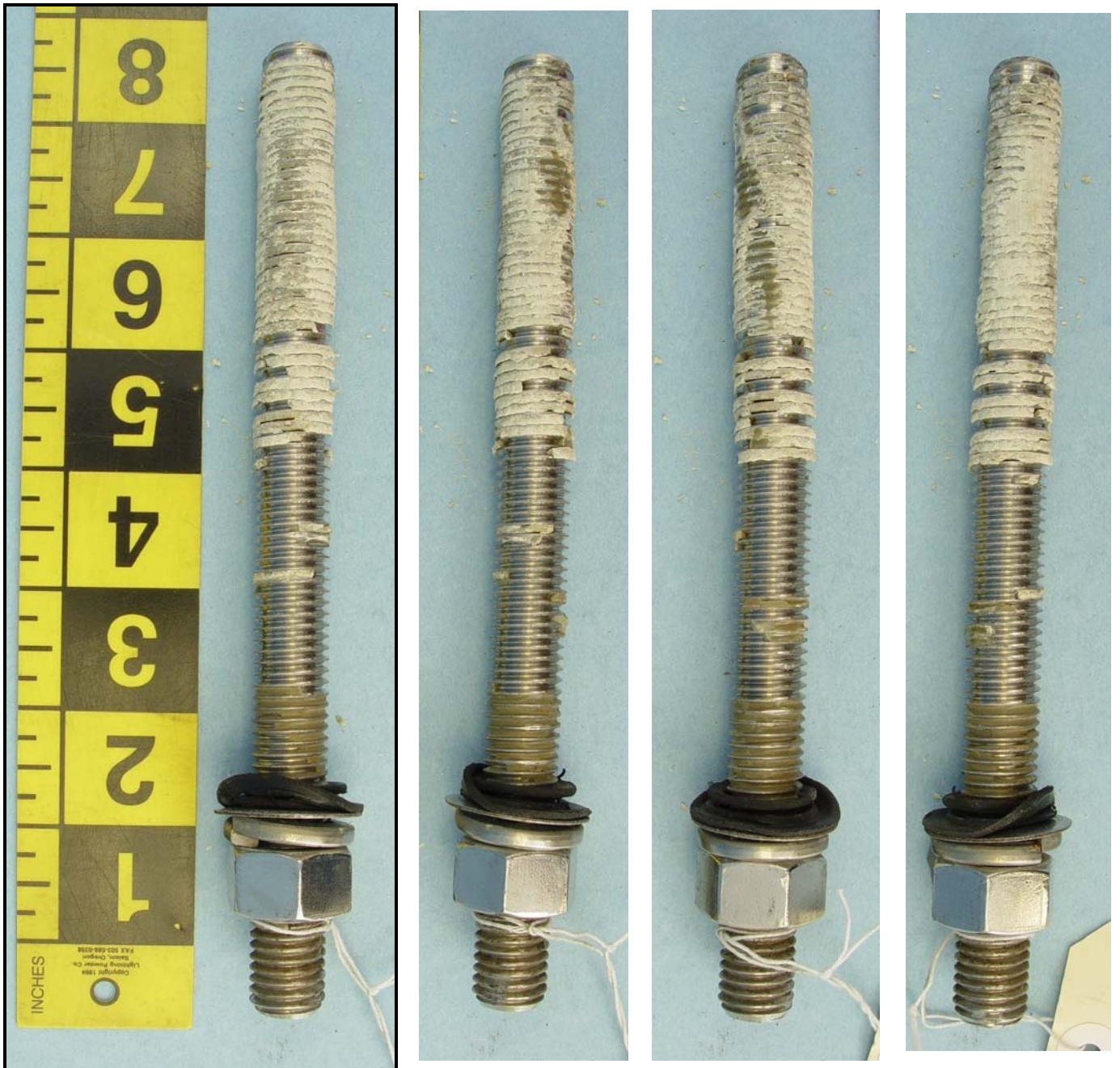
ImageNo: 0609A00130, Project No:2006080009

Four views of anchor 1-1.



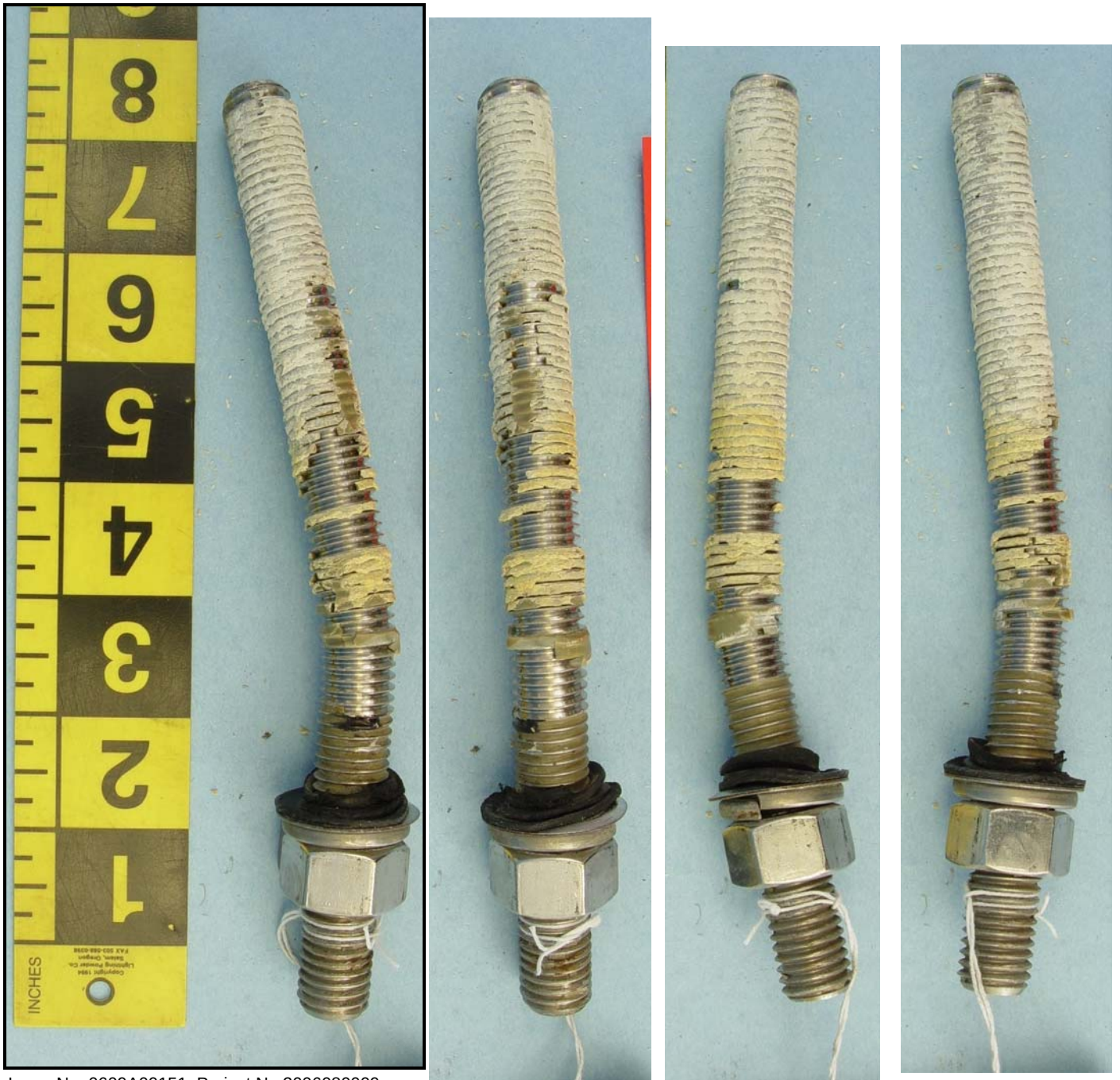
ImageNo: 0609A00135, Project No:2006080009

Four views of anchor 1-2.



ImageNo: 0609A00139, Project No:2006080006

Four views of anchor 1-3



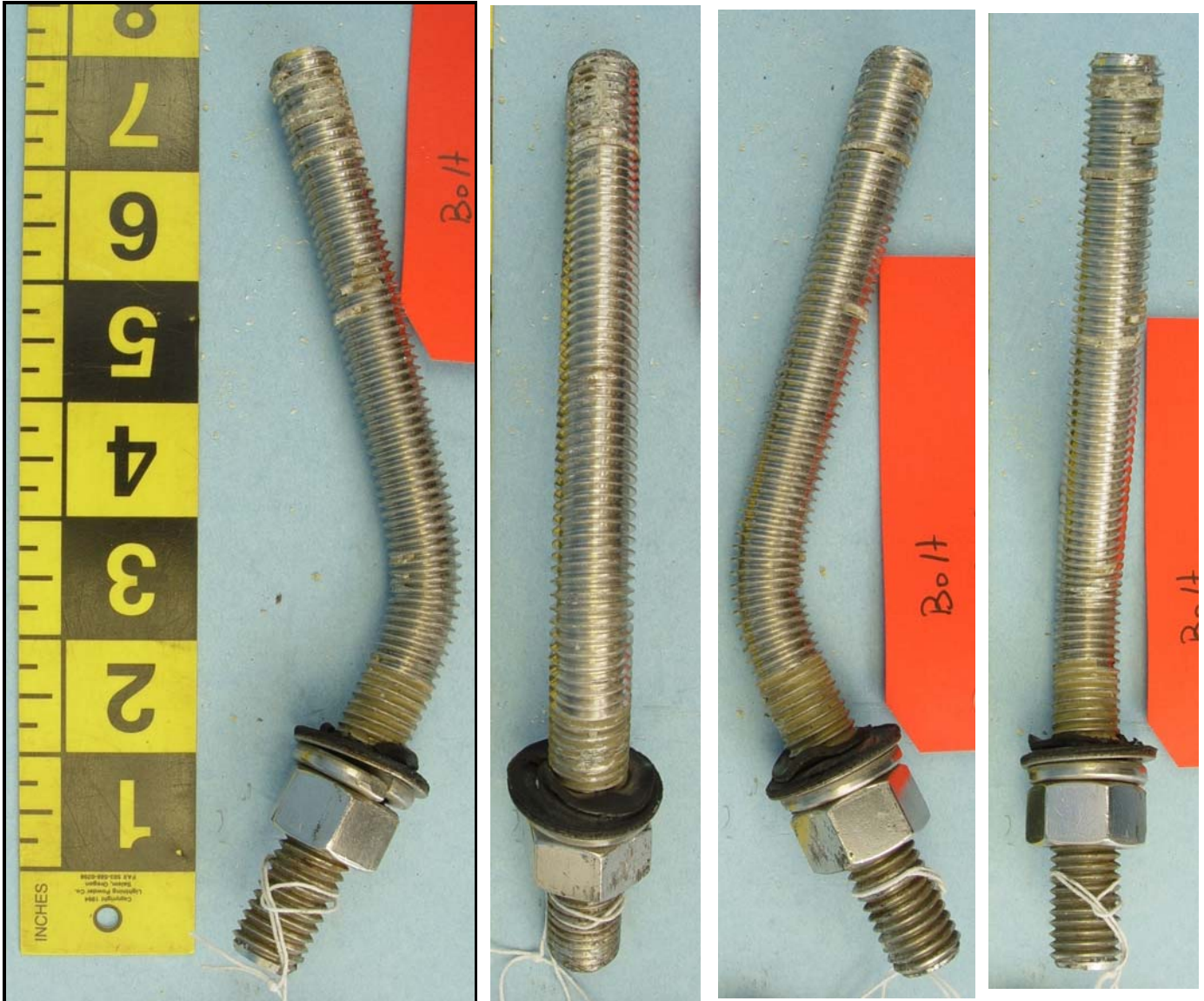
ImageNo: 0609A00151, Project No:2006080009

Four views of anchor 1-4.



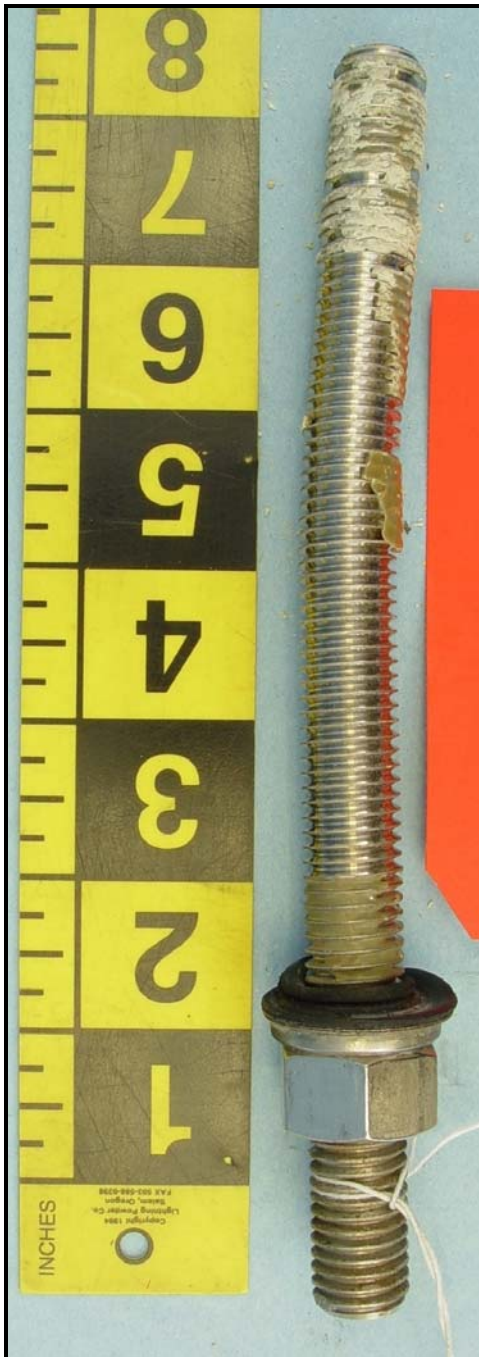
ImageNo: 0609A00155, Project No:2006080

Four views of anchor 2-1.

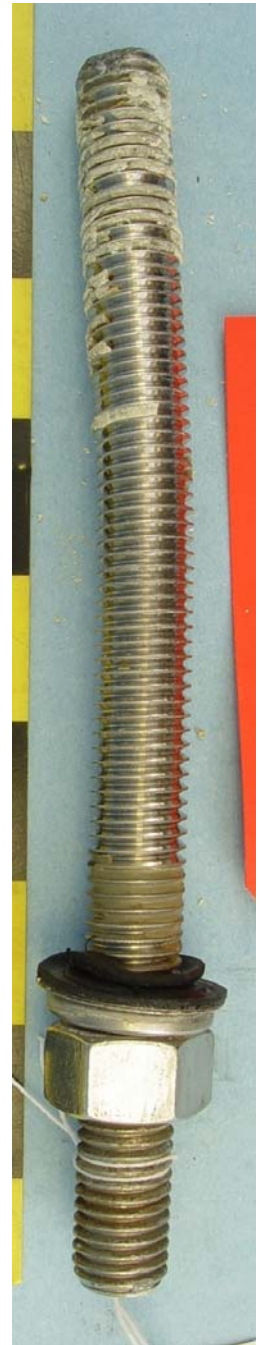


ImageNo: 0609A00163, Project No:2006080009

Four views of anchor 2-2.



ImageNo: 0609A00165, Project No:200608000



Four views of anchor 2-3.



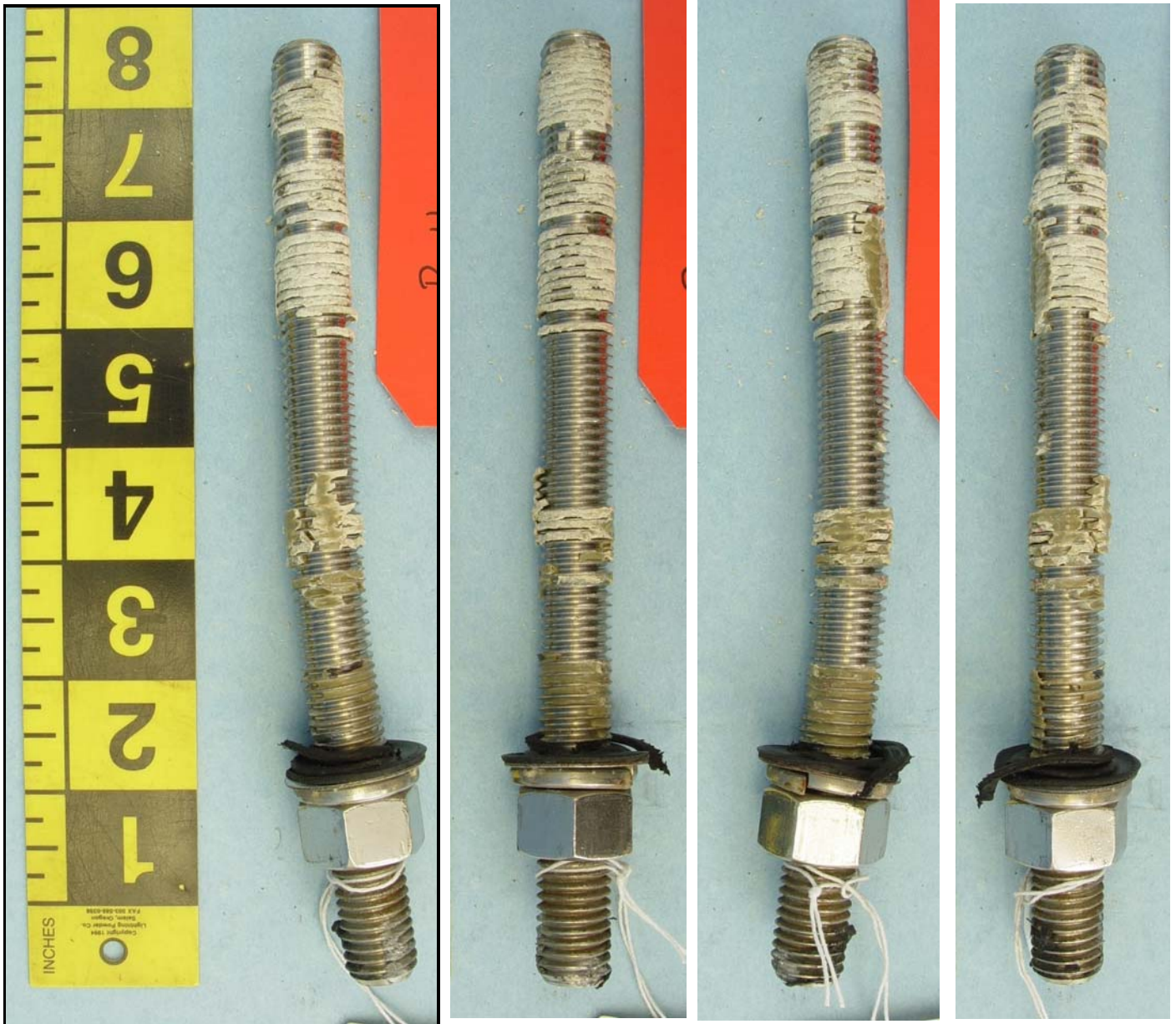
ImageNo: 0609A00170, Project No:2006080009

Four views of anchor 2-4.



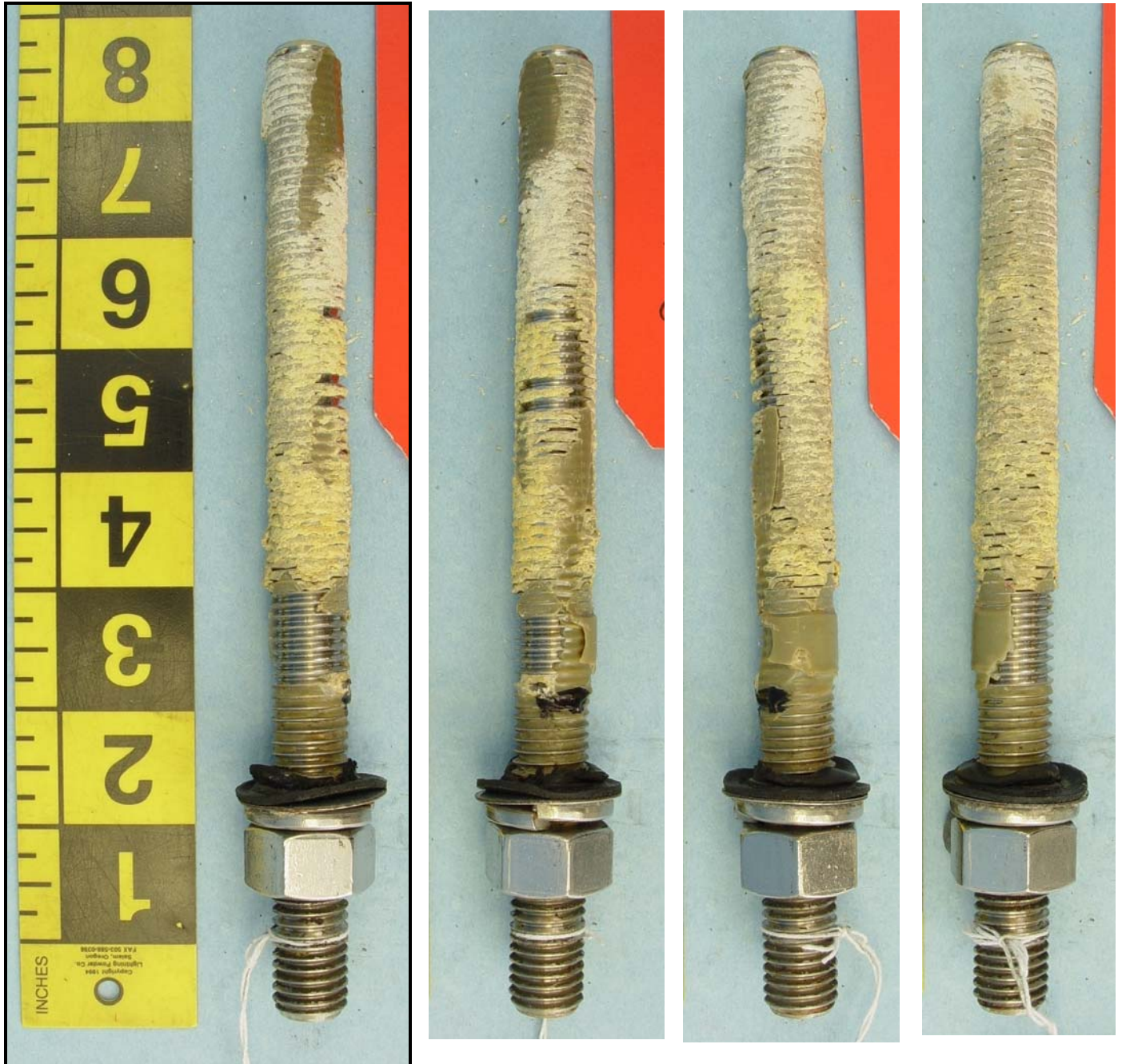
ImageNo: 0609A00174, Project No:2006080009

Four views of anchor 3-1.



ImageNo: 0609A00178, Project No:2006080009

Four views of anchor 3-2.



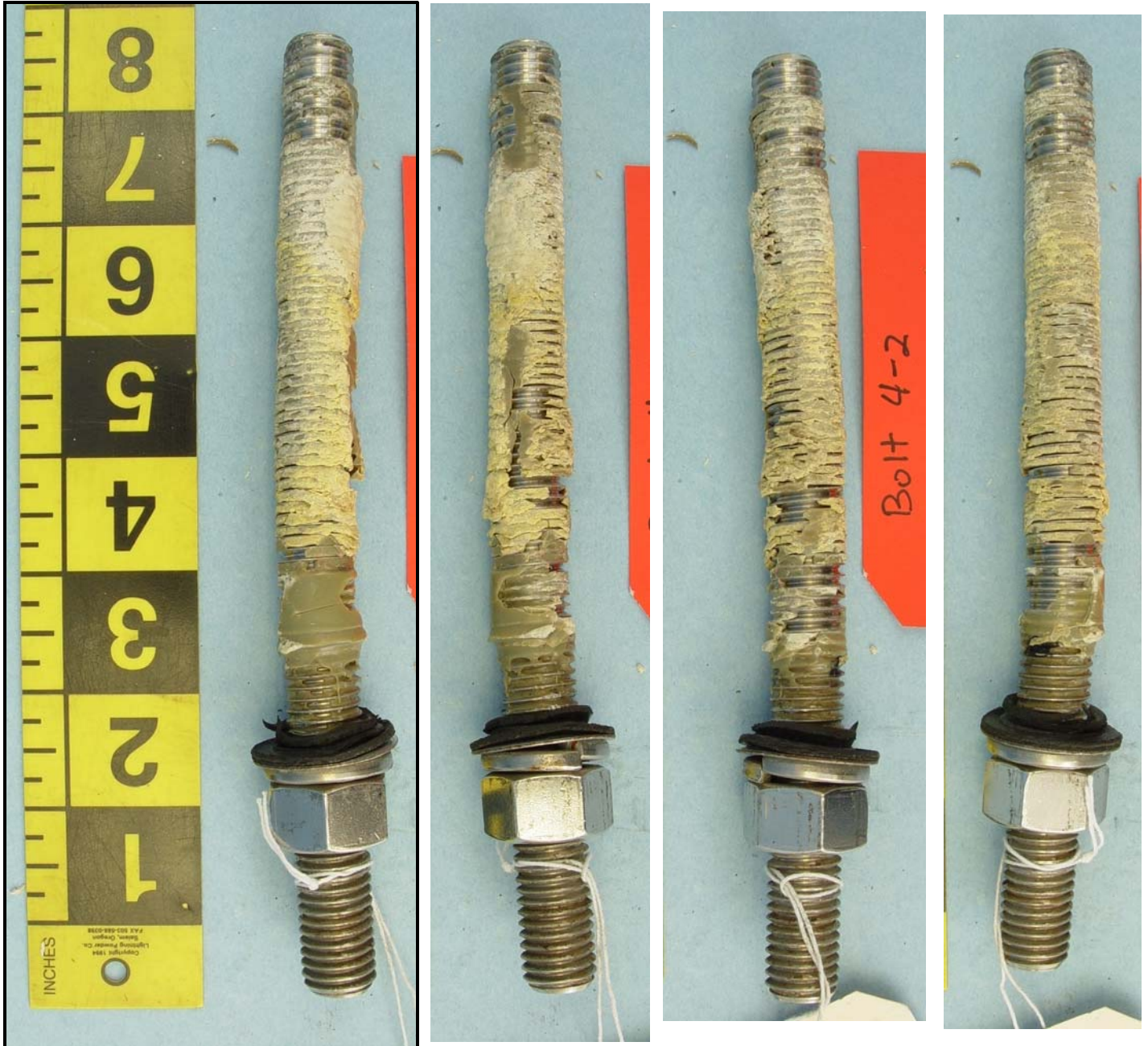
ImageNo: 0609A00182, Project No:2006080009

Four views of anchor 3-3.



ImageNo: 0609A00186, Project No:2006080009

Four views of anchor 3-4.



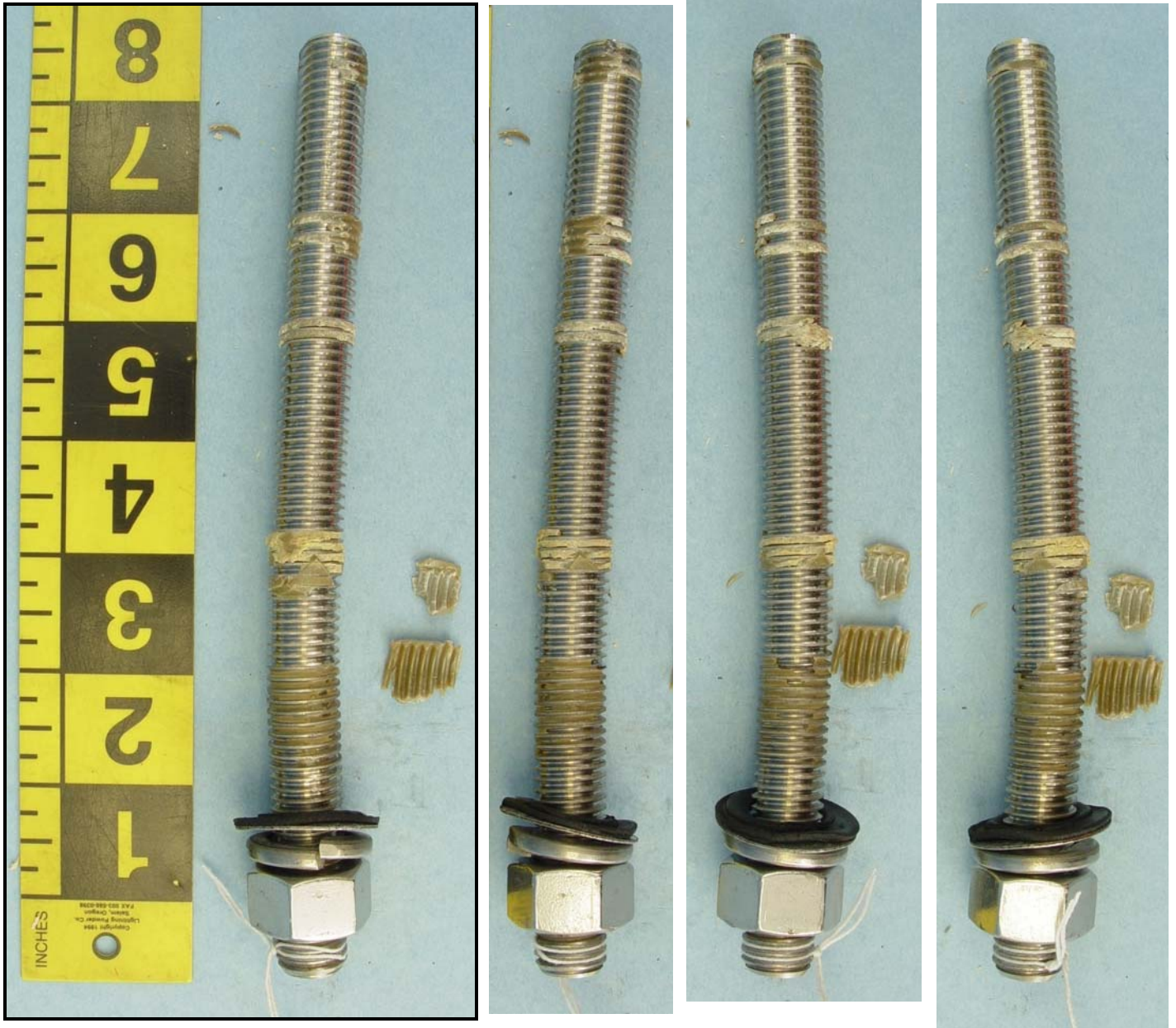
ImageNo: 0609A00203, Project No:2006080009

Four views of anchor 4-2.



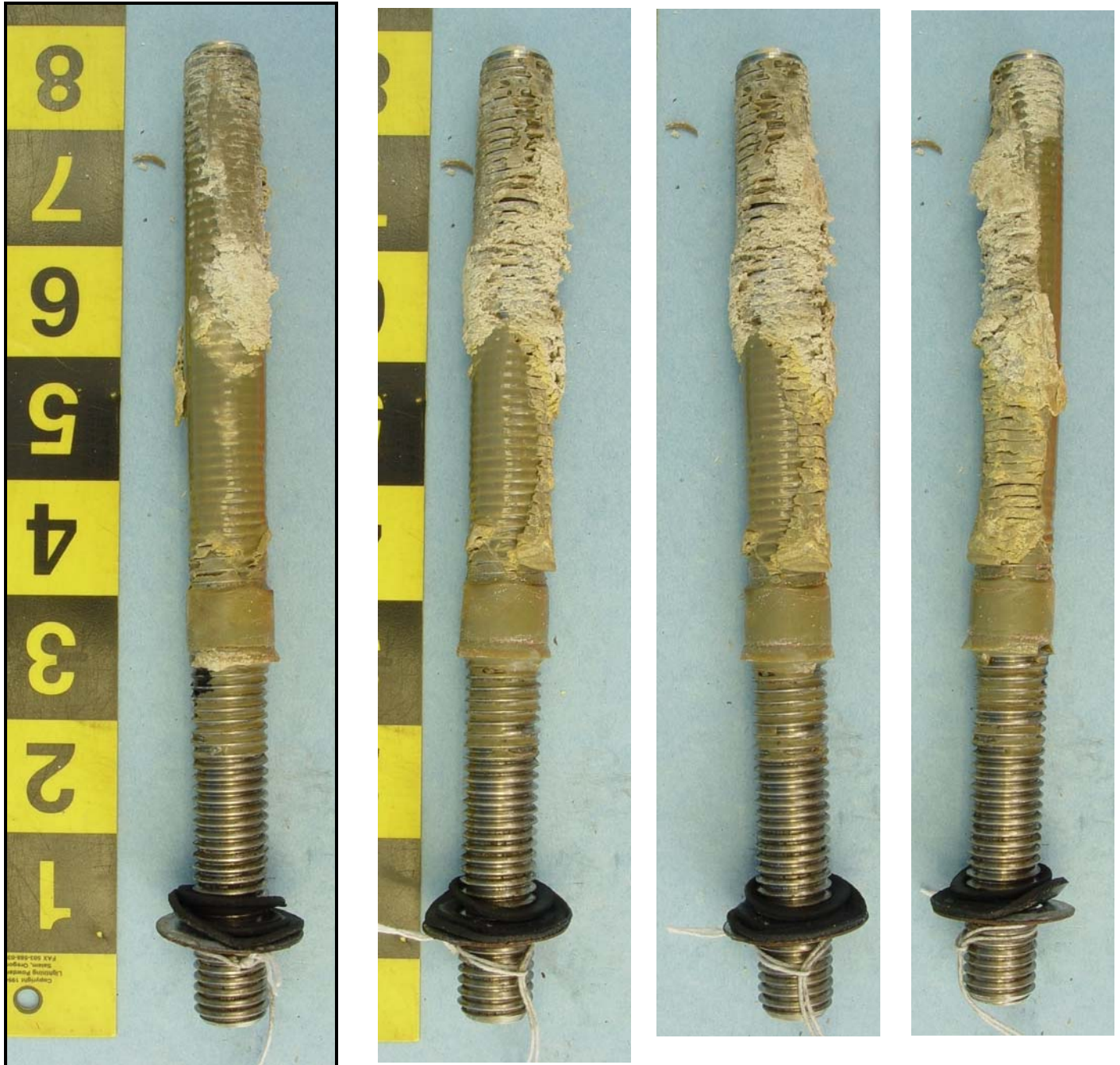
ImageNo: 0609A00222, Project No:2006080009

Four views of anchor 5-1 shown after removal of the red polyethylene cap.



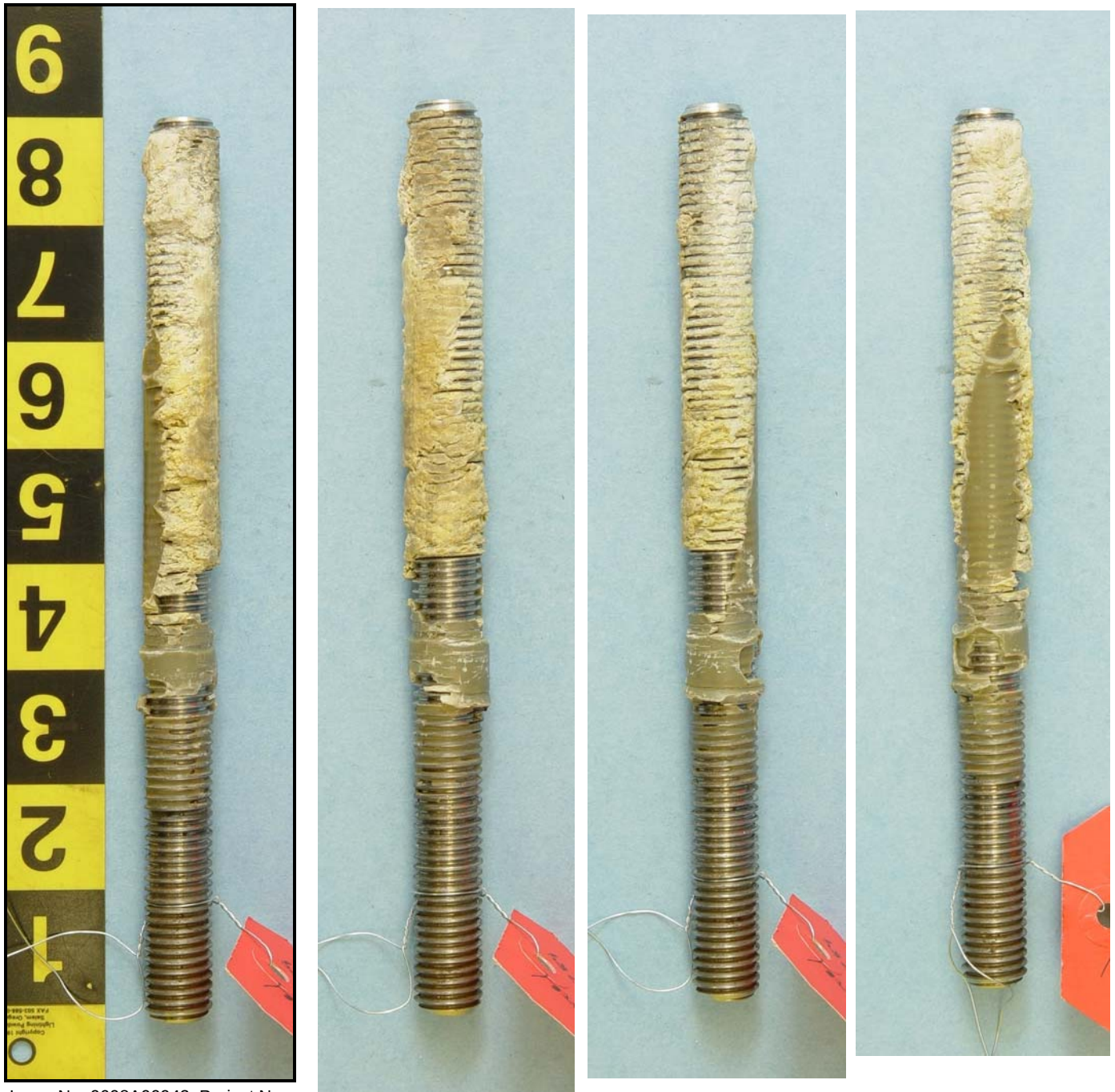
ImageNo: 0609A00227, Project No:2006080009

Four views of anchor 5-2 shown after removal of the red polyethylene cap.



ImageNo: 0609A00235, Project No:2006i

Four views of anchor 5-3 shown after removal of the red polyethylene cap.



ImageNo: 0608A00942, Project No

Four views of anchor 6-1 shown after removal of the red polyethylene cap, sealing washer, lock washer, and nut.



ImageNo: 0611A00125, Project No:2006080

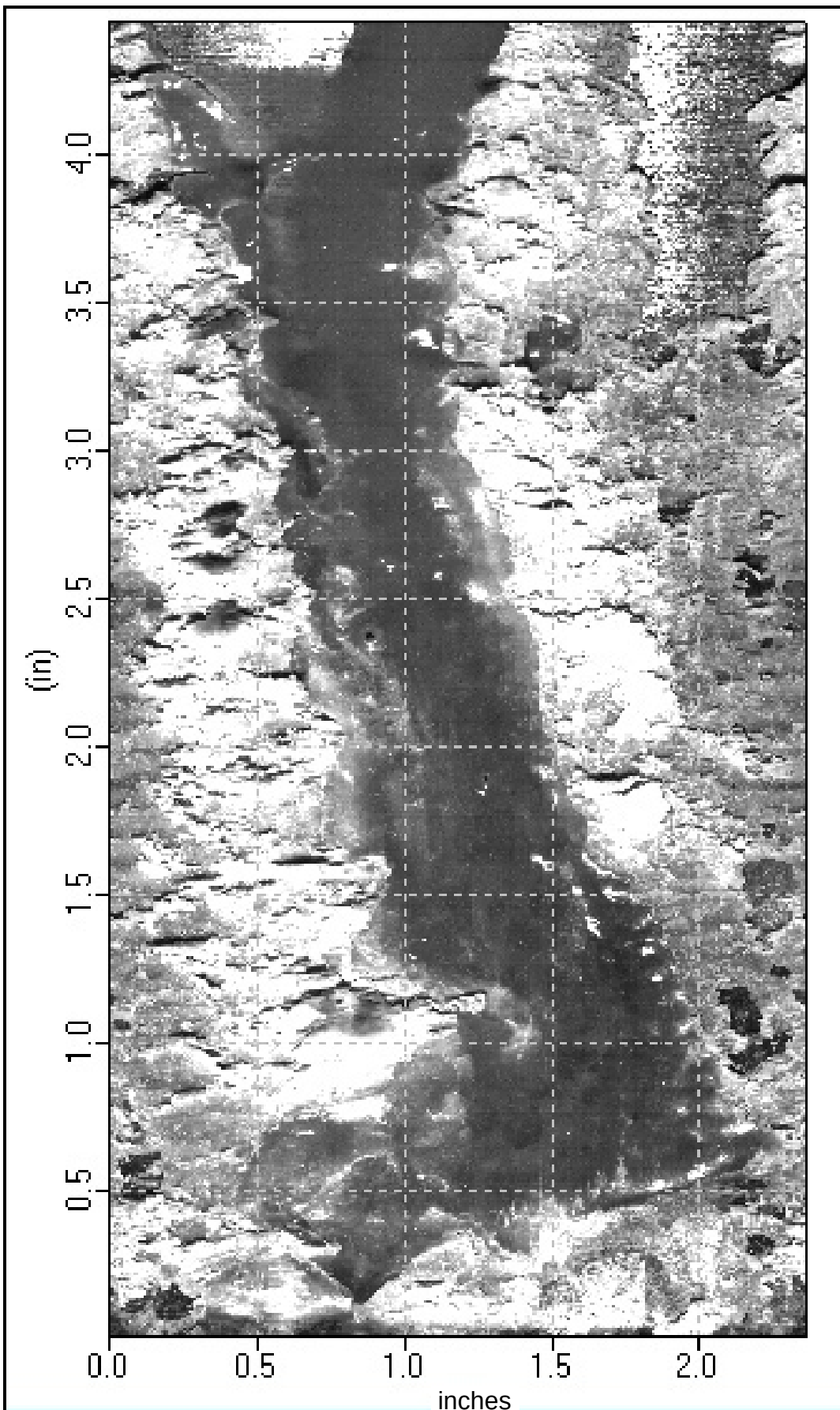
Four views of anchor 6-2 shown after removal of the red polyethylene cap, sealing washer, lock washer, and nut.



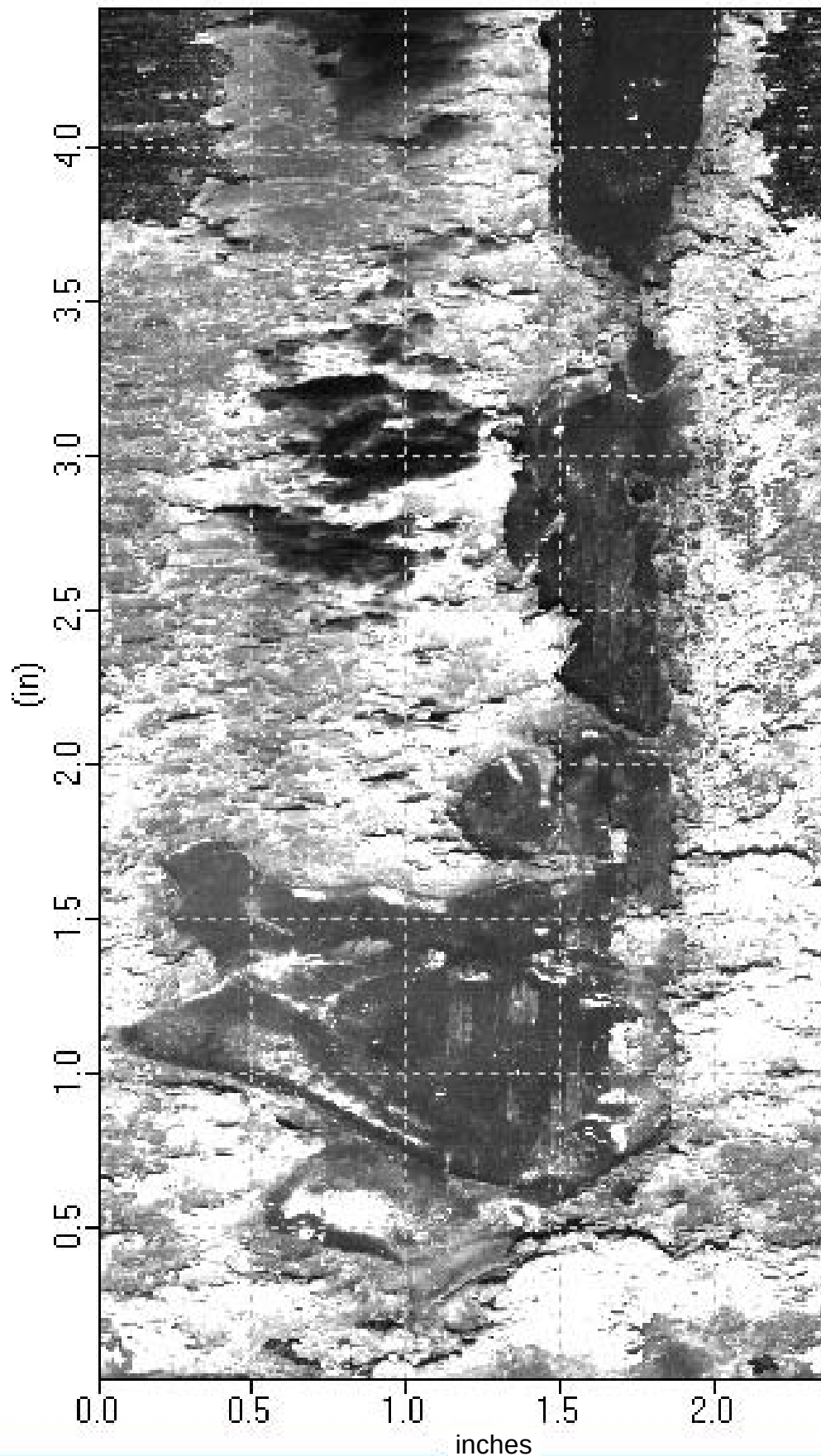
ImageNo: 0609A00124, Project No:20060800

Four views of anchor 6-3 shown after removal of the red polyethylene cap, sealing washer, lock washer, and nut.

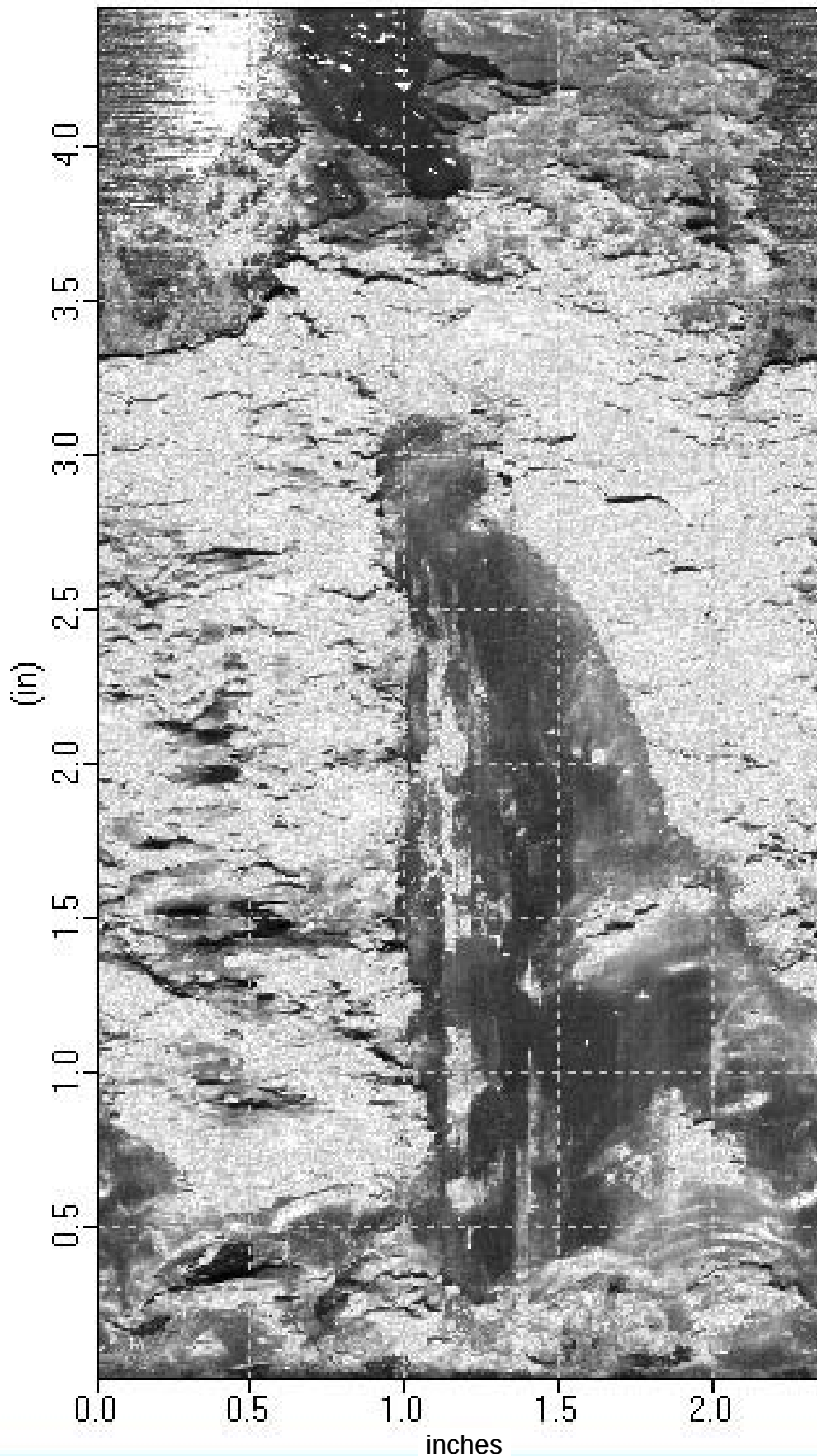
Appendix 2
Laser scanned images of the Failure
Area Anchor Holes



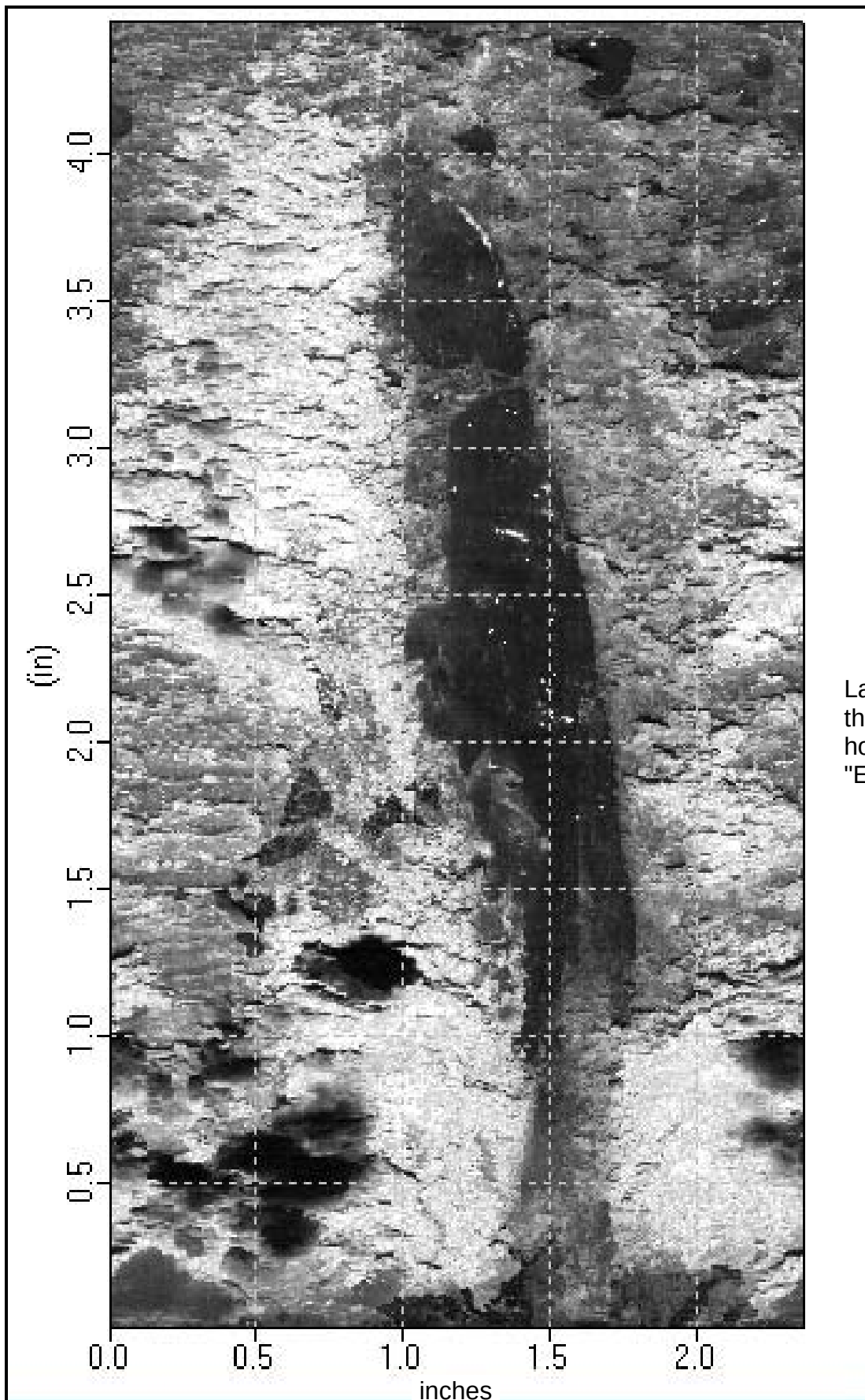
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-1-N".



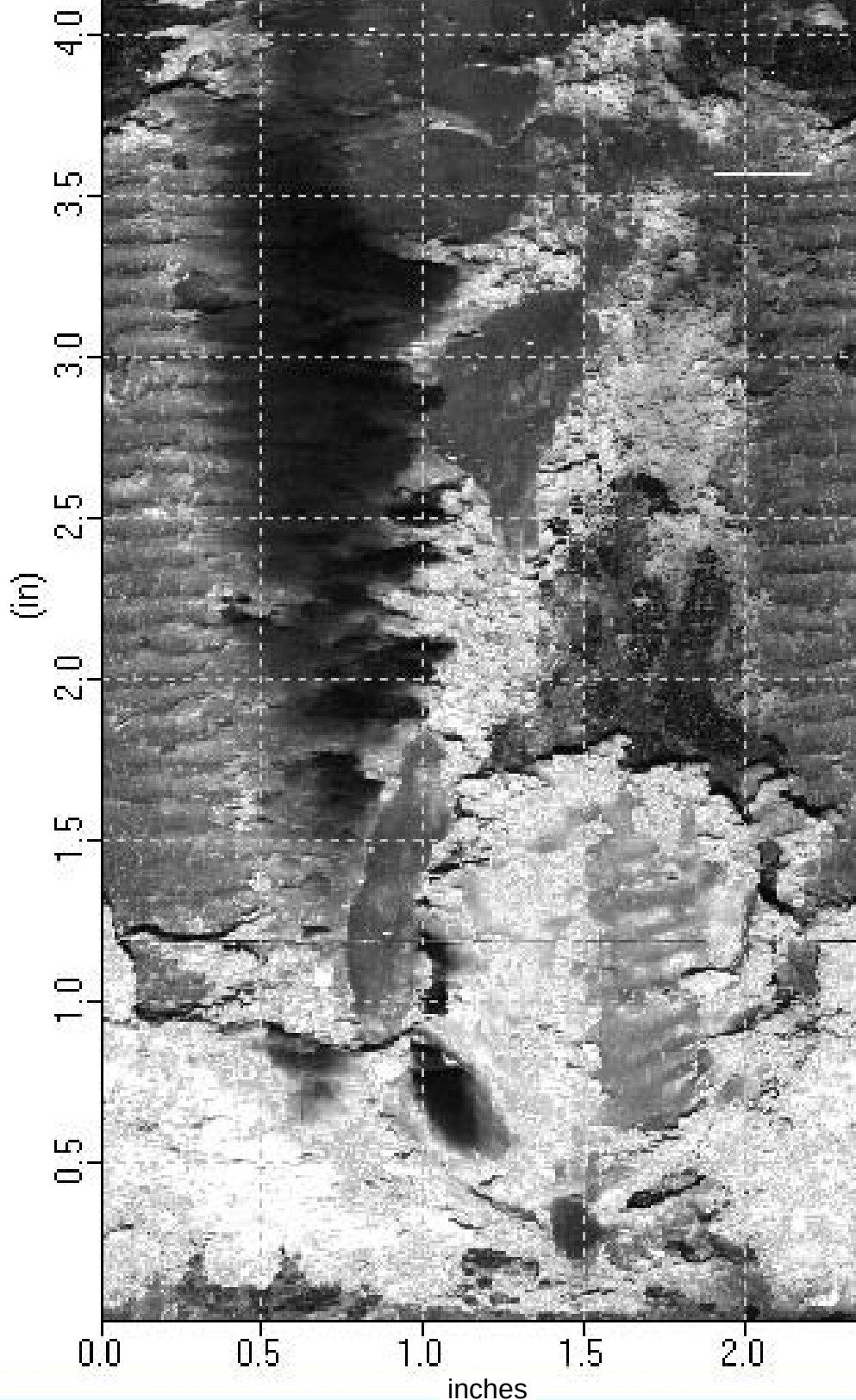
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-1-S".



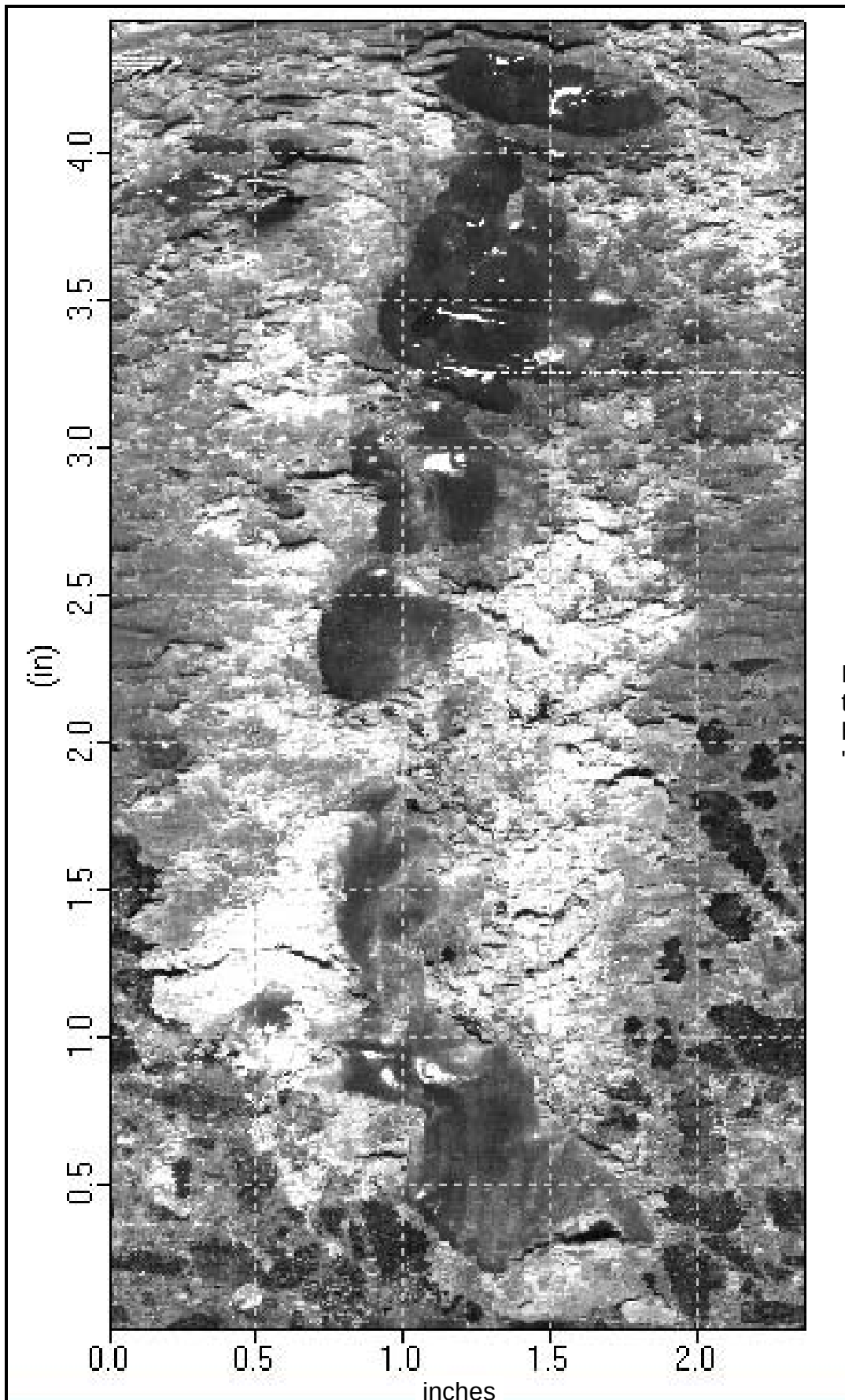
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-2/3/4-NE".



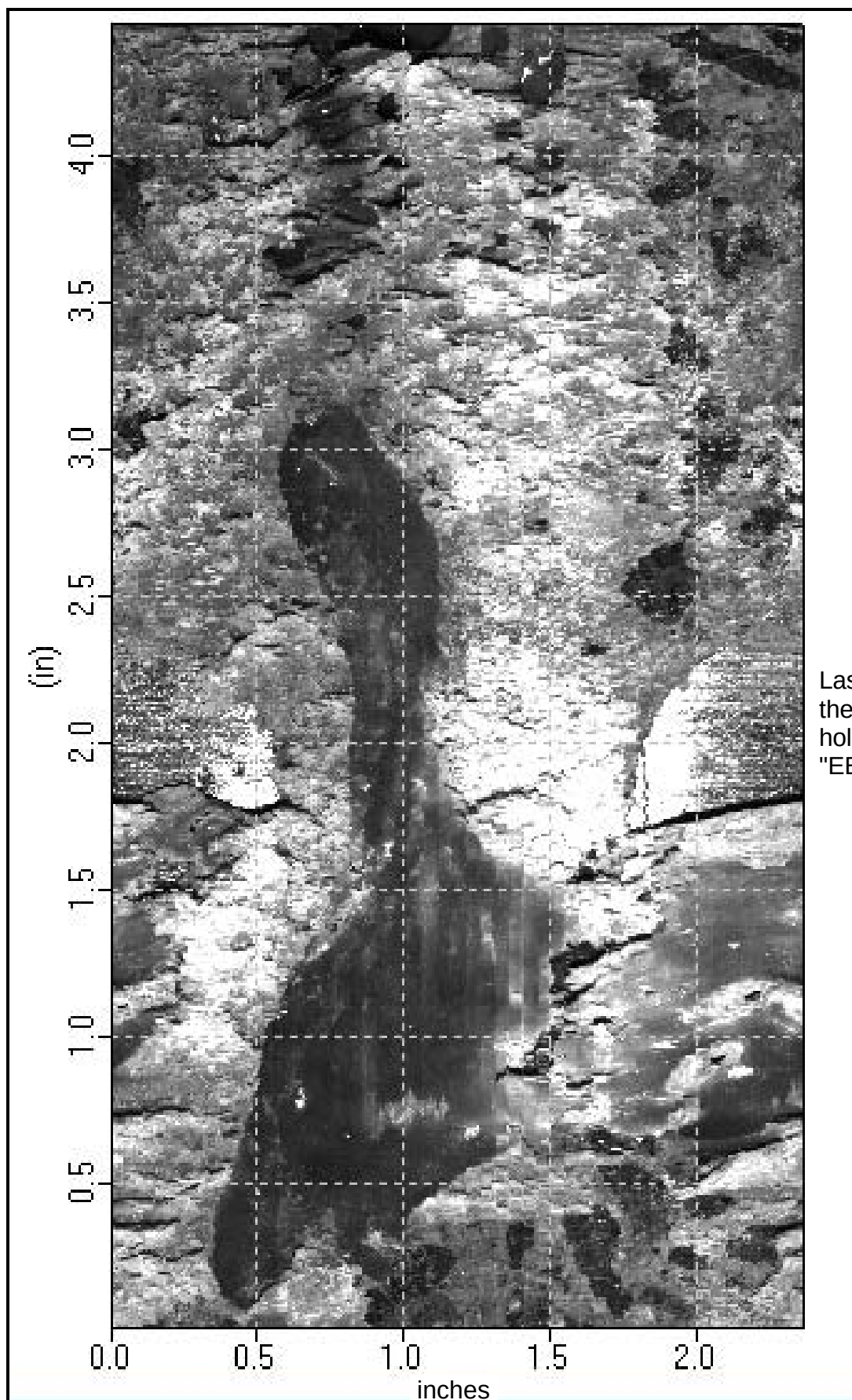
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-2/3/4-NW".



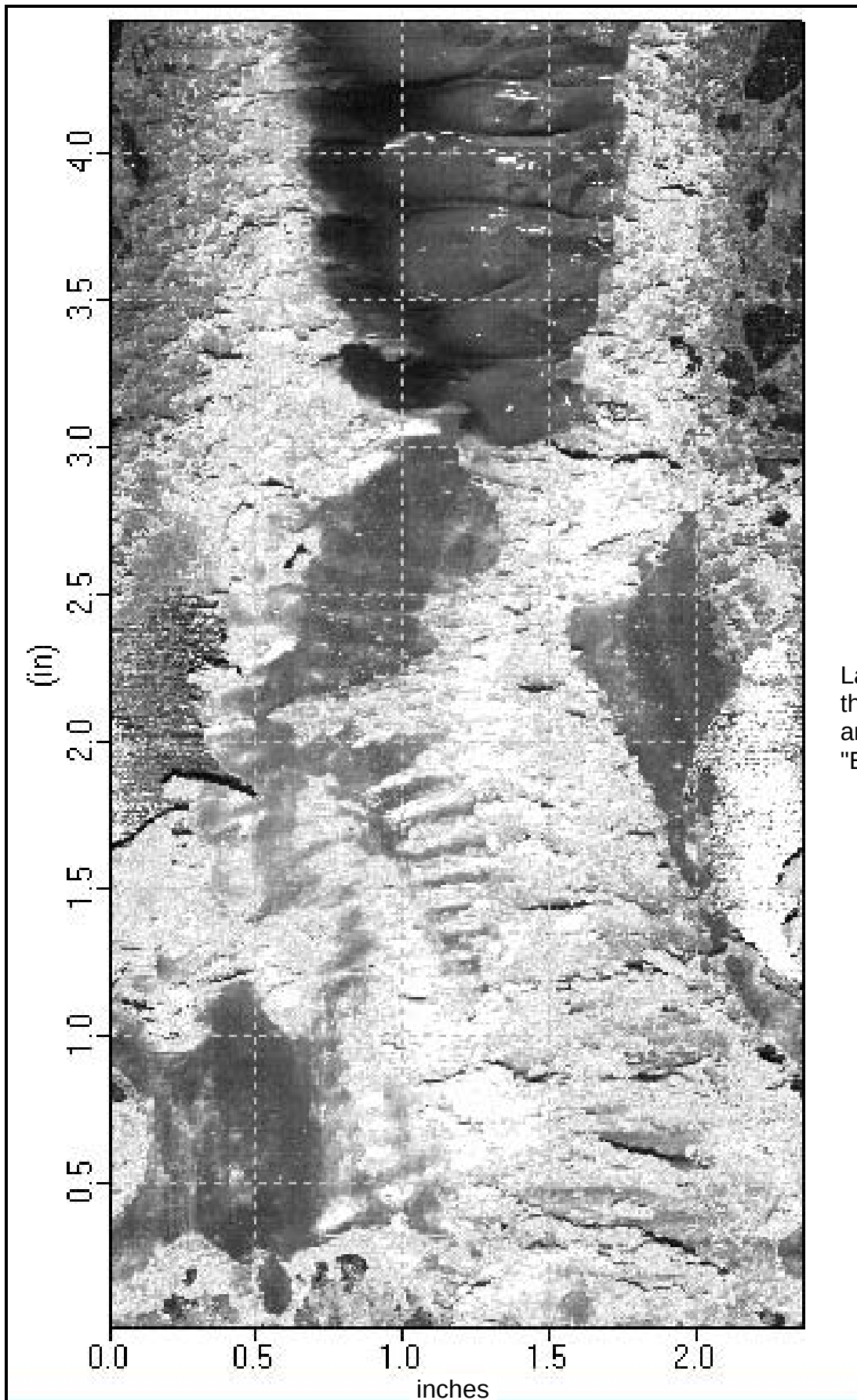
Laser scanned image of
the interior of the
anchor hole at position
"EB081-M1-2/3/4-SE".



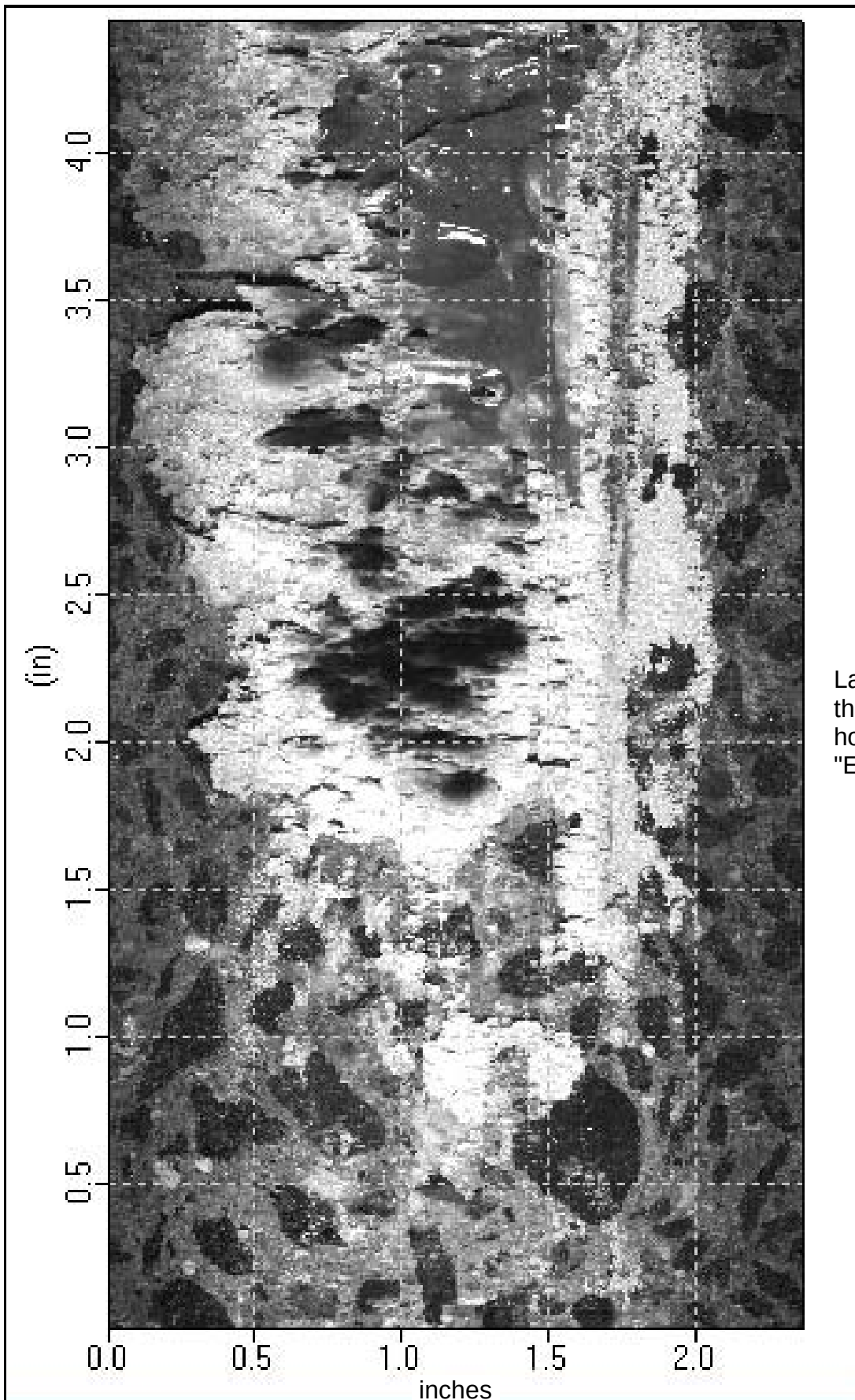
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-2/3/4-SW".



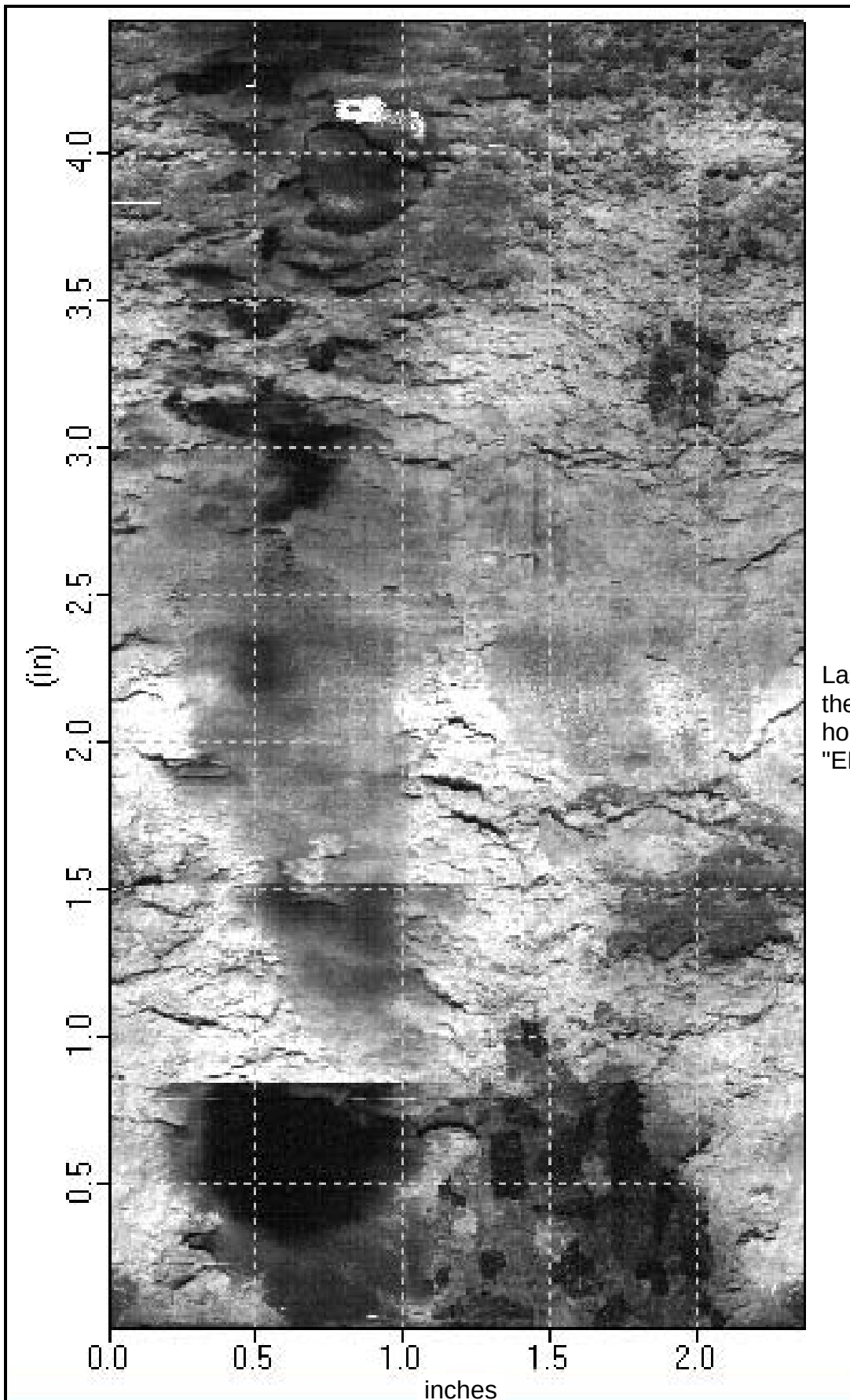
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-5-N".



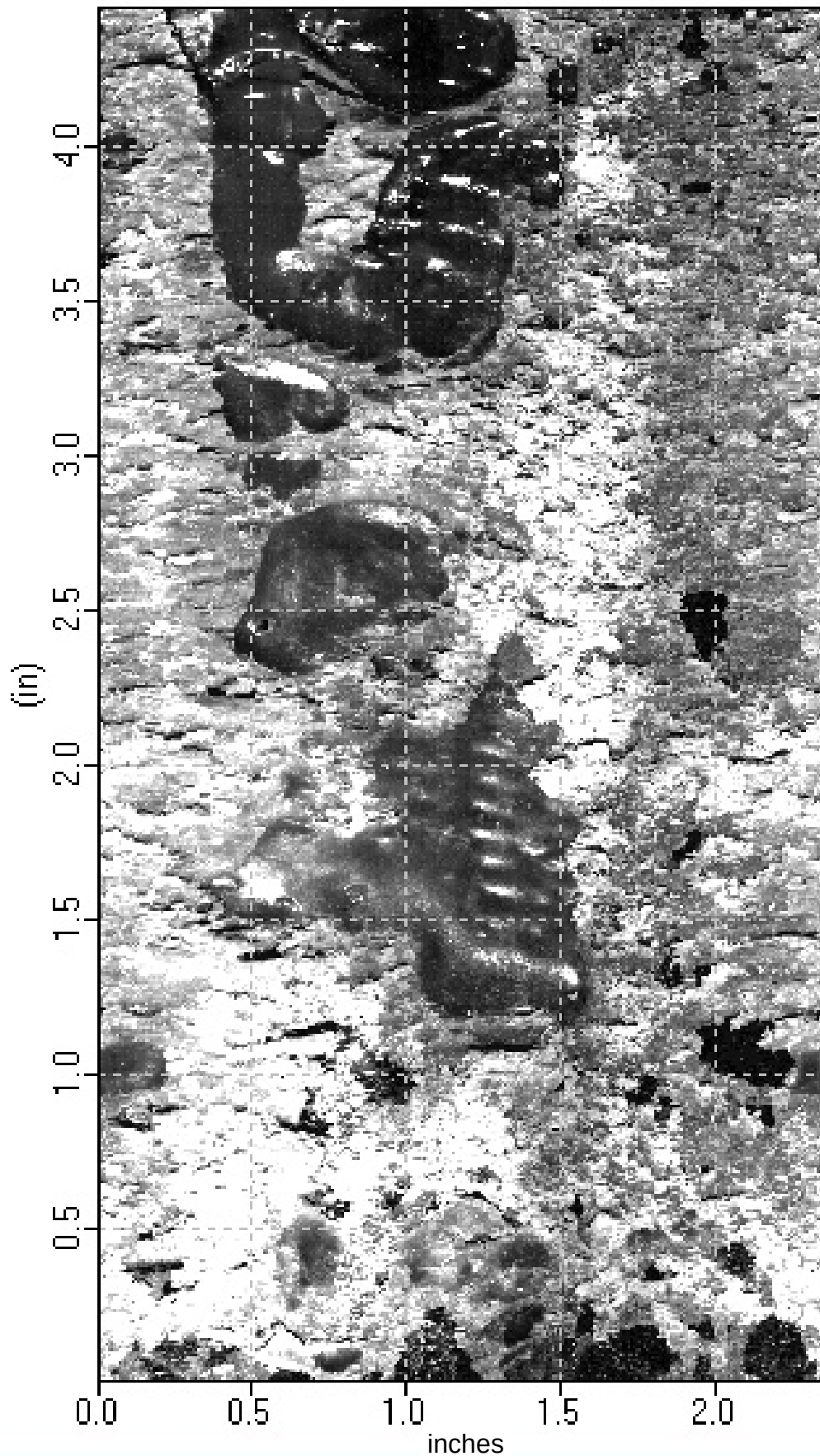
Laser scanned image of
the interior of the
anchor hole at position
"EB081-M1-5-S".



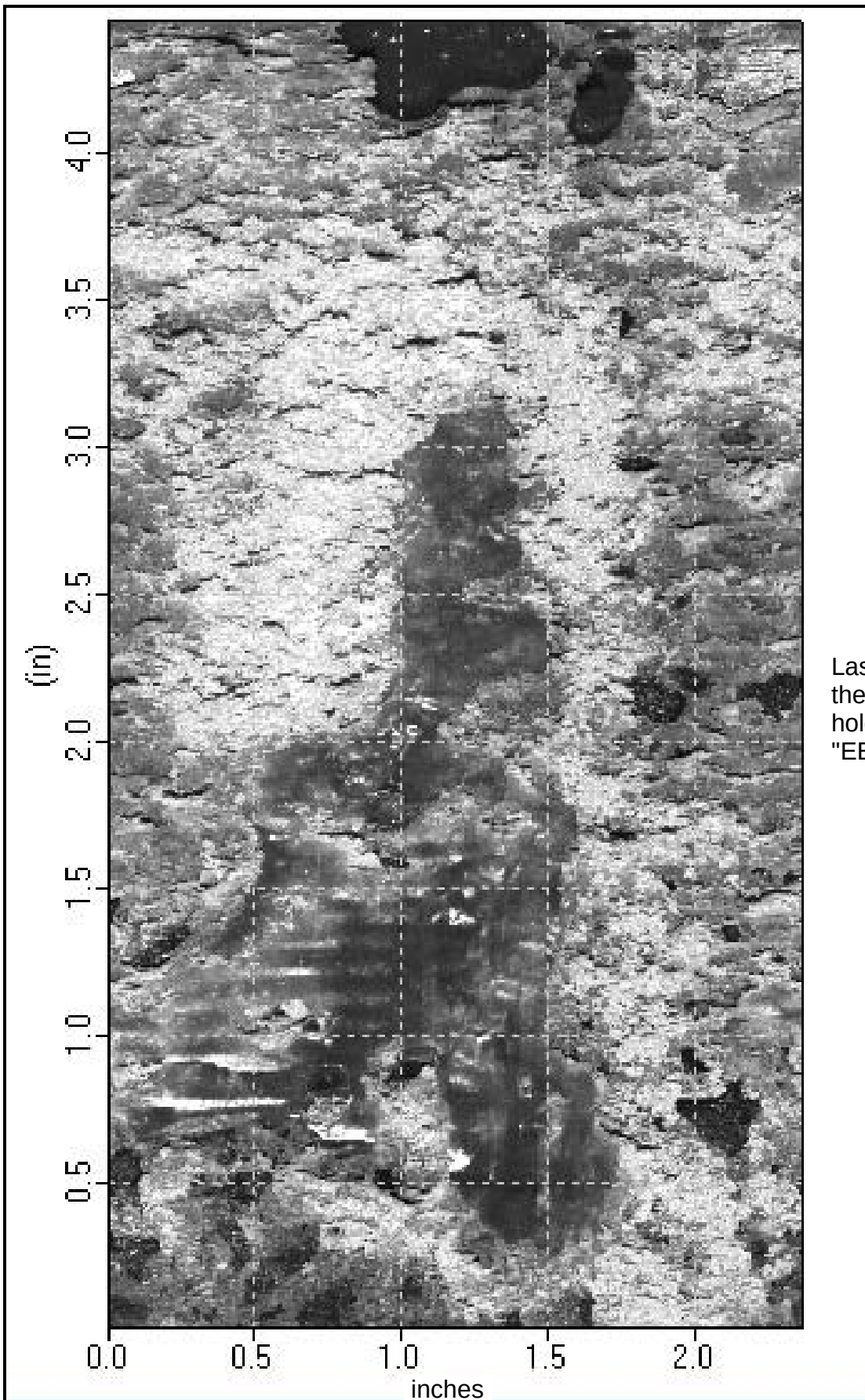
Laser scanned image of the interior of the anchor hole at position "EB081-M1-6-N".



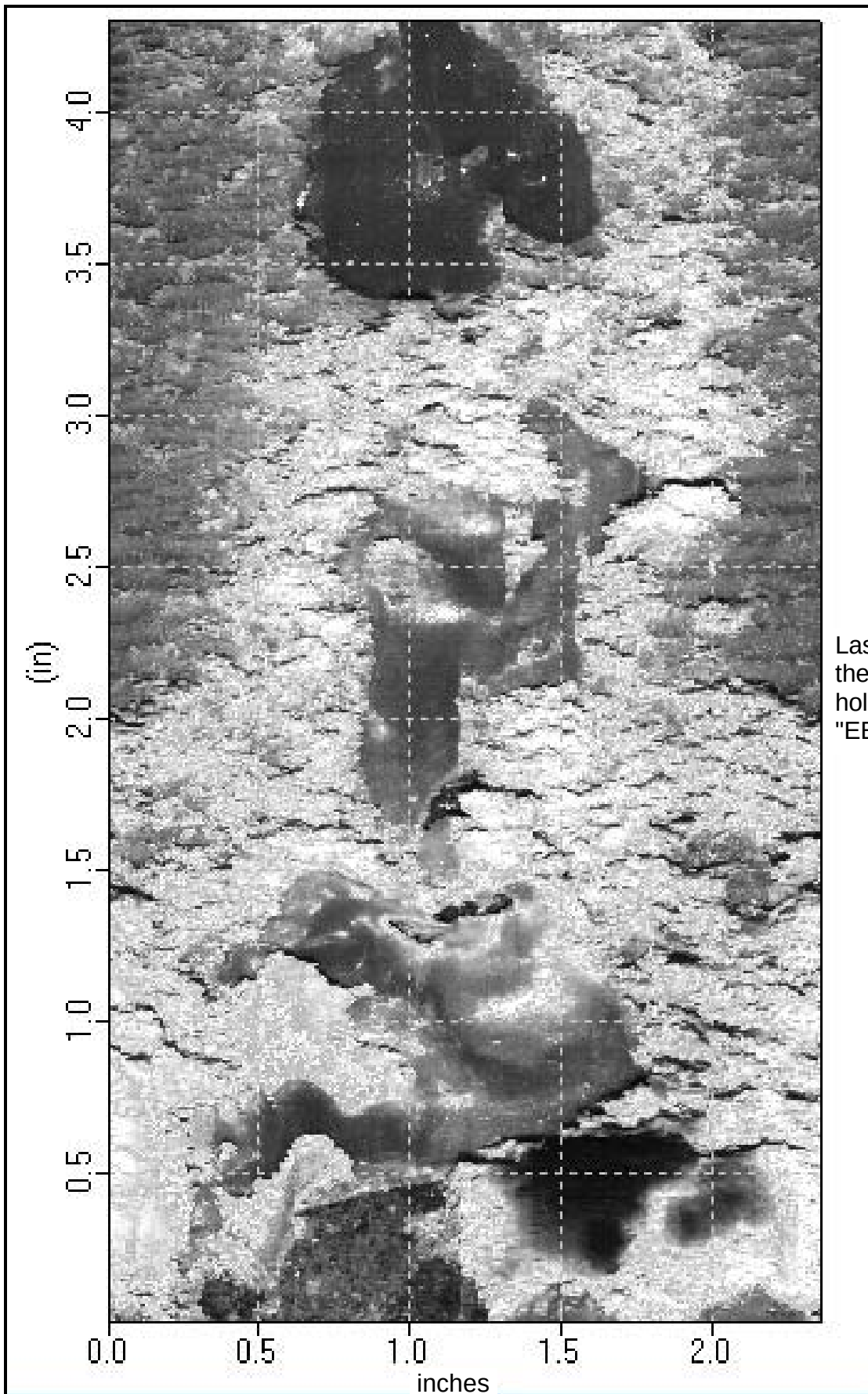
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-6-S".



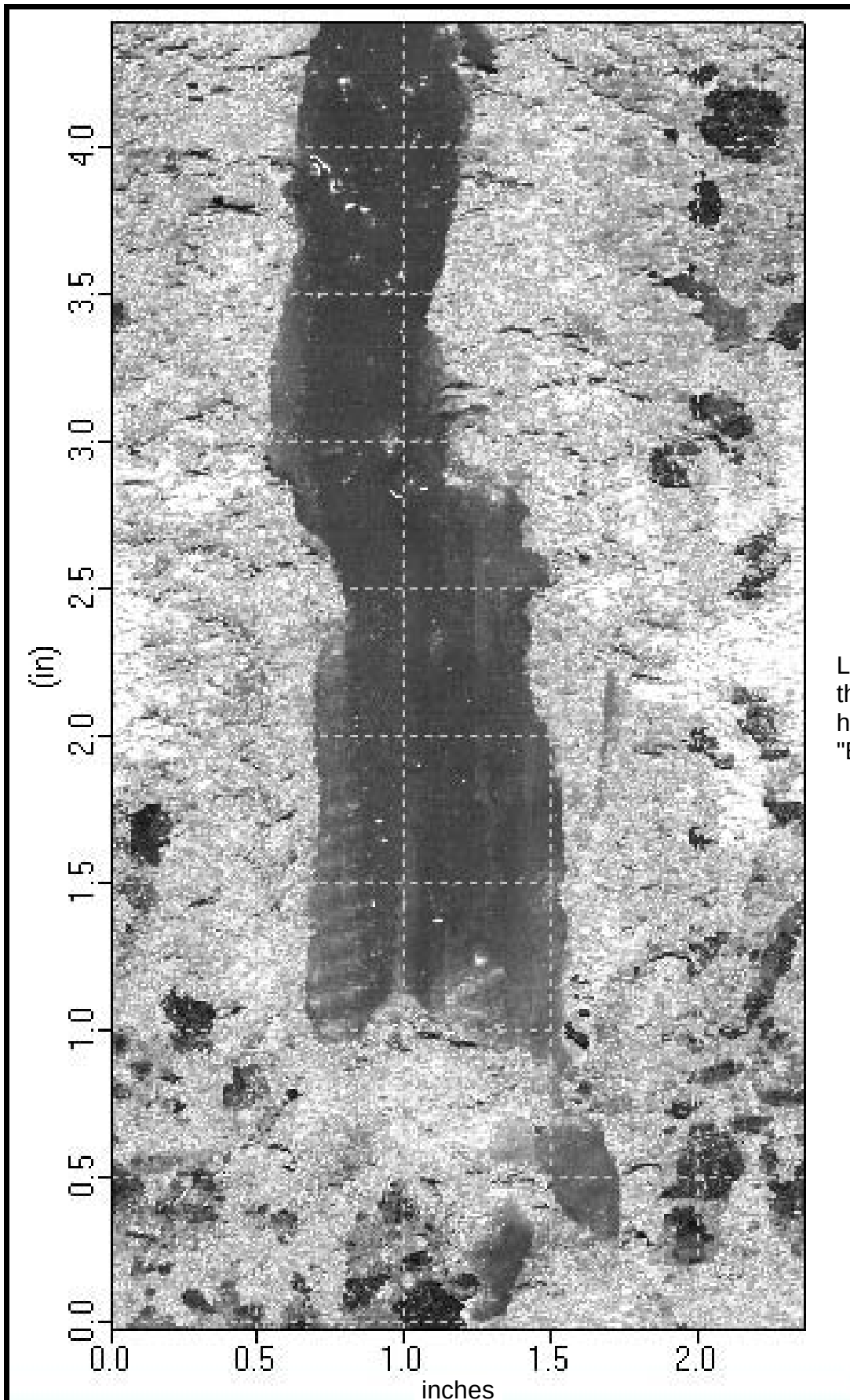
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-7-N".



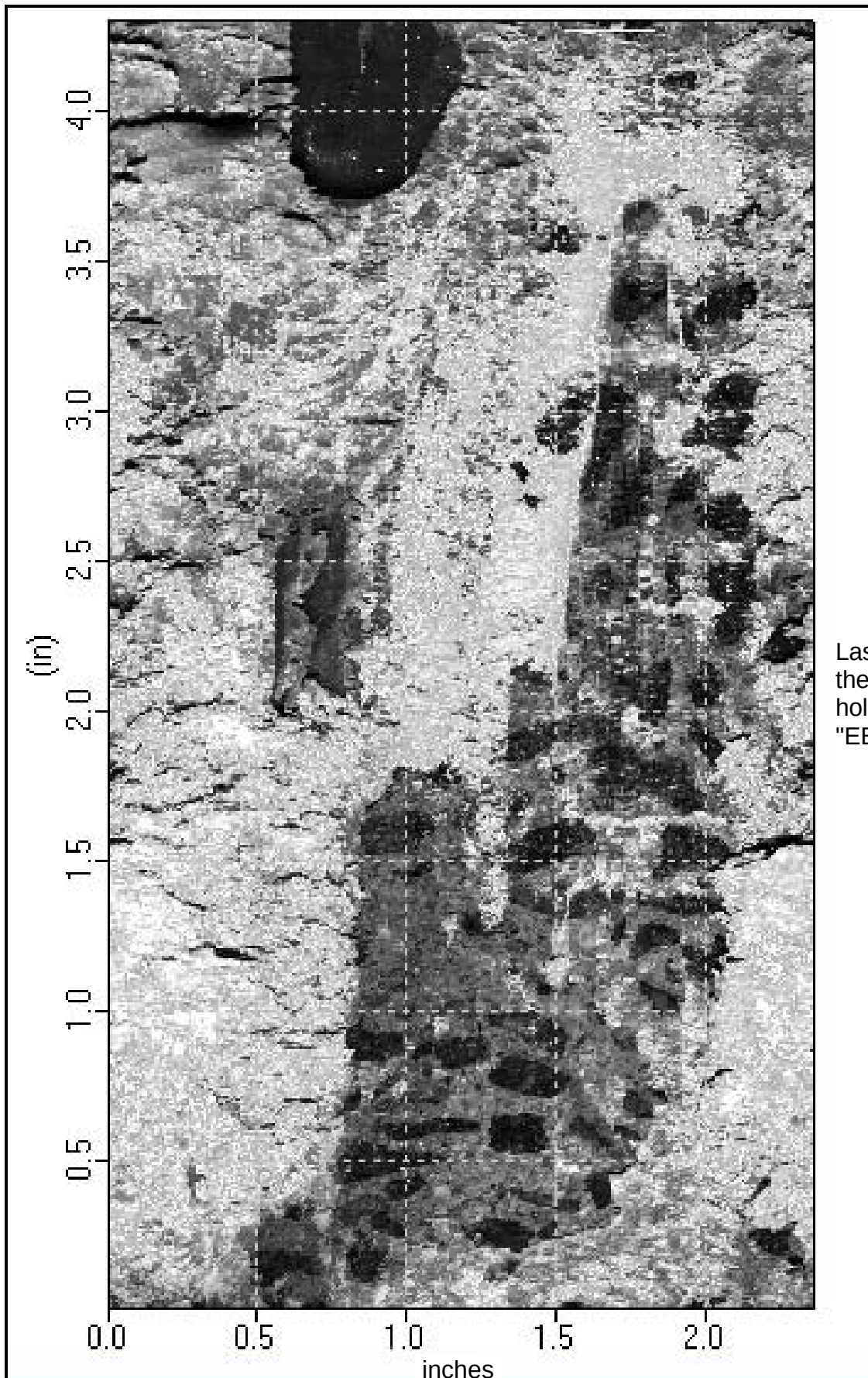
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-7-S".



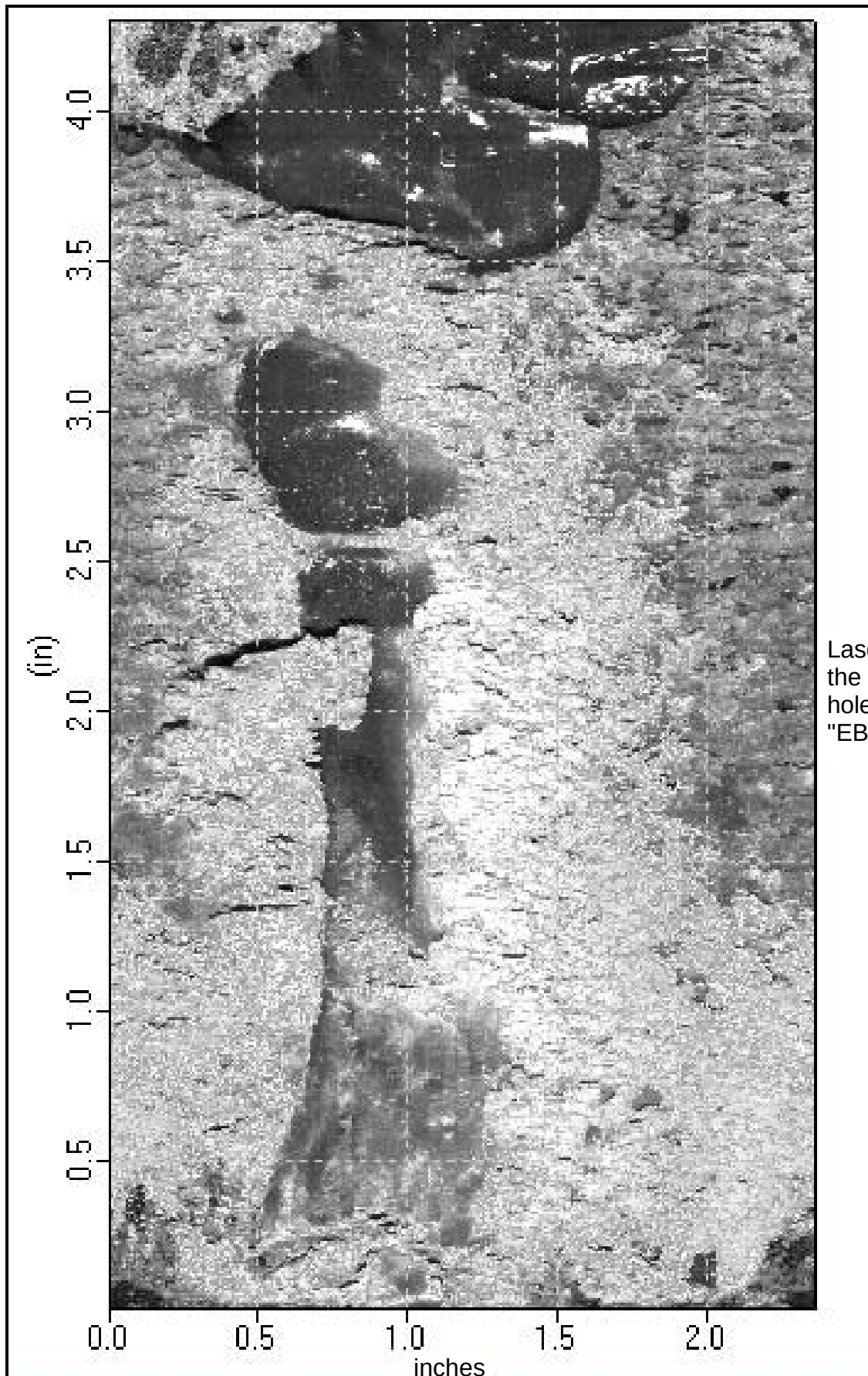
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-8-N".



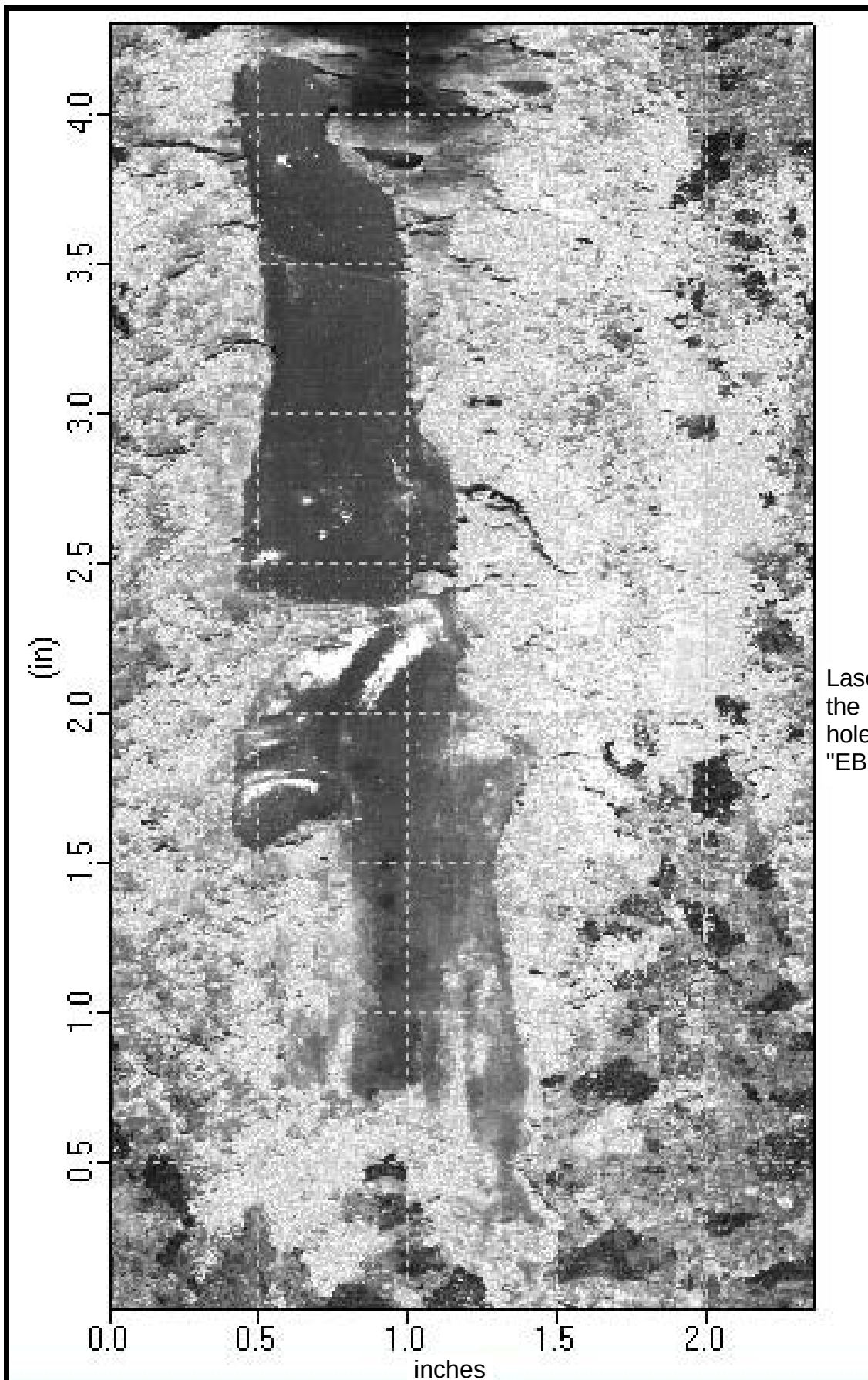
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-8-S".



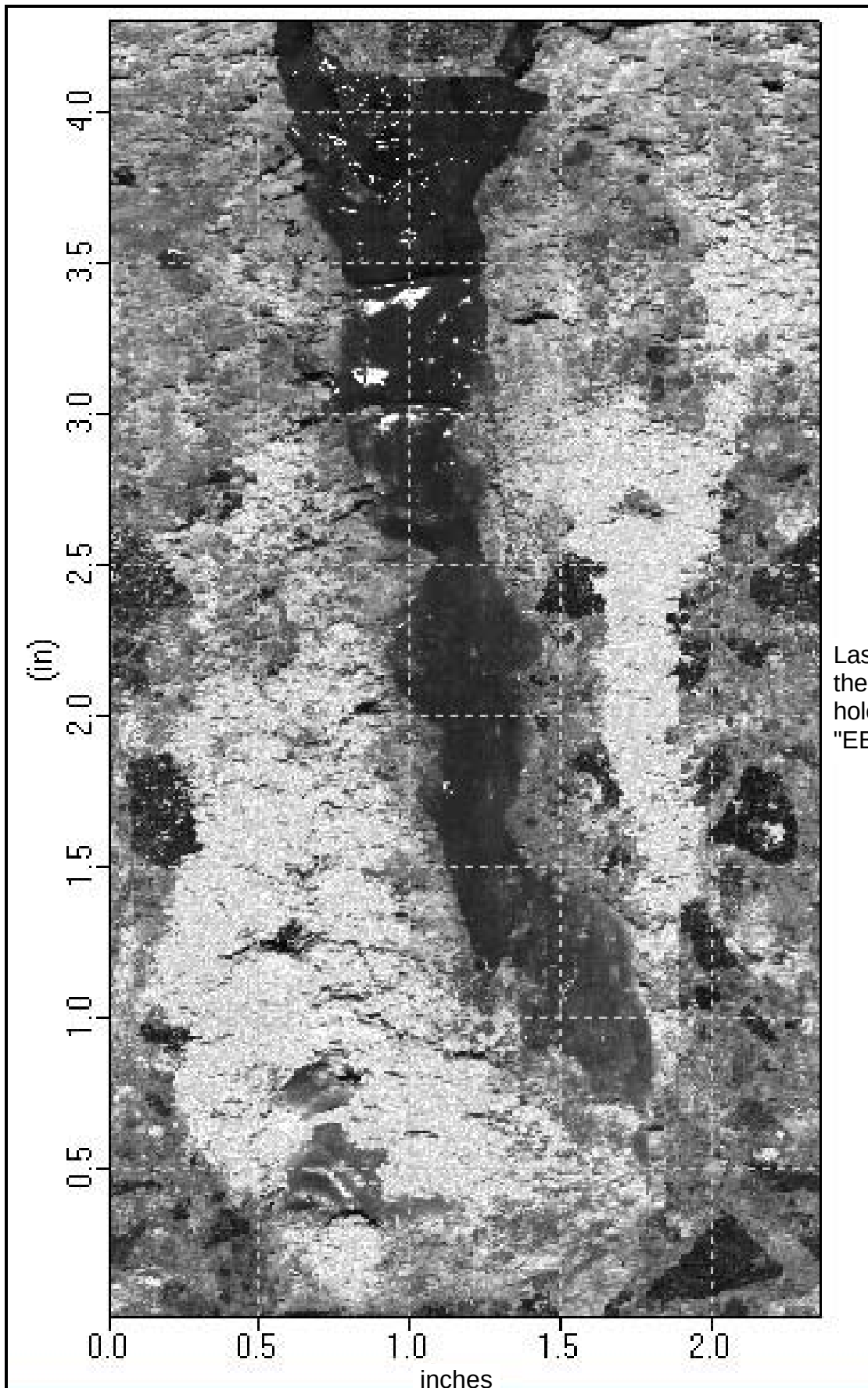
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-9/10/11-NE".



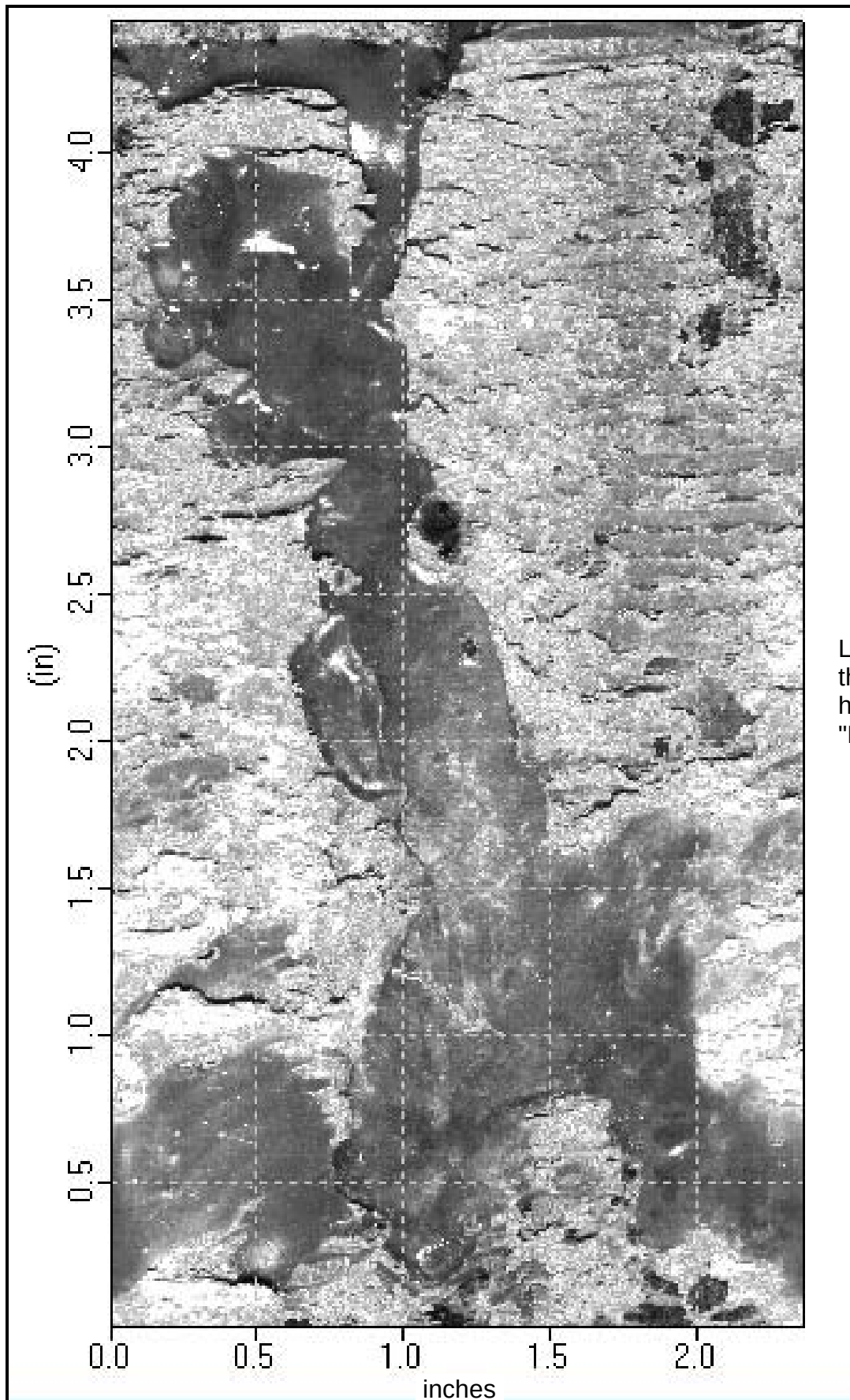
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-9/10/11-NW".



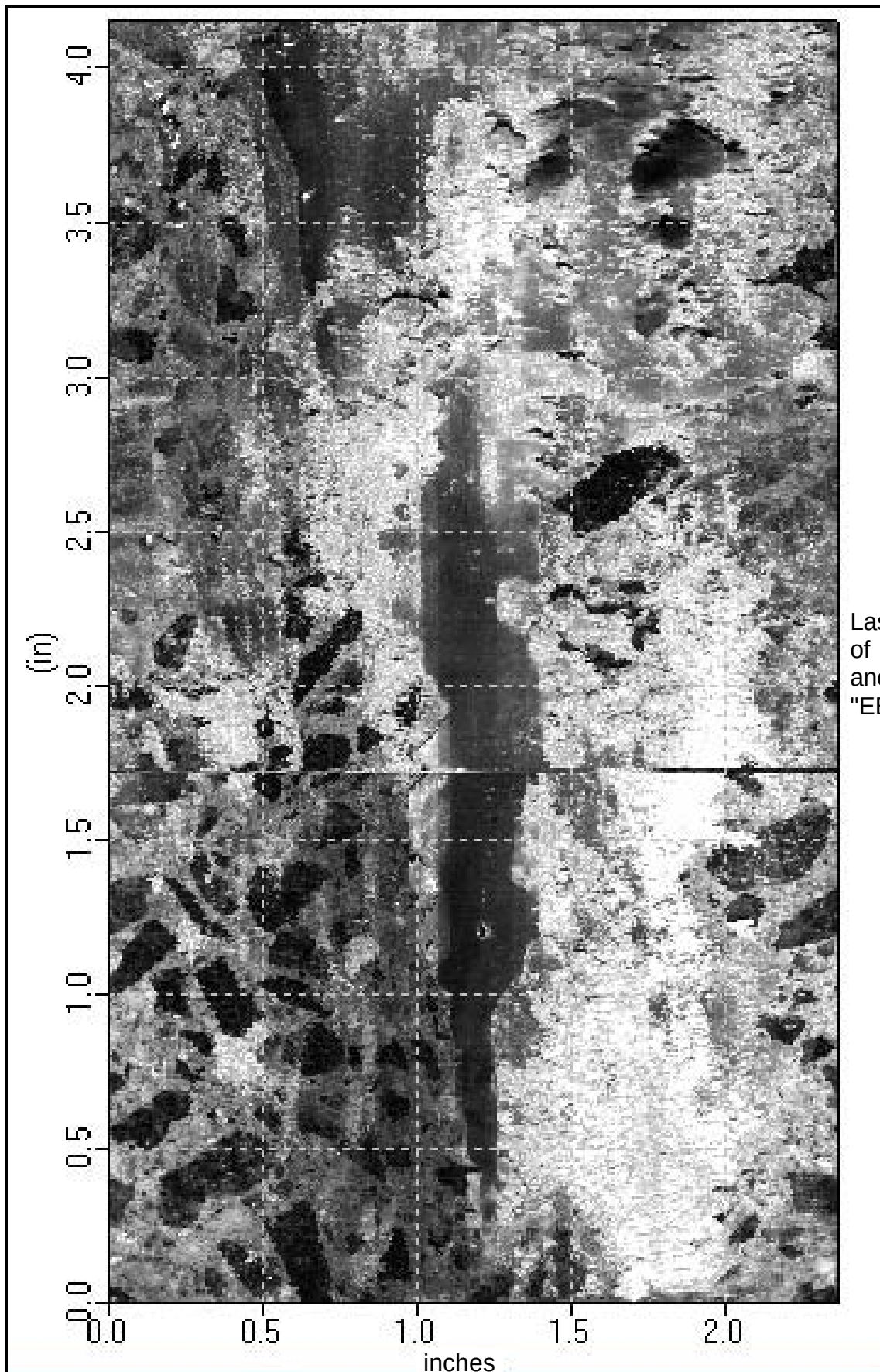
Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-9/10/11-SE"



Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-9/10/11-SW".

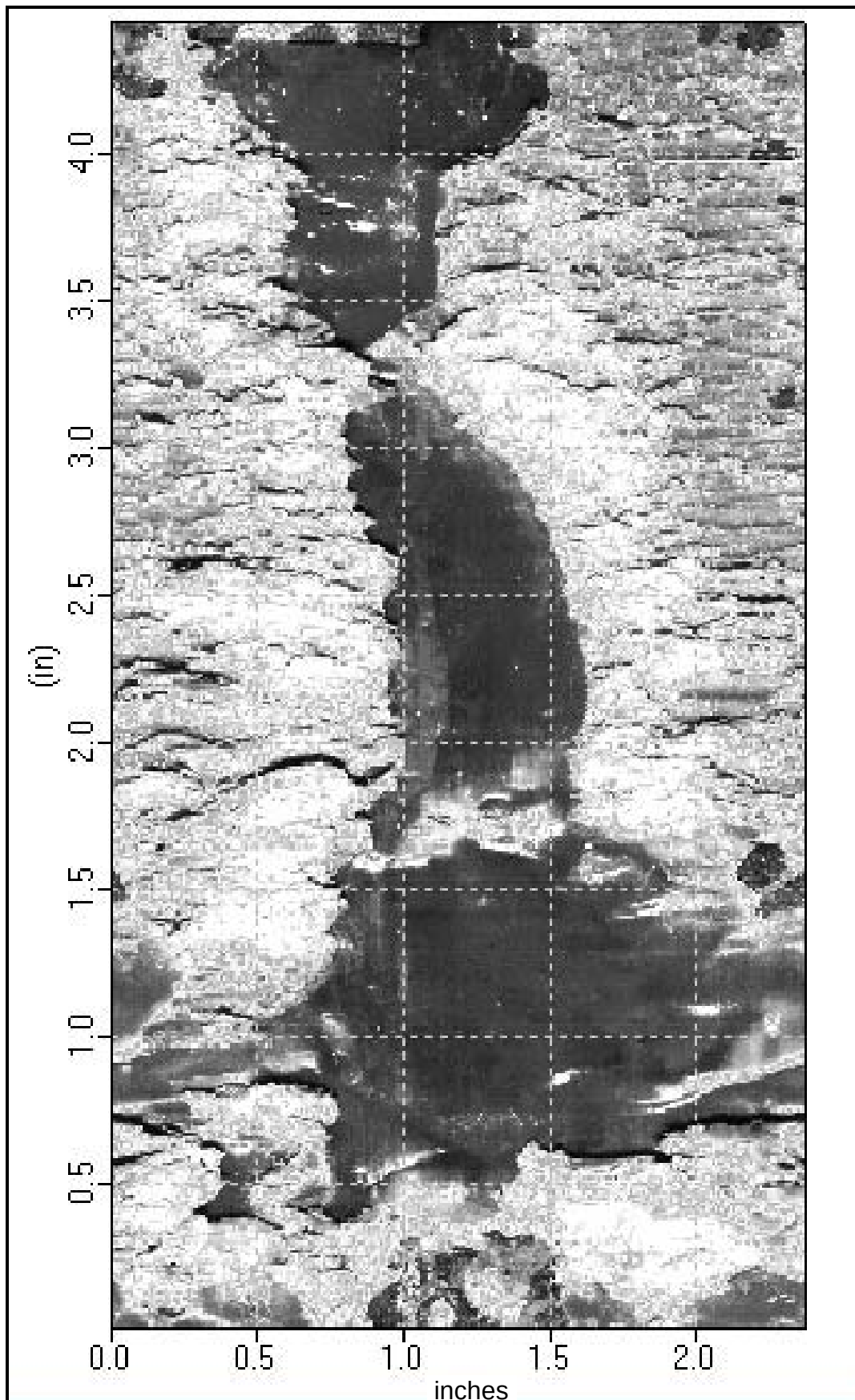


Laser scanned image of
the interior of the anchor
hole at position
"EB081-M1-12-N".

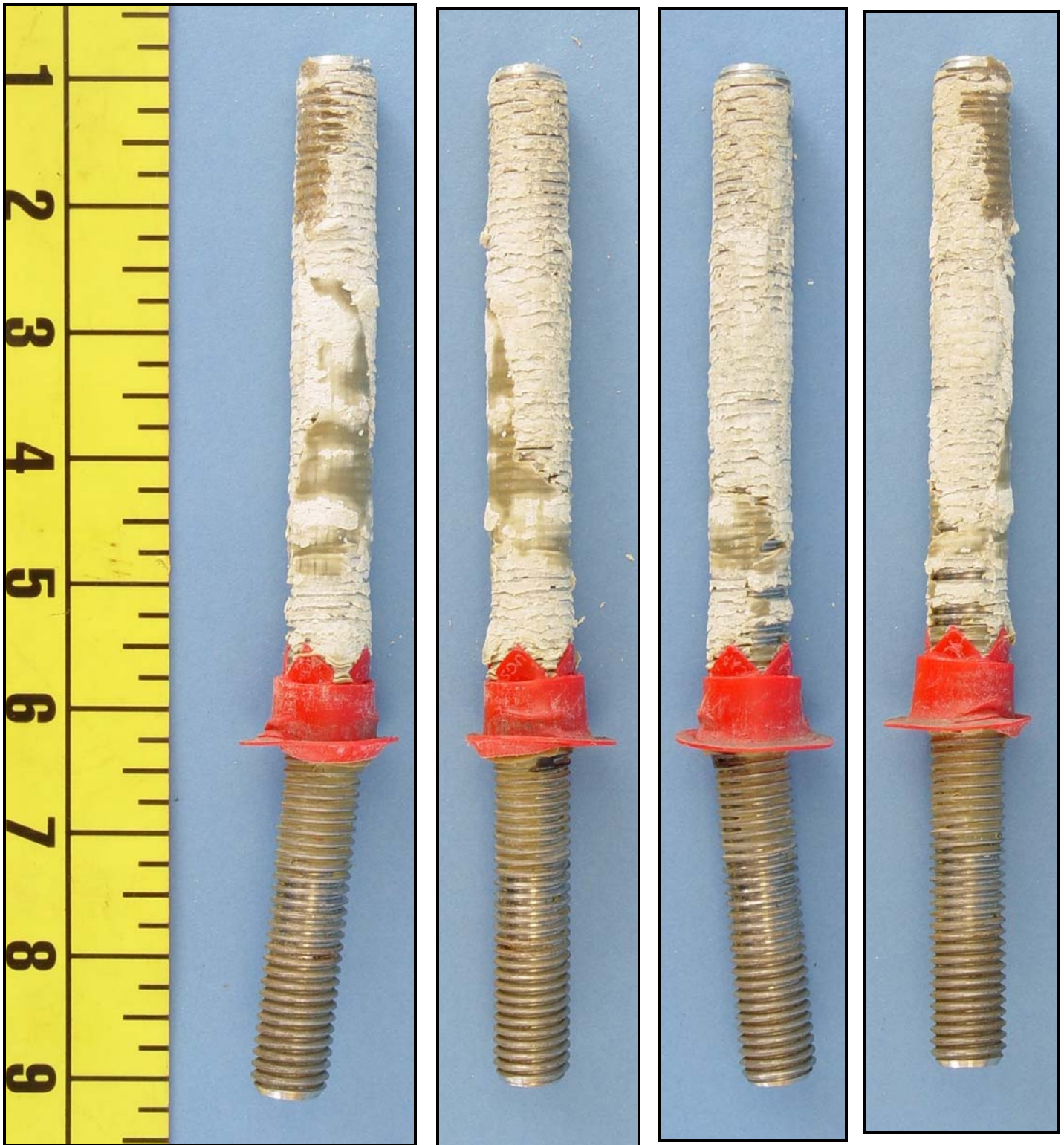


Laser scanned image
of the interior of the
anchor hole at position
"EB081-M1-12-S".

Appendix 3
Laser Scanned Images of Anchor Holes
and Associated Anchors
From Outside the Failure Area

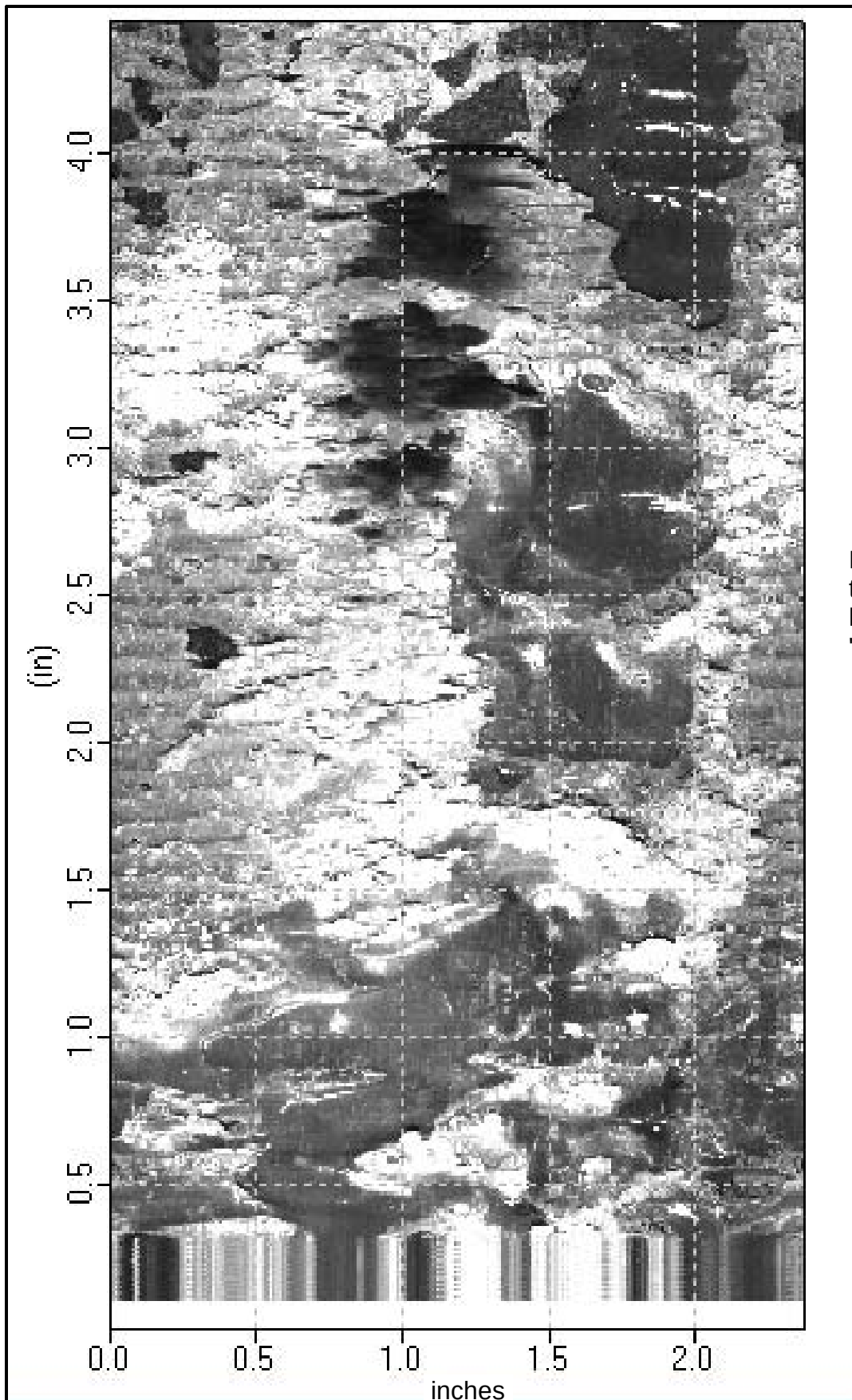


Laser scanned image
of the interior of the
anchor hole at position
"EB082-M1-12-N".



ImageNo: 0612A00404, Project No:2006080009

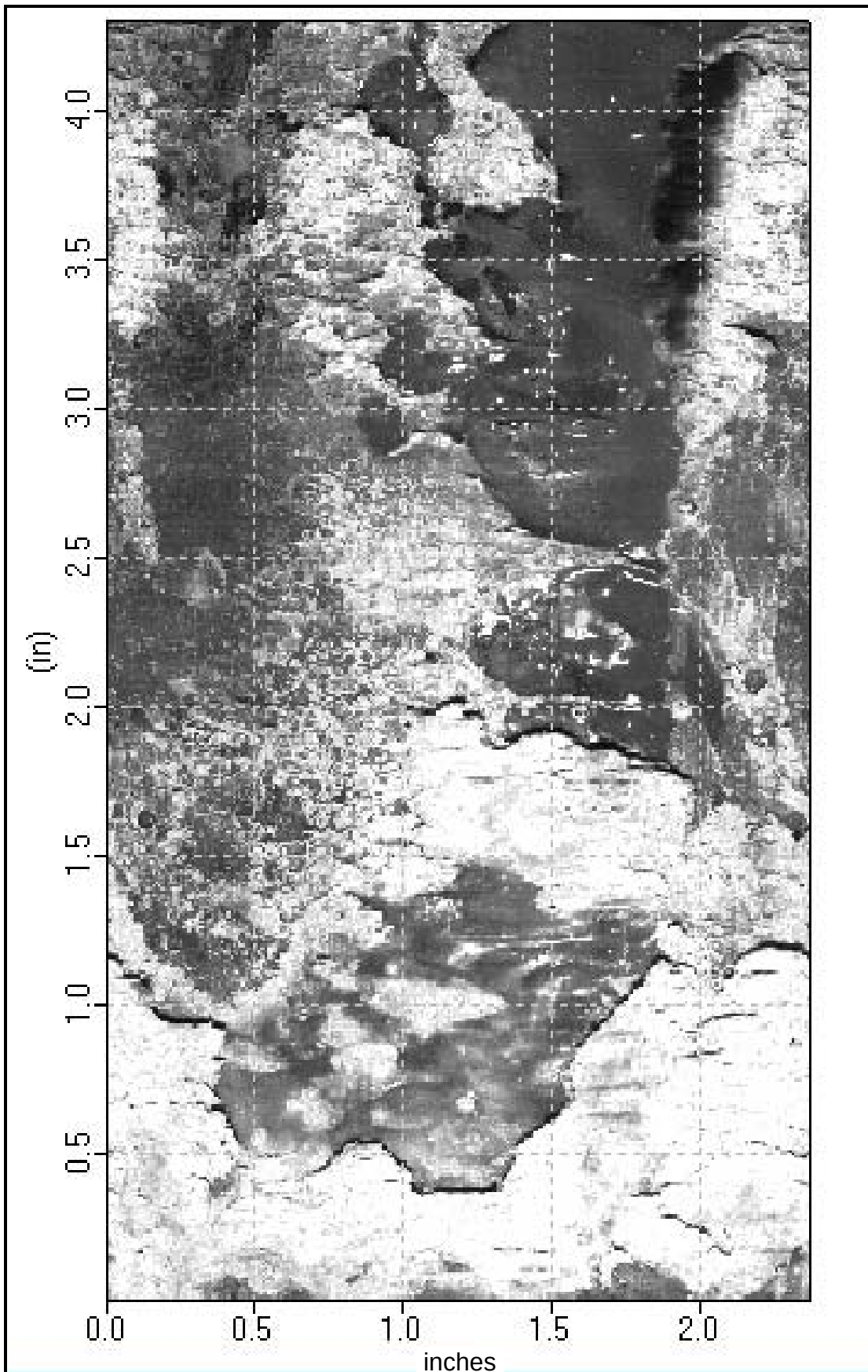
Four views of the anchor at position "EB082-M1-12-N".



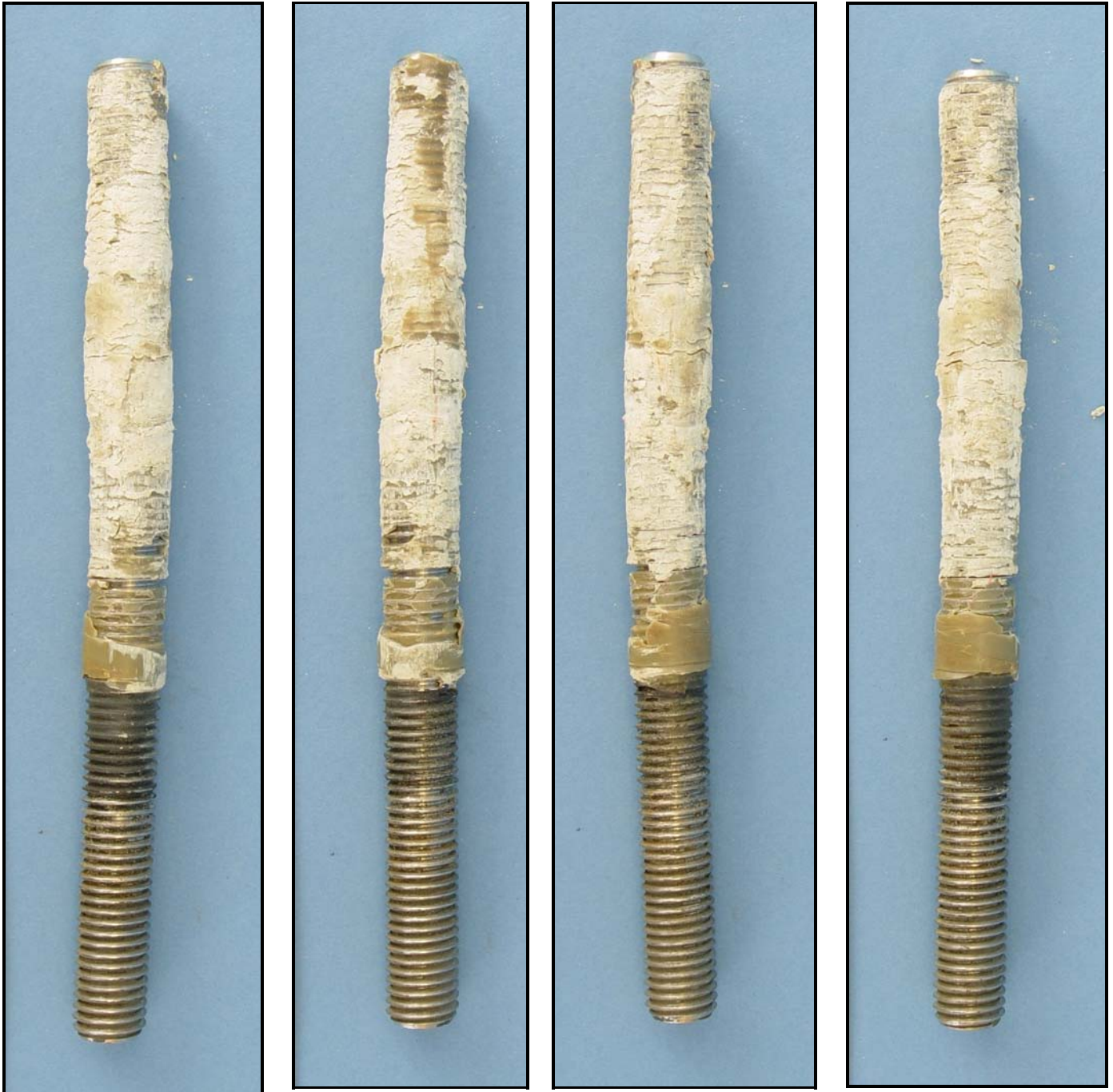
Laser scanned image of
the interior of the anchor
hole at position
"EB082-M1-12-S".



Four views of the anchor at position "EB082-M1-12-S".

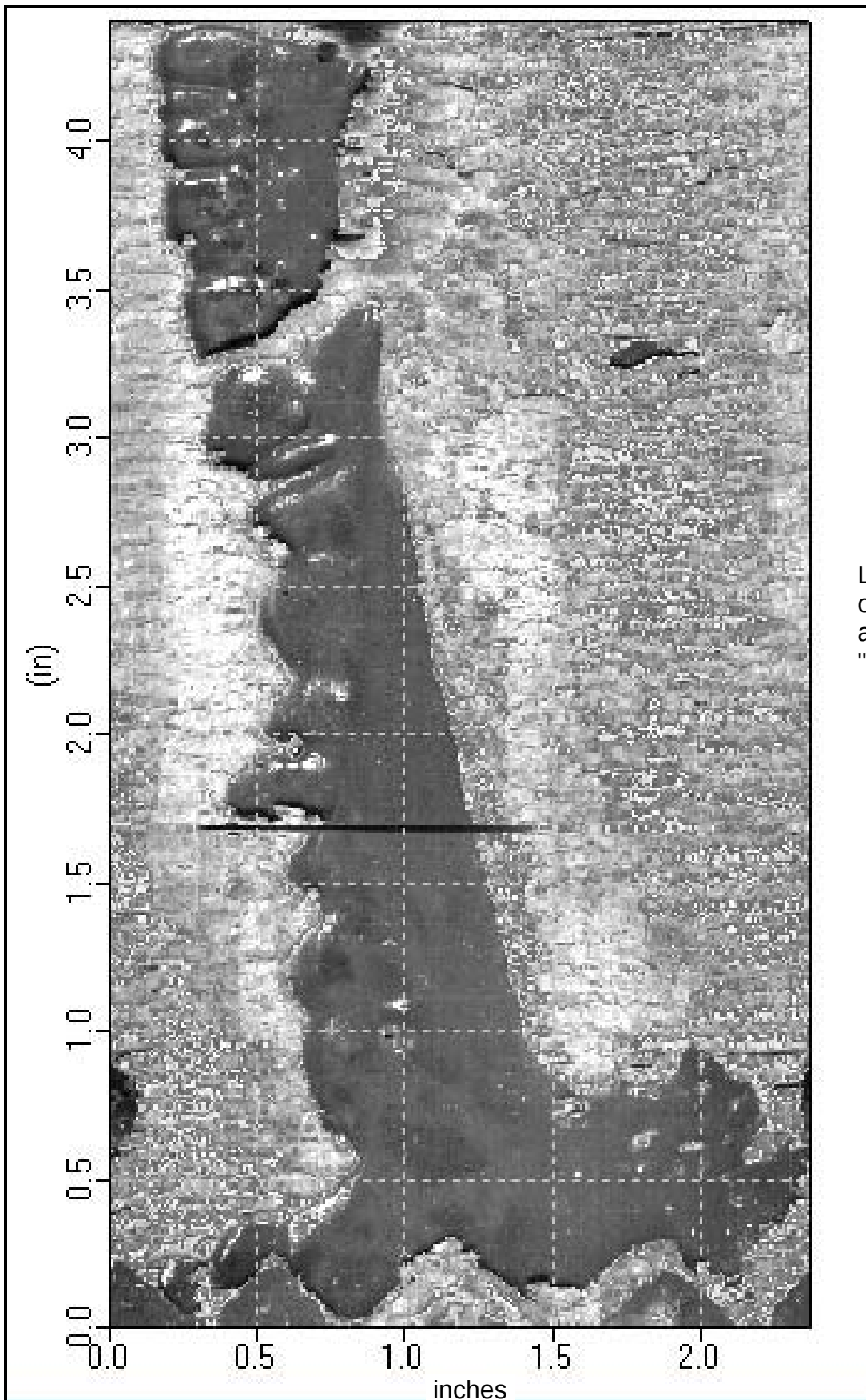


Laser scanned image
of the interior of the
anchor hole at position
"EB081-N-10-N".

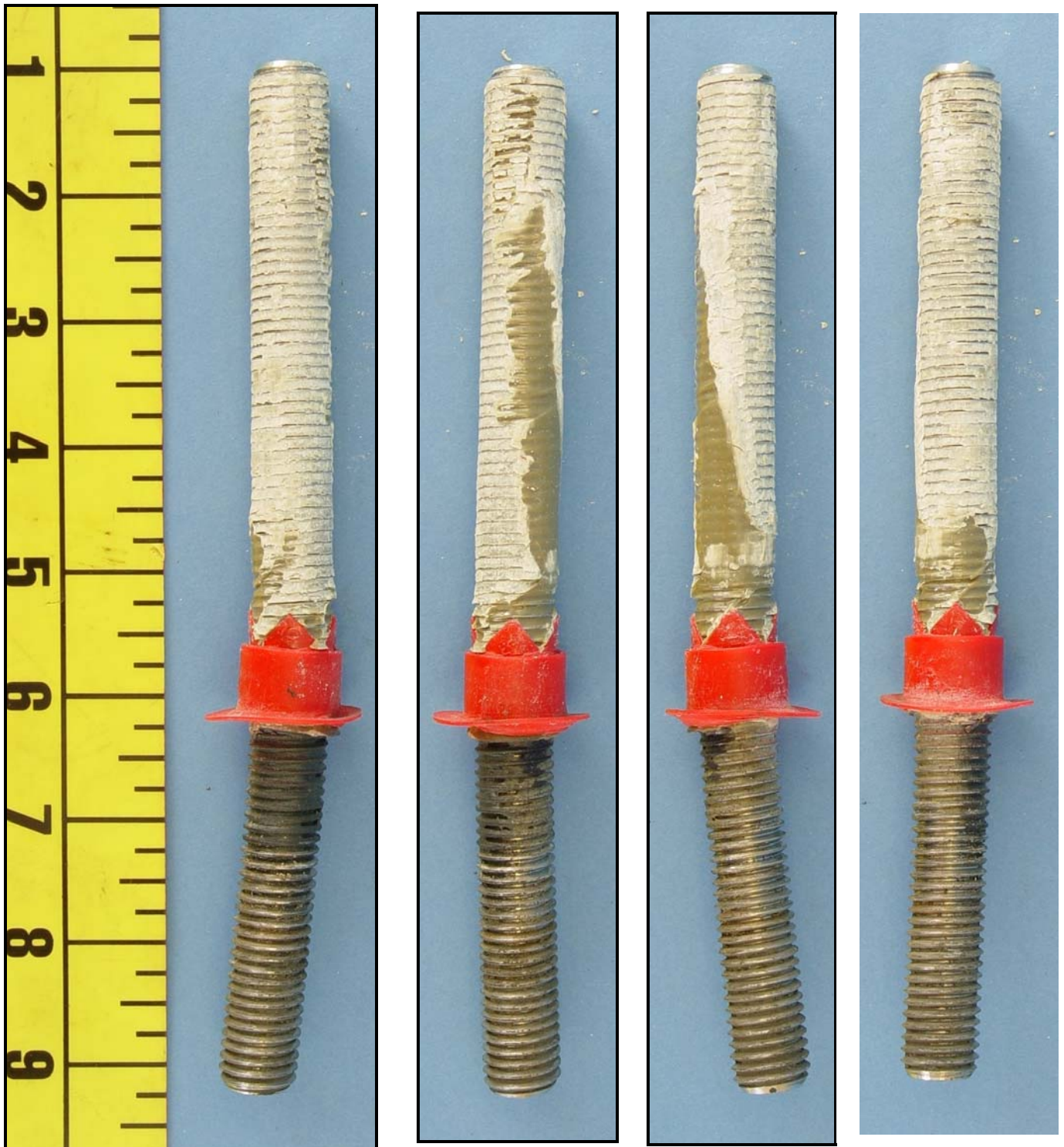


ImageNo: 0612A00409, Project

Four views of the anchor at position "EB081-N-10-N".



Laser scanned image of the interior of the anchor hole at position "EB081-N-10-S".



ImageNo: 0612A00413, Project No:200608000!

Four views of the anchor at position "EB081-N-10-S".